

Predictor Isolated Test Coverage

2026-06-11

1. Overview

Predictor **Isolated Test Coverage** predictor advisory query GNSS LiDAR FIM fusion fallback

isolated GTest system launch planner success /iap/integrity monitor fusion trajectory

test_predictor_module	19	PredictorModule GNSS advisory LiDAR advisory fusion advisory GNSS sweep current/advisory separation fusion gate
test_lidar_observability_fim	15	LiDAR observability/FIM corridor degeneracy primitive generation missing-input diagnostics
test_predicted_araim	5	GNSS predicted ARAIM legacy wrapper clock Schur complement advisory FIM

2. Build

iap

```
source install/setup.bash
colcon build --packages-select iap --cmake-args -DBUILD_TESTING=ON
```

Predictor isolated coverage

```
source install/setup.bash
colcon test --packages-select iap --ctest-args -R
"test_predictor_module|test_lidar_observability_fim|test_predicted_araim" --output-on-failure
```

```
ctest --test-dir build/iap -R 'test_predictor_module|test_lidar_observability_fim|test_predicted_araim' --output-on-failure
./build/iap/test_predictor_module --gtest_list_tests
./build/iap/test_lidar_observability_fim --gtest_list_tests
./build/iap/test_predicted_araim --gtest_list_tests
```

3. Results

```
Starting >>> iap
Finished <<< iap [4.71s]

Summary: 1 package finished [4.77s]
```

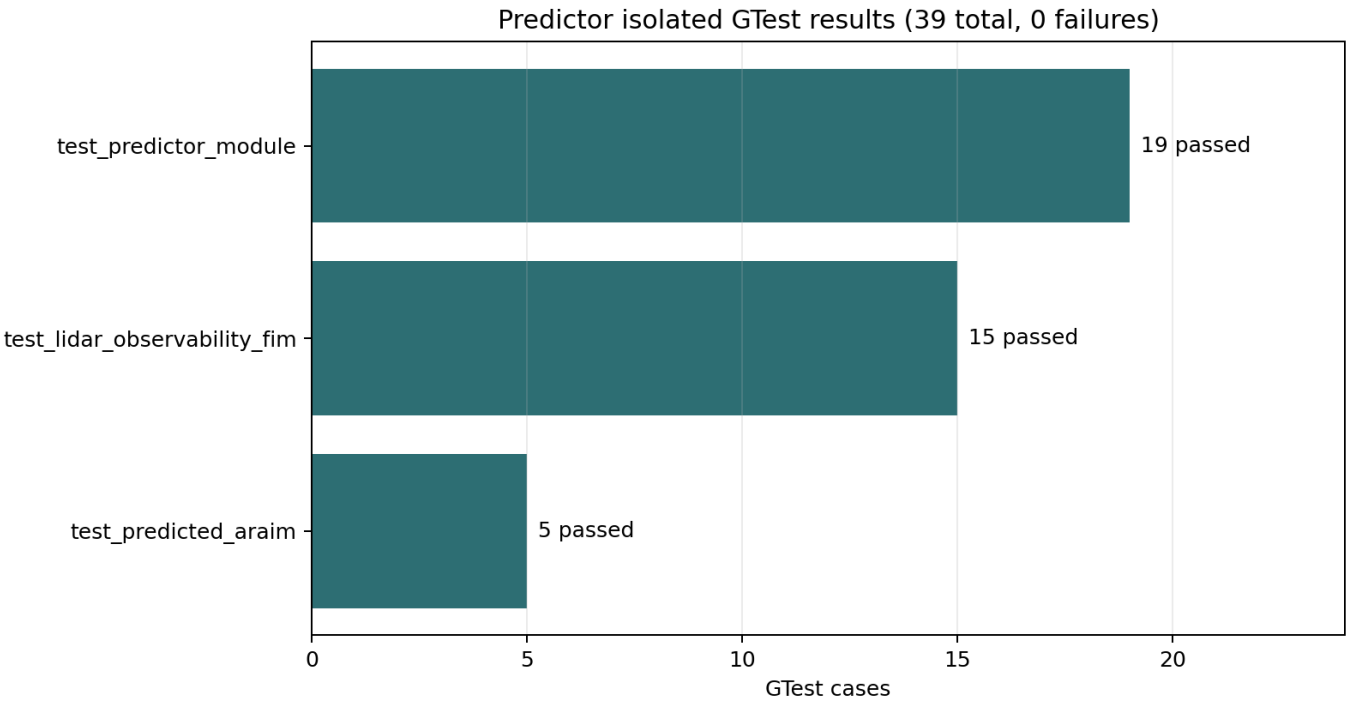
test_predicted_araim	Passed 5 cases	0.05 sec
test_lidar_observability_fim	Passed 15 cases	0.05 sec
test_predictor_module	Passed 19 cases	0.05 sec

```
100% tests passed, 0 tests failed out of 3

Label Time Summary:
gtest      = 0.15 sec*proc (3 tests)
```

Total Test time (real) = 0.15 sec

0000



4. GNSS Open-sky 8 advisory

C++

PredictorModuleTest.GnssOpenSkyProducesFinitePLAndFim
PredictedArimComputerTest.OpenSkyWithoutOccupancyProducesResult

0000

src/iap/test/test_predictor_module.cpp
src/iap/test/test_predicted_araim.cpp

4.1

GNSS predictor open-sky advisory result planner GNSS advisory

- valid == true
- available == true
- fallback == false
- HPL/VPL/PL
- 3x3 position-only GNSS FIM finite
- open-sky epoch

open-sky multi-satellite input returns finite HPL/VPL and valid 3x3 position FIM

4.2

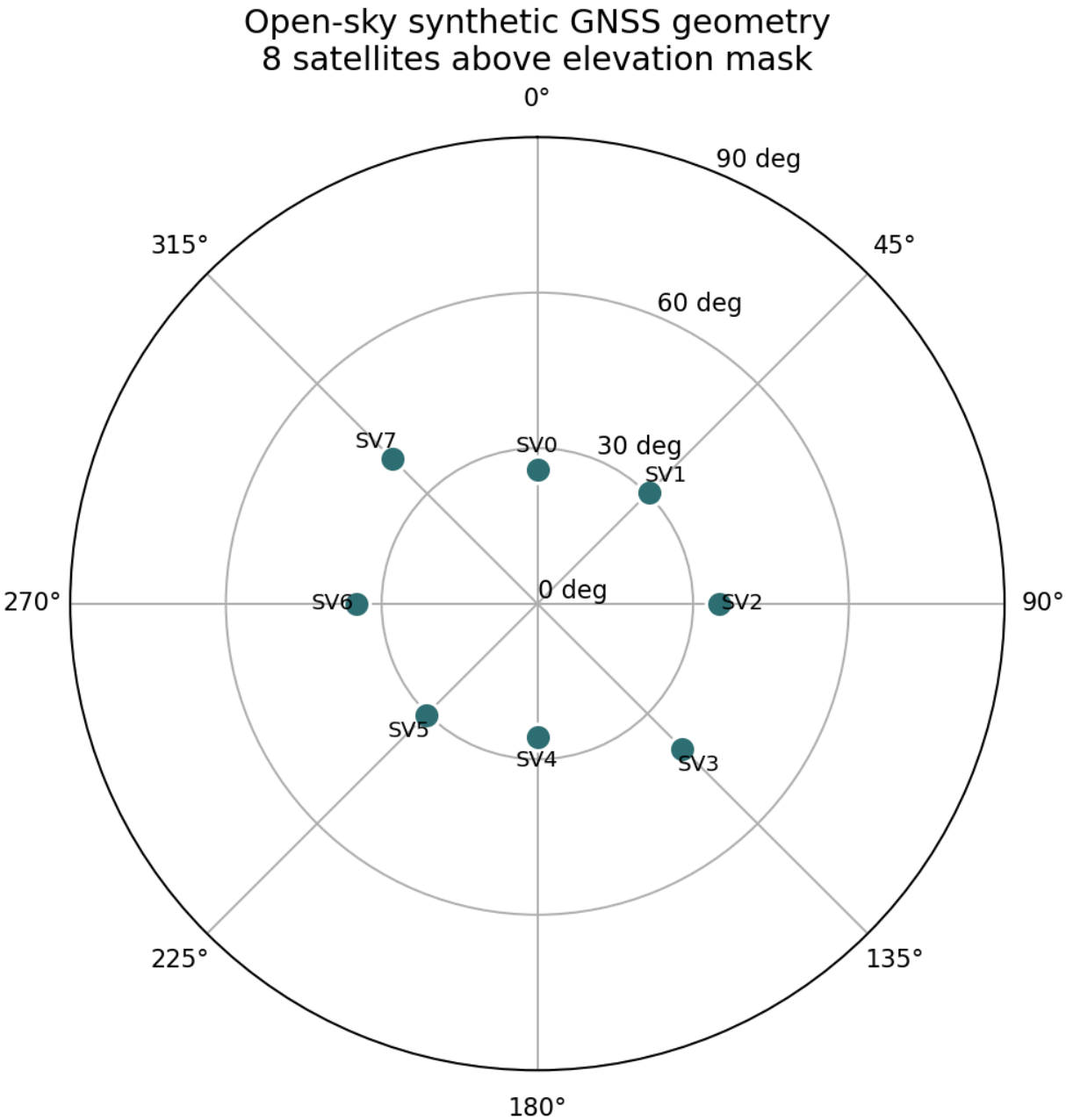
8 GNSS RINEX launch topic

stamp	100.0
gps_sec	2100000.0
sat_id	300 + i * 100 + i
constellation	G
elevation	0.45 + 0.08 * (i % 4) * 0.45 + 0.10 * (i % 4) rad

azimuth	$2\pi i/n_{\text{sats}}$
pr_sigma	$3.0 + (i \% 2) \text{ m}$
excluded	false

open-sky LocalOccupancyGrid visibility predictor elevation mask

8 open-sky sky plot zenith angle elevation



4.3

GNSS predictor dynamic budget

geometry_params.dynamic_budget	false	K factor
geometry_params.K_ff	5.0	fault-free PL scale
geometry_params.K_fa	4.0	false-alarm term scale
geometry_params.K_md	3.0	missed-detection term scale
geometry_params.min_sats	4	4 GNSS geometry advisory

Excluded satellites epoch geometry GNSS predictor visibility geometry excluded satellite epoch 8

5.3

open-sky GNSS

geometry_params.min_sats	4	4 fallback
gnss.fallback_pl	33.0	PredictedAraimComputer fallback PL sentinel
visibility_params.min_elevation	0.1 rad	missing/too-few/excluded mask

5.4

Predictor module

- valid == false
- available == false
- fallback == true
- fallback reason
- hpl/vpl/pl_scalar

Predicted ARAIM legacy

- missing epoch fallback_reason == "no_gnss_epoch"
- 3 fallback_reason == "too_few_sats"
- fallback HPL/VPL fallback sentinel

5.5

```
PredictorModuleTest.GnssMissingEpochIsExplicitFallback RUN COMPLETED
PredictorModuleTest.GnssTooFewSatsDoesNotReturnFiniteFallbackPl RUN COMPLETED
PredictorModuleTest.GnssExcludedSatellitesReduceUsedCountAndFallbackExplicitly RUN COMPLETED
PredictedAraimComputerTest.MissingEpochFallbackReason RUN COMPLETED
PredictedAraimComputerTest.TooFewSatellitesFallsBack RUN COMPLETED
```

5.6

GNSS predictor epoch excluded satellite fallback reason excluded satellites predictor epoch FDE advisory geometry

6. GNSS Map occlusion/skymask

C++

```
PredictorModuleTest.GnssMapOcclusionReducesVisibleCountAndDegradesProtectionLevels
```

```
src/iap/test/test_predictor_module.cpp
```

6.1

predictor GNSS visibility/skymask advisory debug map occlusion protection level

Map occlusion/skymask changes visible satellite count and degrades PDOP/HPL/VPL

6.2

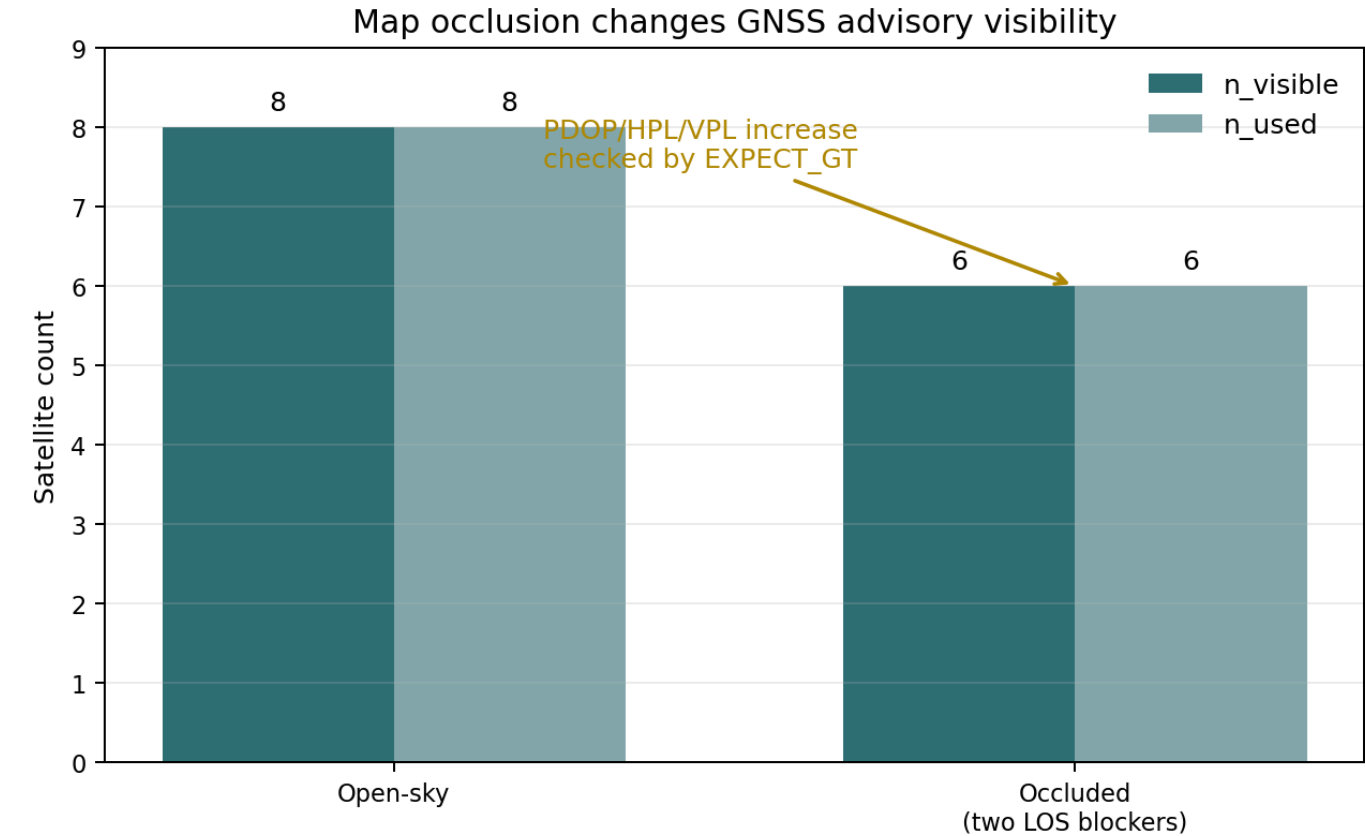
8 GNSS

	8
Query position	(0, 0, 0)

Occupancy mode	LocalOccupancyGrid
Hard occlusion	true
Occlusion range	6.0 m
Ray start offset	0.0 m
Blocker index	{0, 2}
Blocker range	2.0 m 2.25 m 2.5 m along each selected LOS

open-sky baseline index 0 index 2 LOS occupied voxels GNSS advisory

Open-sky occluded case PDOP/HPL/VPL artifacts GTest EXPECT_GT



6.3

visibility_params.hard_occlusion	true	occupancy ray hit
visibility_params.occ_range	6.0	blocker 2-2.5 m ray casting
visibility_params.ray_start_offset	0.0	blocker offset
LocalOccupancyGrid.voxel_size	0.25	blocker spacing voxel hit

6.4

Baseline open-sky

- open_sky.valid == true
- open_sky.n_visible == 8
- open_sky.n_used == 8

Occluded case

- occluded.valid == true
- occluded.fallback == false
- occluded.n_visible < open_sky.n_visible
- occluded.n_used < open_sky.n_used
- occluded.n_visible == 6
- occluded.n_used == 6

- `occluded.pdop > open_sky.pdop`
- `occluded.hpl > open_sky.hpl`
- `occluded.vpl > open_sky.vpl`

6.5

```
PredictorModuleTest.GnssMapOcclusionReducesVisibleCountAndDegradesProtectionLevels RUN COMPLETED
```

6.6

8 LOS blocker predictor 8 6 PDOP HPL VPL open-sky baseline map occlusion/skymask GNSS advisory protection level

6.7 GNSS geometry sweep

C++

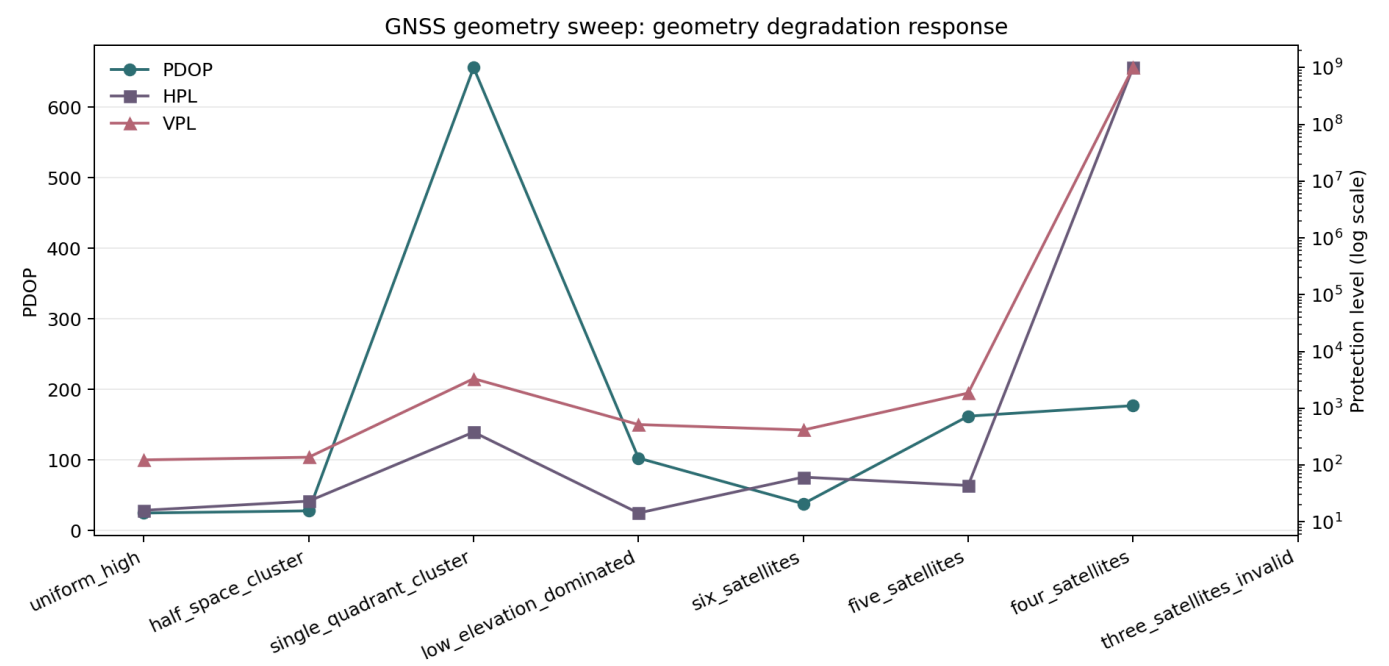
```
PredictorModuleTest.GnssGeometryDegradationSweepIncreasesPl
```

```
gnss_geometry_sweep.csv
```

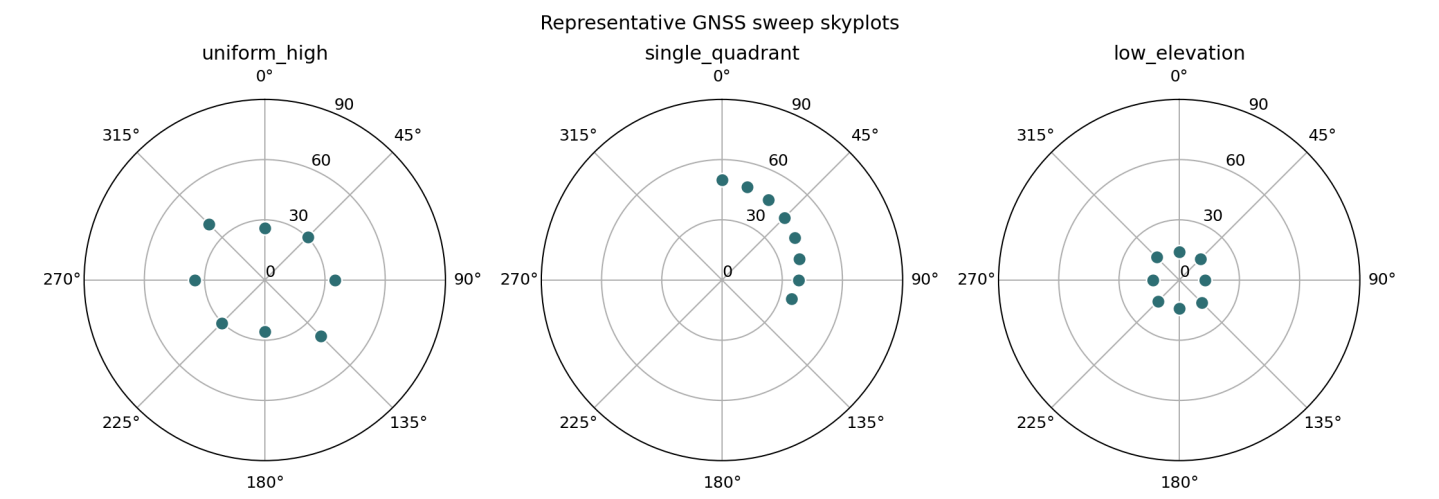
fixed synthetic GNSS epoch 8 geometry case 6 5 4 3 invalid case

```
case_id,n_used,azimuth_spread_deg,elevation_mean_deg,elevation_min_deg,
pdop,hdop,vdop,hpl,vpl,pl_e,pl_n,pl_u,
lambda_gnss_min_eig,lambda_gnss_condition,valid,fallback_reason
```

- `n_used >= min_sats` valid case finite HPL/VPL/FIM diagnostics
- `three_satellites_invalid` too_few_sats
- 4 geometry HPL/VPL open-sky baseline



skyplot



GNSS predictor geometry degradation 4 case PDOP/PL min_sats PL fallback

6.8 B GNSS sigma inflation sweep

C++

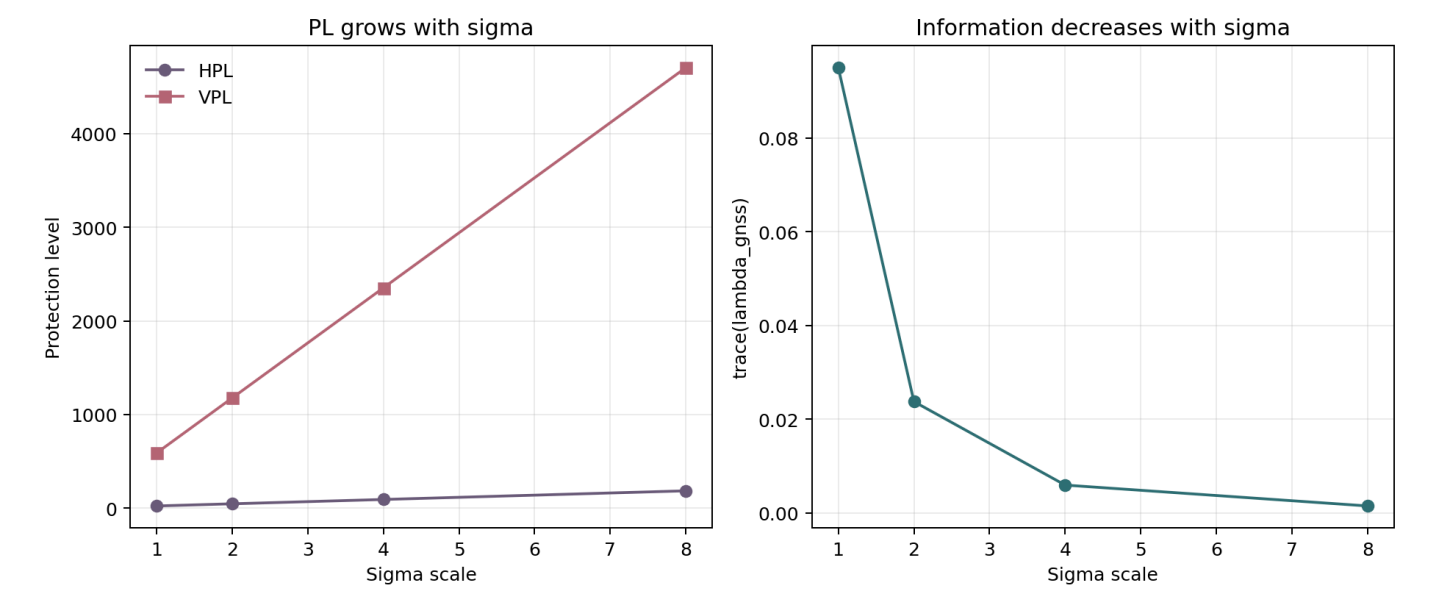
```
PredictorModuleTest.GnssSigmaInflationIncreasesPl
```

```
gnss_sigma_sweep.csv
```

8 geometry open-sky visibility/canopy noise model sigma

```
sigma_scale = 1x, 2x, 4x, 8x
```

- HPL/VPL sigma scale
- lambda_gnss_trace sigma scale
- case valid NaN/Inf



geometry measurement noise GNSS protection level predictor visible count effective sigma

7. GNSS Clock Schur complement advisory FIM

C++


```
PredictedAraimComputerTest.AdvisoryFimUsesClockSchurComplement
PredictorModuleTest.GnssOpenSkyProducesFinitePLAndFim
```

000000

```
src/iap/test/test_predicted_araim.cpp
src/iap/test/test_predictor_module.cpp
```

7.1 00000

GNSS 00 normal matrix 00 3D position 0 receiver clock state0Predictor fusion 000 map/ENU 3D position-only information000 GNSS predictor 000000
clock000 3x3 `lambda_gnss`0

0000000

- position-clock 4D normal matrix 0 Schur complement 00 clock
- 00 `lambda` 00000000
- 00 trace 00
- 000000000000 PSD 00000000

7.2 00000

00 8 000 GNSS 0000000000 4 0 open-sky 0000000000

```
predictor.predict_advisory_fim(Eigen::Vector3d::Zero())
```

0000 4x4 normal matrix `h_full`00000

- `h_pp` = `h_full.block<3,3>(0,0)`
- `h_pc` = `h_full.block<3,1>(0,3)`
- `h_cp` = `h_full.block<1,3>(3,0)`
- `h_cc` = `h_full(3,3)`

7.3 00000

00	0	00
<code>fim_clock_epsilon</code>	<code>1.0e-6</code>	00 clock block 00000
<code>min_sats</code>	<code>4</code>	8 00000 FIM 00000

7.4 0000

00000000000000

```
expected = h_pp - (h_pc * h_cp) / (h_cc + fim_clock_epsilon)
```

000000

- `result.valid` == `true`
- `result.n_used` == 8
- `norm(result.lambda - expected)` <= `1e-10`
- `result.trace` > 0.0
- `result.min_eig` >= `-1e-9`

7.5 0000000

000000

```
PredictedAraimComputerTest.AdvisoryFimUsesClockSchurComplement RUN COMPLETED
```

7.6 00000

GNSS advisory FIM 0 clock elimination 0 Schur complement 000000000000 finite00 trace 0 PSD 000000000000 Predictor fusion 000 GNSS `lambda_gnss`
0 position-only map/ENU information0000000 receiver clock 0 4D 000

8. LiDAR 00000Feature-rich 3D primitives FIM

C++

```
PredictorModuleTest.LidarRichPrimitivesProducesValidFim
LidarObservabilityFimTest.RichCloudProducesValidInformation
LidarObservabilityFimTest.AdvisoryFimReflectsNormalAnisotropy
```

#####

```
src/iap/test/test_predictor_module.cpp
src/iap/test/test_lidar_observability_fim.cpp
```

8.1 #####

LiDAR predictor feature-rich 3D geometry ##### FIM primitive normals ##### FIM

8.2 #####

primitives

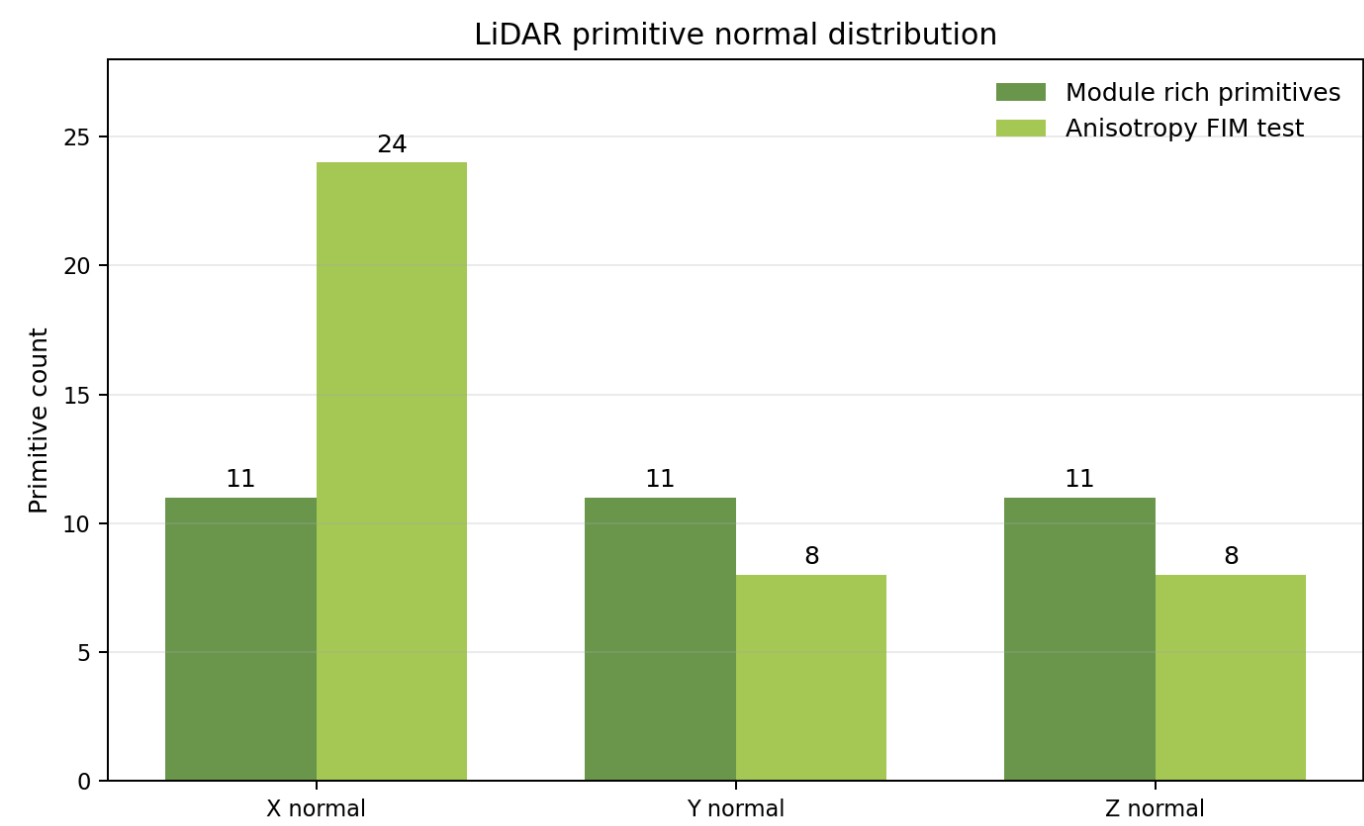
Primitive group	center	normal	
X normal	(0.4*i, 0, 0)	UnitX()	11
Y normal	(0, 0.4*i, 0)	UnitY()	11
Z normal	(0, 0, 0.4*i)	UnitZ()	11

LidarObservabilityFimTest.RichCloudProducesValidInformation dense 3D point cloud

X	ix=-3..3	0.7
Y	iy=-3..3	0.6
Z	iz=-2..2	0.5

rich cloud $7*7*5 - 1 = 244$

Feature-rich module test anisotropy test primitive normal



8.3 #####

Parameter	Value	Description
<code>fim_radius_m</code>	<code>10.0</code>	Radius of FIM primitives in meters
<code>fim_min_voxels</code>	<code>6</code>	Minimum number of voxels per primitive
<code>fim_range_sigma_base</code>	<code>1.0</code>	Base sigma for LiDAR range noise
<code>fim_condition_max</code>	<code>1.0e8</code>	Maximum FIM condition value
<code>enable_legacy_observability</code>	<code>false</code>	Whether to enable legacy observability checks

8.4

Checks

- `result.valid == true`
- `result.available == true`
- `result.fim_valid == true`
- `lambda_lidar` is 3x3 and finite
- `lambda_trace > 0.0`
- `n_primitives > 0`

Normal anisotropy

- X normal primitives `lambda(0,0)` `lambda(1,1)` `lambda(2,2)`
- `n_valid_normals == primitives.size()`

8.5

Tests

```
PredictorModuleTest.LidarRichPrimitivesProducesValidFim RUN COMPLETED
LidarObservabilityFimTest.RichCloudProducesValidInformation RUN COMPLETED
LidarObservabilityFimTest.AdvisoryFimReflectsNormalAnisotropy RUN COMPLETED
```

8.6

Feature-rich LiDAR primitives position-only LiDAR FIM Normal anisotropy FIM geometry normal distribution

9. LiDAR missing input diagnostics

C++

```
LidarObservabilityFimTest.TooFewPointsFallbackIsExplicit
LidarObservabilityFimTest.DegenerateGeometryLowersAlpha
LidarObservabilityFimTest.PoorCurrentTdopAndExclusionsDownweightAlpha
LidarObservabilityFimTest.MissingCloudFallbackIsFinite
LidarObservabilityFimTest.InvalidParamsFallbackIsFinite
PredictorModuleTest.LidarMissingPrimitivesIsExplicitFallback
LidarObservabilityFimTest.AdvisoryFimRequiresNormals
LidarObservabilityFimTest.EmptyPrimitiveGenerationReportsMissingNormals
```

Files

```
src/iap/test/test_lidar_observability_fim.cpp
src/iap/test/test_predictor_module.cpp
```

9.1

LiDAR predictor diagnostics FIM

-
-
- LiDAR TDOP trunk exclusion observability
- LiDAR map
-
- normals/primitives

9.2

9.5

```
LidarObservabilityFimTest.TooFewPointsFallbackIsExplicit RUN COMPLETED
LidarObservabilityFimTest.DegenerateGeometryLowersAlpha RUN COMPLETED
LidarObservabilityFimTest.PoorCurrentTdopAndExclusionsDownweightAlpha RUN COMPLETED
LidarObservabilityFimTest.MissingCloudFallbackIsFinite RUN COMPLETED
LidarObservabilityFimTest.InvalidParamsFallbackIsFinite RUN COMPLETED
PredictorModuleTest.LidarMissingPrimitivesIsExplicitFallback RUN COMPLETED
LidarObservabilityFimTest.AdvisoryFimRequiresNormals RUN COMPLETED
LidarObservabilityFimTest.EmptyPrimitiveGenerationReportsMissingNormals RUN COMPLETED
```

9.6

LiDAR predictor fallback diagnostics finite LiDAR advisory FIM failure mode FIM PL

9.7 DCorridor / one-sided LiDAR degeneracy

C++

```
LidarObservabilityFimTest.CorridorGeometryWeakensAlongCorridorInformation
```

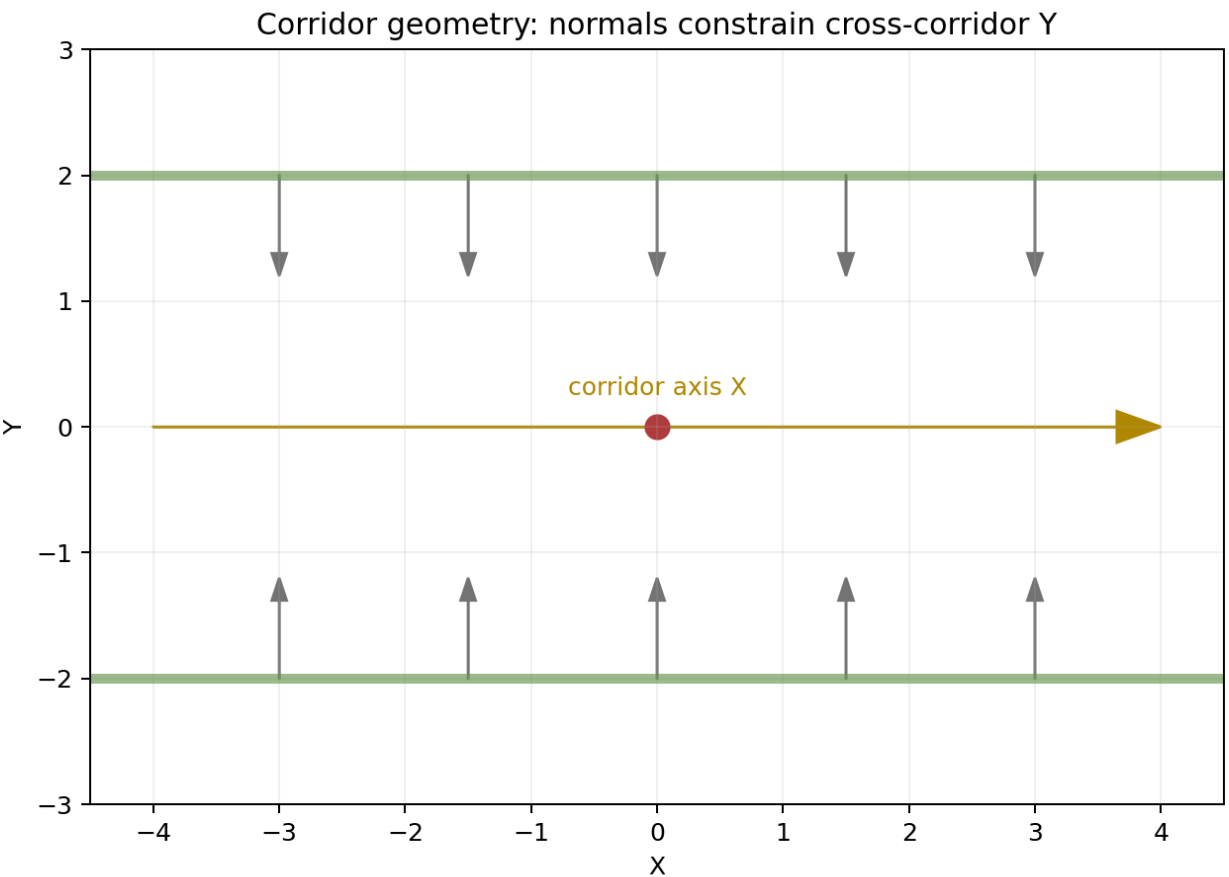
```
lidar_corridor_degeneracy.csv
```

LiDAR FIM primitives

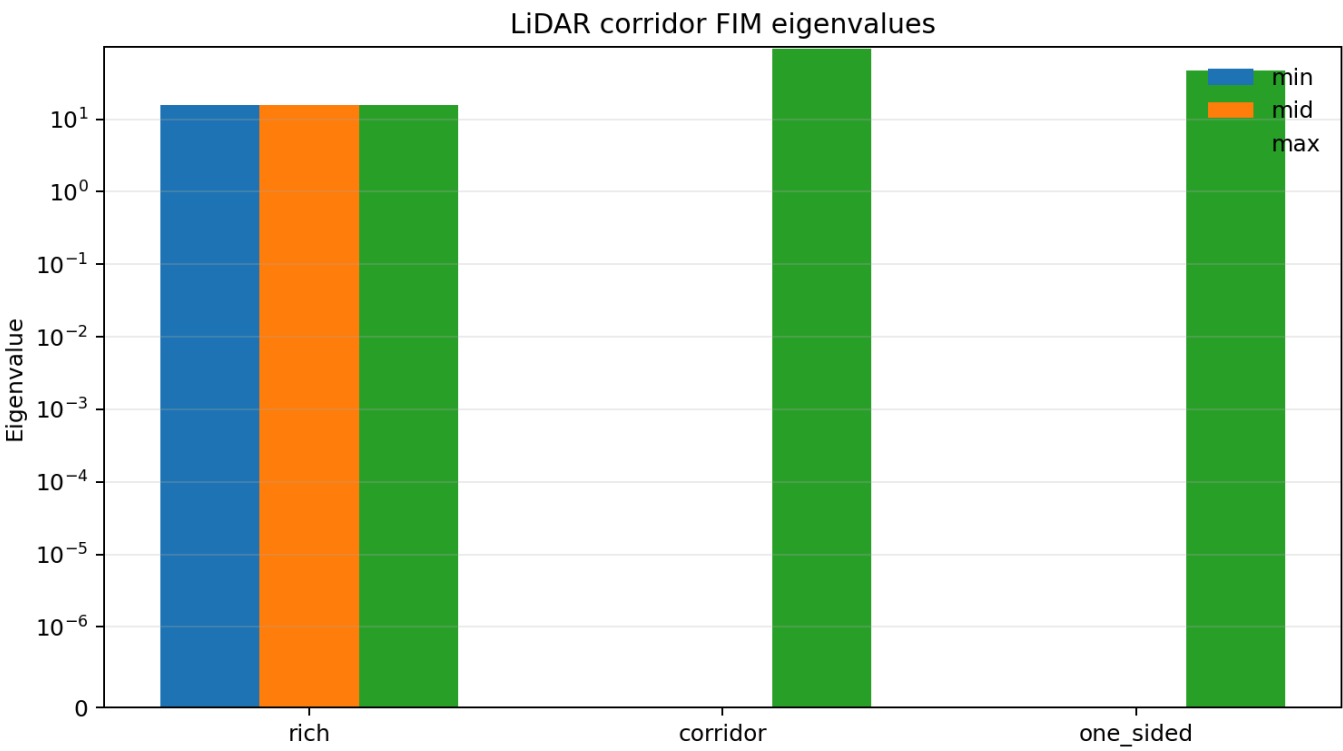
Case		
rich	X/Y/Z normals	feature-rich baseline
corridor	normal $\pm Y$ X	
one_sided	normal	

```
case_id,n_primitives,lambda_xx,lambda_yy,lambda_zz,
min_eig,mid_eig,max_eig,condition,tdop,normal_diversity_score,
along_corridor_sigma,degeneracy_score,allowed_for_fusion,fallback_reason
```

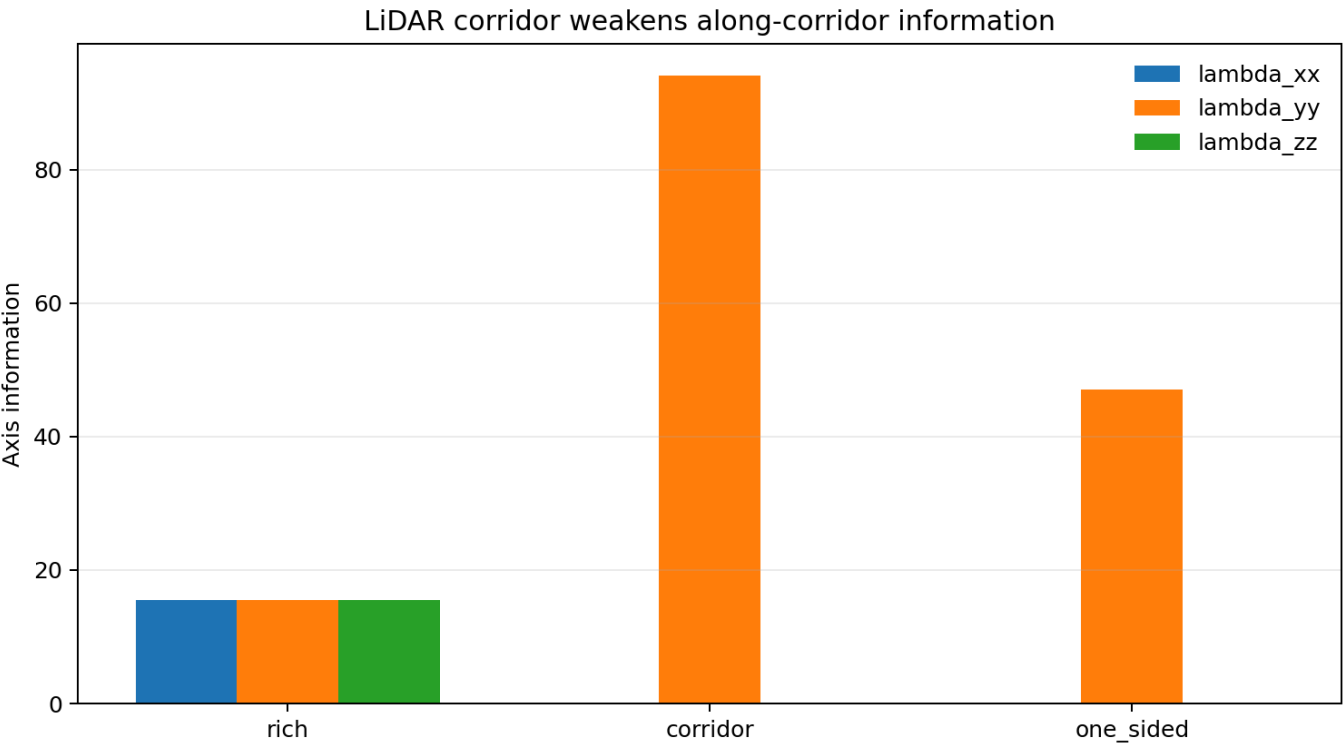
Top-down geometry



Eigenvalue



Axis information



Corridor is one-sided geometry. λ_{xx} is 0 condition. σ is sigma. σ is sigma. "LiDAR PL" is isolated.

10. LiDAR Primitive generation

C++

```
LidarObservabilityFimTest.PcaRadiusChangesPrimitiveCount
LidarObservabilityFimTest.PcaMinSupportBoundaryIsInclusive
LidarObservabilityFimTest.VoxelSamplingReducesDenseClusterDuplicates
LidarObservabilityFimTest.CloudProvidedNormalsAreUsedBeforePca
```

```
src/iap/test/test_lidar_observability_fim.cpp
```

10.1

LiDAR FIM primitive generation primitive

- PCA radius primitive count
- $pca_min_support$ inclusive
- voxel sampling dense cluster duplicate primitives
- cloud normals PCA fallback

10.2

PCA radius

half_width	3
spacing	0.4
	49
small radius	0.45
large radius	0.75

Min-support 6

```
(-1,-1,0), (-1,1,0), (1,-1,0), (1,1,0), (0,0,0), (0.4,-0.2,0)
```

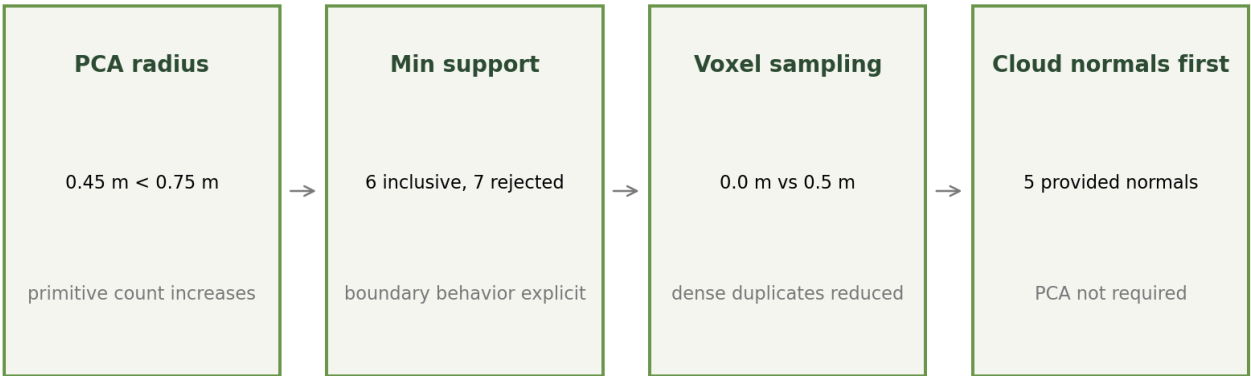
Voxel sampling

- dense plane grid half_width=6 spacing=0.02
- sparse plane grid half_width=1 spacing=0.2 offset (2,0,0)
- pca_voxel_sample_m=0.0 0.5

Cloud normals 5 UnitX() normal

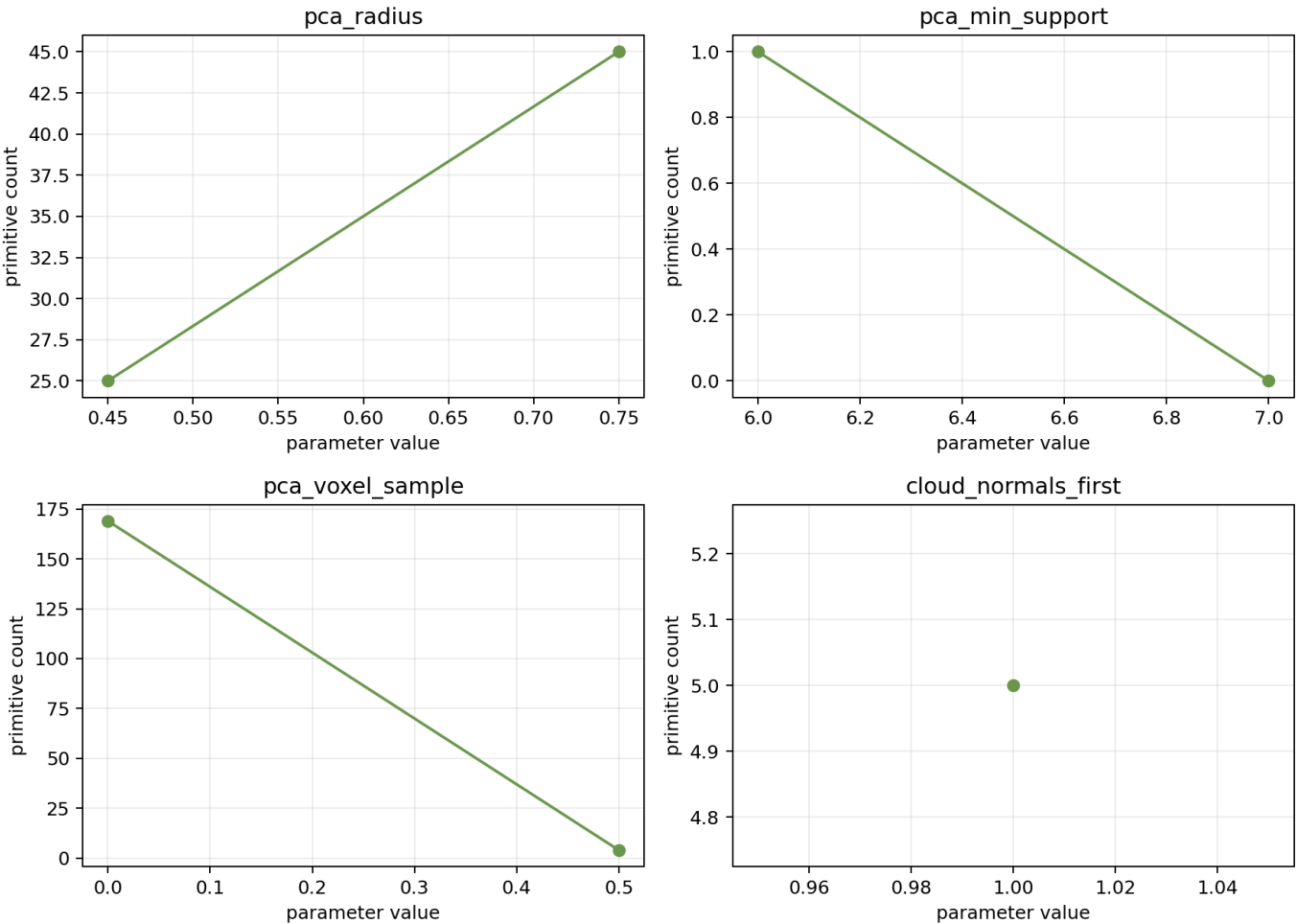
Primitive generation

LiDAR primitive generation parameter-boundary checks



Each box is backed by a fixed GTest input and an explicit assertion.

```
lidar_primitive_generation_parameters.csv
```

10.3

PCA radius	<code>pca_radius_m=0.45 vs 0.75</code>	large radius primitive count
Min support	<code>pca_min_support=6 vs 7</code>	6 missing 7
Voxel sampling	<code>pca_voxel_sample_m=0.5</code>	dense cluster duplicate
Cloud normals first	<code>use_cloud_normals_first=true</code>	primitive

10.4

- `small->size() < large->size()`
- `pca_min_support=6` 1 primitive
- `pca_min_support=7` missing_lidar_normals
- voxel sampling dense center primitive 20 4
- cloud-provided normals `support_count == 1` normal `UnitX()`

10.5

```
LidarObservabilityFimTest.PcaRadiusChangesPrimitiveCount RUN COMPLETED
LidarObservabilityFimTest.PcaMinSupportBoundaryIsInclusive RUN COMPLETED
LidarObservabilityFimTest.VoxelSamplingReducesDenseClusterDuplicates RUN COMPLETED
LidarObservabilityFimTest.CloudProvidedNormalsAreUsedBeforePca RUN COMPLETED
```

10.6

Primitive generation PCA radiussupport thresholdvoxel sampling cloud normals LiDAR advisory FIM primitives

11. Fusion GNSS-only advisory missing prior

C++

```
PredictorModuleTest.FusionGnssOnlyProducesFiniteAdvisoryPl
```

#####

```
src/iap/test/test_predictor_module.cpp
```

11.1 #####

fusion predictor ##### LiDAR source##### valid prior ##### GNSS advisory ##### fused output##### predictor advisory ##### prior ##### GNSS

11.2 #####

GNSS	open-sky 8 #####valid
LiDAR	##### LidarAdvisoryResult
Prior	has_lambda_base=false
Fusion params	fim_epsilon=1e-6#####K_H_adv=5.0#####K_V_adv=5.0

11.3 #####

- result.valid == true
- result.available == true
- result.fallback == false
- result.gnss_used == true
- result.lidar_used == false
- result.prior_valid == false
- fallback_reason ##### missing_prior
- hpl/vpl finite

11.4 #####

#####

```
PredictorModuleTest.FusionGnssOnlyProducesFiniteAdvisoryPl RUN COMPLETED
```

11.5 #####

Fusion predictor ##### missing prior ##### GNSS-only advisory ##### fallback reason chain ##### prior ##### missing prior ##### explainability diagnostic##### invalid #####

11.6 ##### C#Current/advisory separation guard

C++

```
PredictorModuleTest.CurrentIntegrityDoesNotOverrideAdvisoryPrediction
```

#####

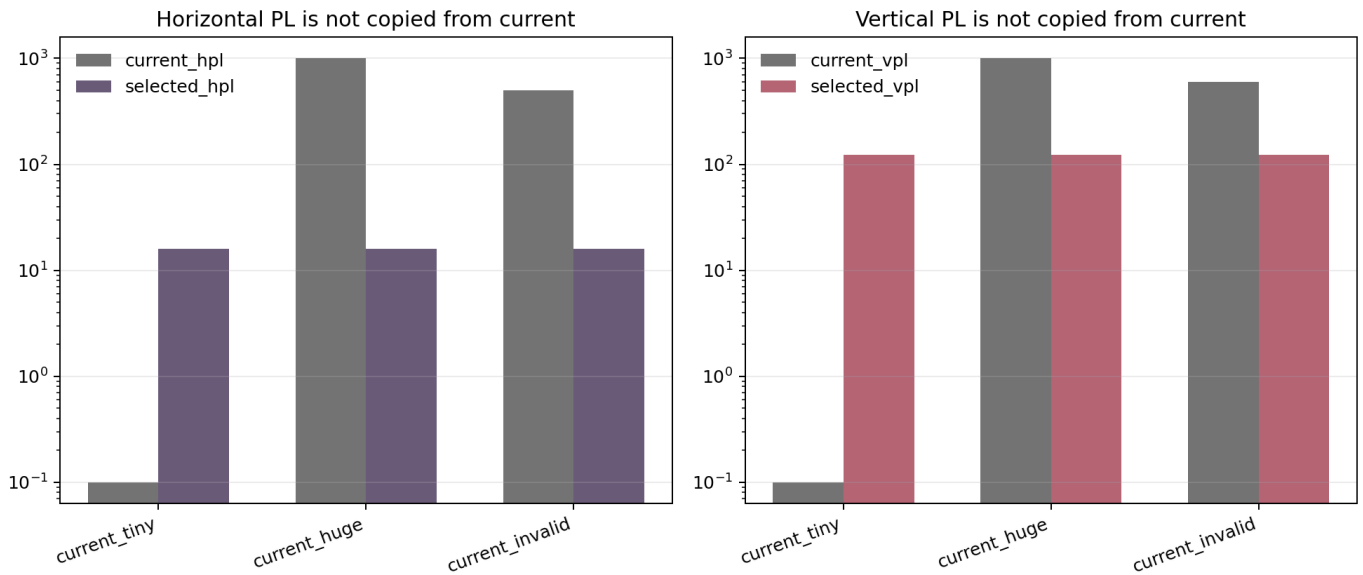
```
current_advisory_separation.csv
```

GNSS epoch##### snapshot ##### current certified monitor fields#####

Case	current_hpl	current_vpl	current.valid
current_tiny	0.1	0.1	true
current_huge	1000	1000	true
current_invalid	500	600	false

#####

- GNSS advisory HPL/VPL
- selected/fused advisory HPL/VPL
- `copied_current_flag == 0`
- fallback reason current HPL/VPL



Predictor `/iap/integrity` current monitor HPL/VPL advisory Current fields snapshot diagnostic/current-state context independent advisory prediction

12. Fusion GNSS + LiDAR FIM additive fusion

C++

```
PredictorModuleTest.ModuleFusesGnssAndLidarWithoutGridFields
```

```
src/iap/test/test_predictor_module.cpp
```

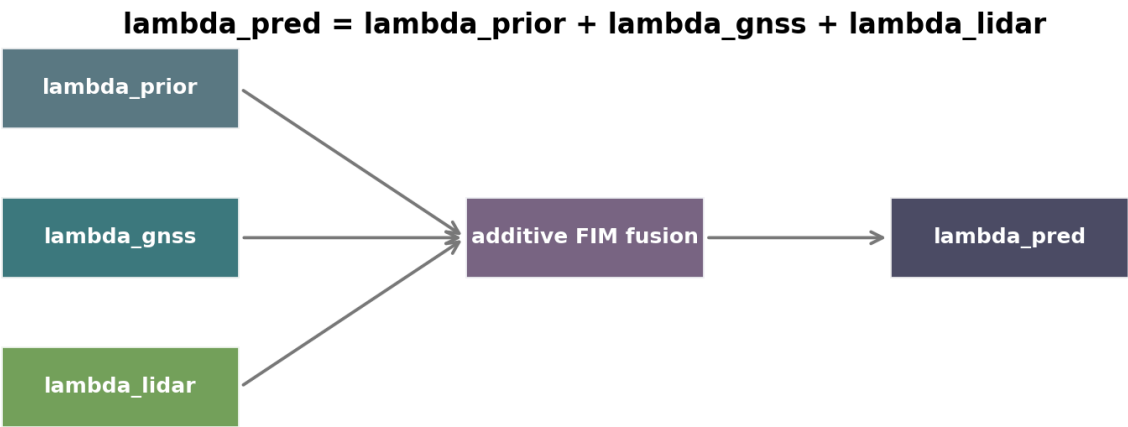
12.1

`PredictorModule::query()` GNSS LiDAR prior fused advisory FIM additive fusion

```
lambda_pred = lambda_prior + lambda_gnss + lambda_lidar
```

query metadata source flags fused protection level

Fusion additive FIM artifacts `lambda_*` `isApprox(..., 1e-9)`



Matrix equality is verified by `isApprox(..., 1e-9)`; numeric lambda matrices were not exported in current artifacts.

12.2 数据

键	值
Query position	(1.0, 2.0, 3.0)
Query time	123.5
Horizon	2.5
Frame	map
GNSS epoch	open-sky 8
LiDAR primitives	X/Y/Z normals 33
Prior lambda	0.25 * I ₃

12.3 融合

Fusion 数据

键	值
fim_epsilon	1.0e-6
K_H_adv	5.0
K_V_adv	5.0

12.4 输出

输出数据

- Query metadata 位置时间horizonframe
- gnss.valid == true
- lidar.valid == true
- fused.valid == true
- fused.gnss_used == true
- fused.lidar_used == true
- fused.prior_valid == true
- lambda_pred 精度 1e-9
- fused.hpl/vpl finite
- source flags validavailableGNSS/LiDAR/Fusion validprior validGNSS/LiDAR usedregularized

12.5 附录

附录

```
PredictorModuleTest.ModuleFusesGnssAndLidarWithoutGridFields RUN COMPLETED
```

12.6

`PredictorModule::query()` GNSSLiDAR prior additive FIM fusion source flags fused advisory source fusion common 3D position state

12.7 Fusion lambda matrix artifact export

C++

```
PredictorModuleTest.ModuleFusesGnssAndLidarWithoutGridFields
```

```
fusion_lambda_matrices.json
```

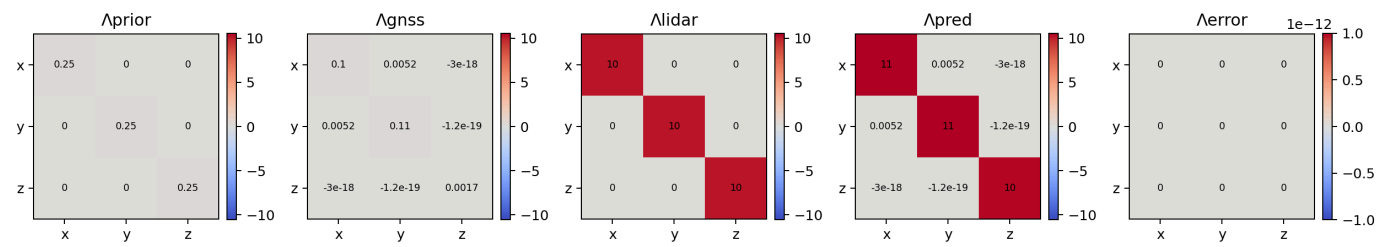
artifact fusion 3x3 information matrices

```
lambda_prior
lambda_gnss
lambda_lidar
lambda_pred
lambda_sum = lambda_prior + lambda_gnss + lambda_lidar
lambda_error = lambda_pred - lambda_sum
lambda_error_norm
eig_prior/eig_gnss/eig_lidar/eig_pred
```

C++ GTest

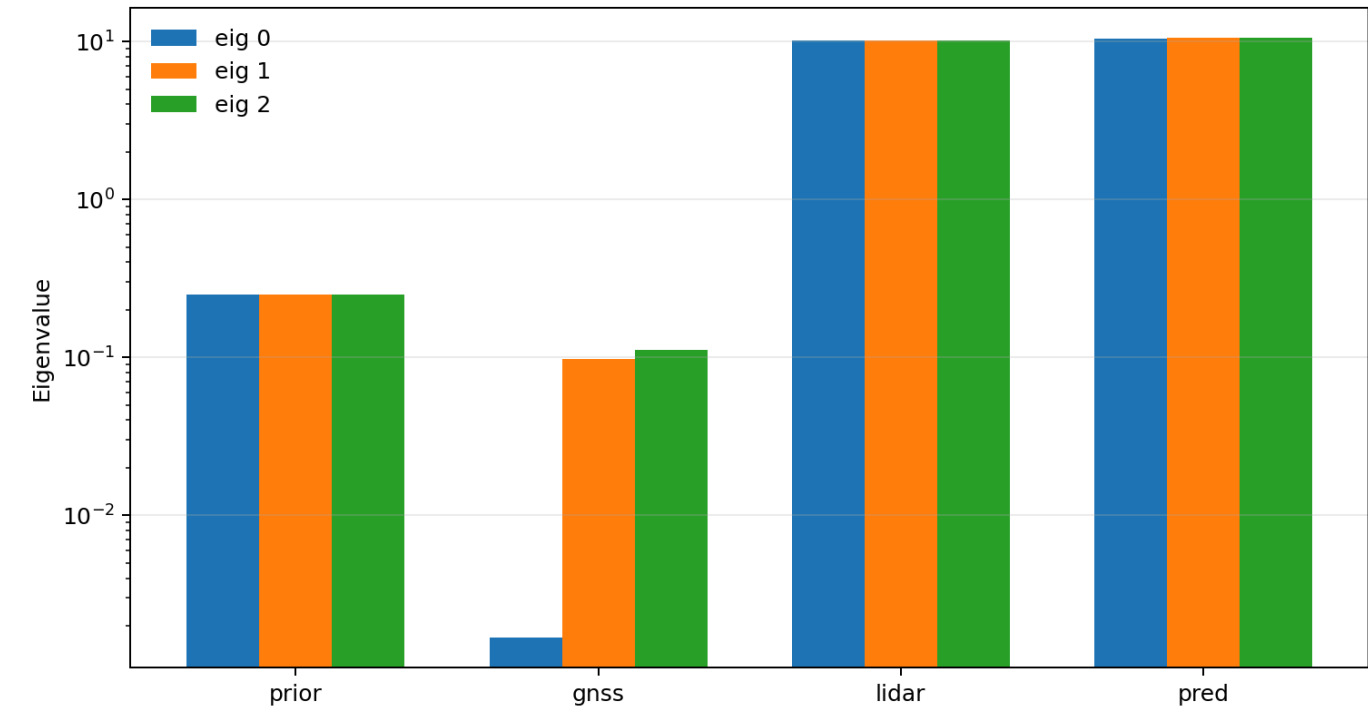
```
result.fused.lambda_pred.isApprox(expected_lambda, 1.0e-9)
```

heatmap



Eigenvalue

Fusion lambda eigenvalues



lambda_error_norm 0 lambda_pred Heatmap eigenvalue schematic priorGNSSLiDAR fused information

13. Fusion Invalid prior no-source negative

C++

```
PredictorModuleTest.FusionRejectsIndefinitePriorButKeepsGnssOnly
PredictorModuleTest.ModuleNoGnssNoLidarIsUnavailableWithNanPl
```

```
src/iap/test/test_predictor_module.cpp
```

13.1

fusion predictor

- Prior indefinite prior GNSS advisory
- GNSSLiDAR source unavailable combined fallback reason

13.2

Invalid prior

GNSS	open-sky 8 valid
LiDAR	
Prior	I_3 lambda_base_pos(0,0) = -1.0

No-source

GNSS epoch	missing
LiDAR primitives	missing
Prior	valid 0.25 * I_3
Query position	(0,0,0)

13.3

Invalid prior

- result.valid == true
- result.available == true
- result.gnss_used == true
- result.lidar_used == false
- result.prior_valid == false
- lambda_prior
- fallback reason invalid_prior_position_information
- lambda_pred == lambda_gnss

No-source negative

- result.valid == false
- result.available == false
- result.fallback == true
- fallback reason gnss:no_gnss_epoch
- gnss.available == false
- lidar.available == false
- fused.available == false
- fused HPL/VPL/PL NaN finite PL
- source flags available fallback

13.4

```
PredictorModuleTest.FusionRejectsIndefinitePriorButKeepsGnssOnly RUN COMPLETED
PredictorModuleTest.ModuleNoGnssNoLidarIsUnavailableWithNanPl RUN COMPLETED
```

13.5

Fusion predictor indefinite prior GNSS advisory advisory source fusion unavailable combined fallback reason prior fused PL

13.6 Fusion conservative gate

C++

```
PredictorModuleTest.DegenerateLidarDoesNotReduceSelectedPl
```

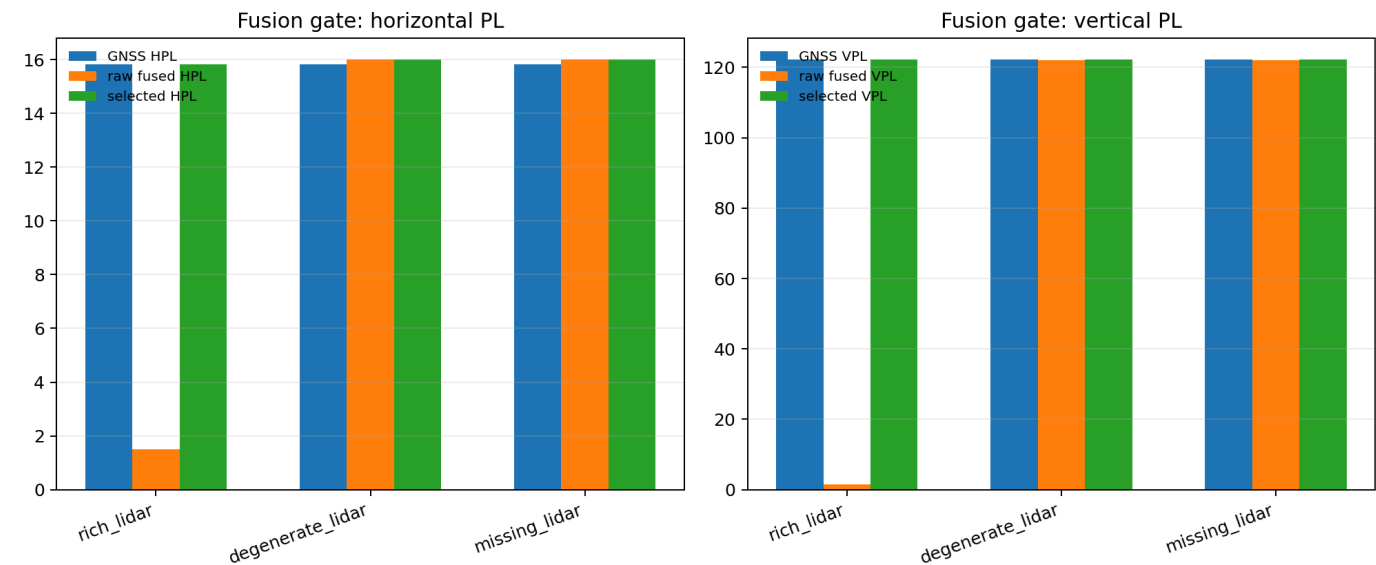
```
fusion_gate_safety.csv
```

GNSS_ONLY/FIM_ADD_DEBUG/CONSERVATIVE_MAX

gnss_hpl/vpl	GNSS-only reference GNSS advisory result
fused_hpl/vpl	default FIM-add raw fused output
selected_hpl/vpl	conservative_max_with_gnss=true selected output

case

Case	LiDAR	
rich_lidar	X/Y/Z feature-rich primitives	LiDAR fusion raw fused PL selected PL GNSS reference
degenerate_lidar	normals LiDAR FIM	LiDAR
missing_lidar	LiDAR source	selected output GNSS reference reason



Raw FIM-add rich LiDAR PL conservative_max_with_gnss selected HPL/VPL GNSS reference LiDAR case selected output GNSS protection level

14. Fusion Query guard

C++

```
PredictorModuleTest.InvalidPositionKeepsQueryMetadataAndFallsBack
PredictorModuleTest.InvalidQueryTimeFallsBackBeforePrediction
PredictorModuleTest.InvalidHorizonFallsBackBeforePrediction
PredictorModuleTest.UnsupportedFrameFallsBackBeforePrediction
```

```
src/iap/test/test_predictor_module.cpp
```

14.1

PredictorModule::query() GNSS/LiDAR/Fusion query input query predictor source flags

14.2

		fallback reason
Invalid position	query_position_map.x = NaN	invalid_position
Invalid query time	query_time_s = NaN	invalid_query_time
Invalid horizon	horizon_s = -0.1	invalid_horizon
Unsupported frame	frame_id = "world"	unsupported_query_frame

14.3

guard

- valid == false
- available == false
- fallback == true
- fallback reason
- GNSS/LiDAR/Fusion valid
- source flags GNSS/LiDAR/Fusion valid

Invalid position query metadata

14.4


```
PredictorModuleTest.InvalidPositionKeepsQueryMetadataAndFallsBack RUN COMPLETED
PredictorModuleTest.InvalidQueryTimeFallsBackBeforePrediction RUN COMPLETED
PredictorModuleTest.InvalidHorizonFallsBackBeforePrediction RUN COMPLETED
PredictorModuleTest.UnsupportedFrameFallsBackBeforePrediction RUN COMPLETED
```

14.5

PredictorModule query input horizon frame advisory GNSS/LiDAR/Fusion

15. System Predictor GNSS open-sky only

system experiment preset

```
predictor_gnss_open_sky_only
```

```
src/iap/results/predictor_validation/predictor_system_predictor_gnss_open_sky_only_run1/
```

15.1

predictor system launch /iap/integrityodomGNSS epoch GNSS-only advisory query IAP log

planner success

- predictor probe /iap/integrity epoch
- predictor_output_mode=gnss_only selected source GNSS
- open-sky GNSS advisory result valid
- query set 6 query
- CSVJSONprofiling figures
- advisory output current /iap/integrity monitor output

15.2

run launch

experiment	predictor_gnss_open_sky_only
araim_experiment	gnss_open_sky
predictor_output_mode	gnss_only
probe_query_set	e1_fixed_neighborhood
predictor_enable_debug_log	true
validator_require_debug_logs	true
start_planner	false predictor launch include ARAIM launch
record_bag	false

query set /iap/integrity epoch 6

Query label	Offset (x,y,z) m
p0	(0, 0, 0)
p_forward	(5, 0, 0)
p_back	(-5, 0, 0)
p_left	(0, 5, 0)
p_right	(0, -5, 0)
p_up	(0, 0, 2)

15.3

1 analyzer

項目	値	目標値
GNSS selected source ratio	> 95%	100%
GNSS valid ratio	> 95%	100%
selected fallback ratio	< 5%	0%
module_total_us p95	< 2000 us	34.21 us
valid rows NaN/Inf	0	0
current/advisory copy ratio	< 5%	0%

15.4 検証結果

Analyzer 結果

```
passed: true
query_count: 2748
selected_source_GNSS_ratio: 1.0
gnss_valid_ratio: 1.0
fallback_ratio: 0.0
module_total_us p50/p95/max: 26.31 / 34.21 / 56.55 us
copied_current_flag_ratio: 0.0
```

Query 結果

Query label	Rows
p0	458
p_forward	458
p_back	458
p_left	458
p_right	458
p_up	458

2748 = 458 * 6 predictor query CSV 76.17 s 6 query/epoch /iap/integrity 6.01 Hz

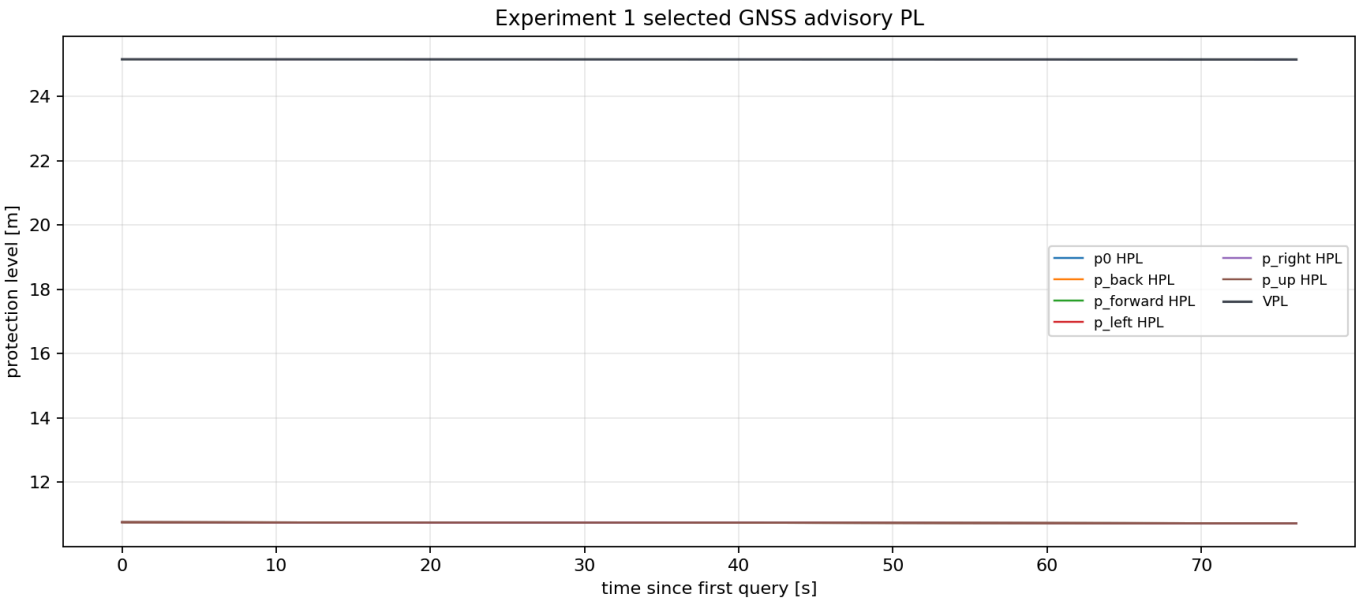
GNSS open-sky geometry

項目	値
gnss_n_visible	41
gnss_n_used	41
gnss_pdop	4.6597 - 4.6639 4.6618
selected_hpl	10.7246 - 10.7570 m 10.7407 m
selected_vpl	25.1454 - 25.1517 m 25.1489 m
module_total_us	20.43 - 56.55 us 26.95 us

run export/test_predictor_validation_summary.json predictor_el_analysis_summary.json test_predictor_query_probe_summary.json run validator summary system validator

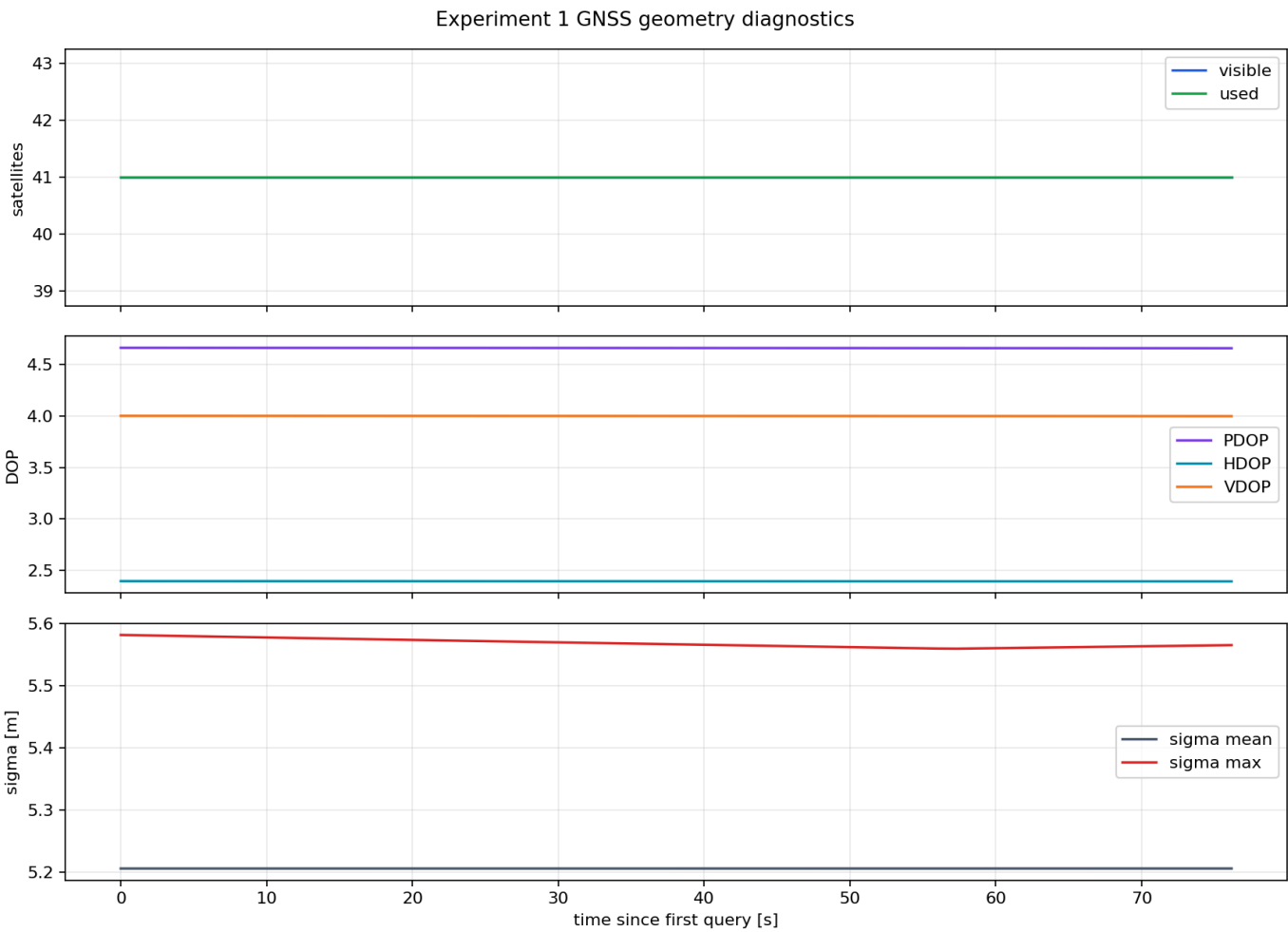
15.5 検証

Selected PL timeline



6 query selected HPL/VPL open-sky fallback HPL 10.74 mVPL 25.15 m GNSS-only selected output run

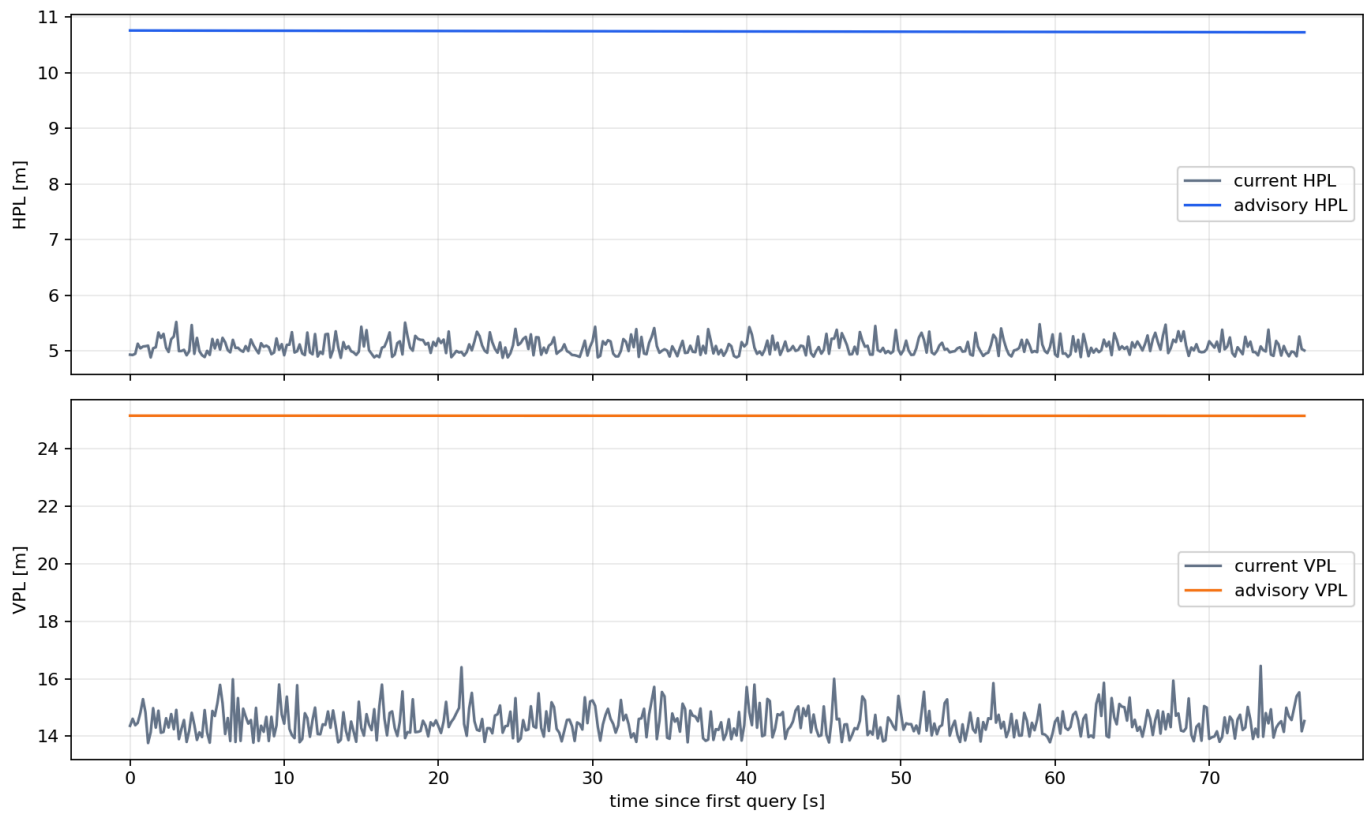
GNSS geometry timeline



GNSS visible/used 41PDOP 4.66 open-sky preset visibility geometry geometry

Current vs advisory

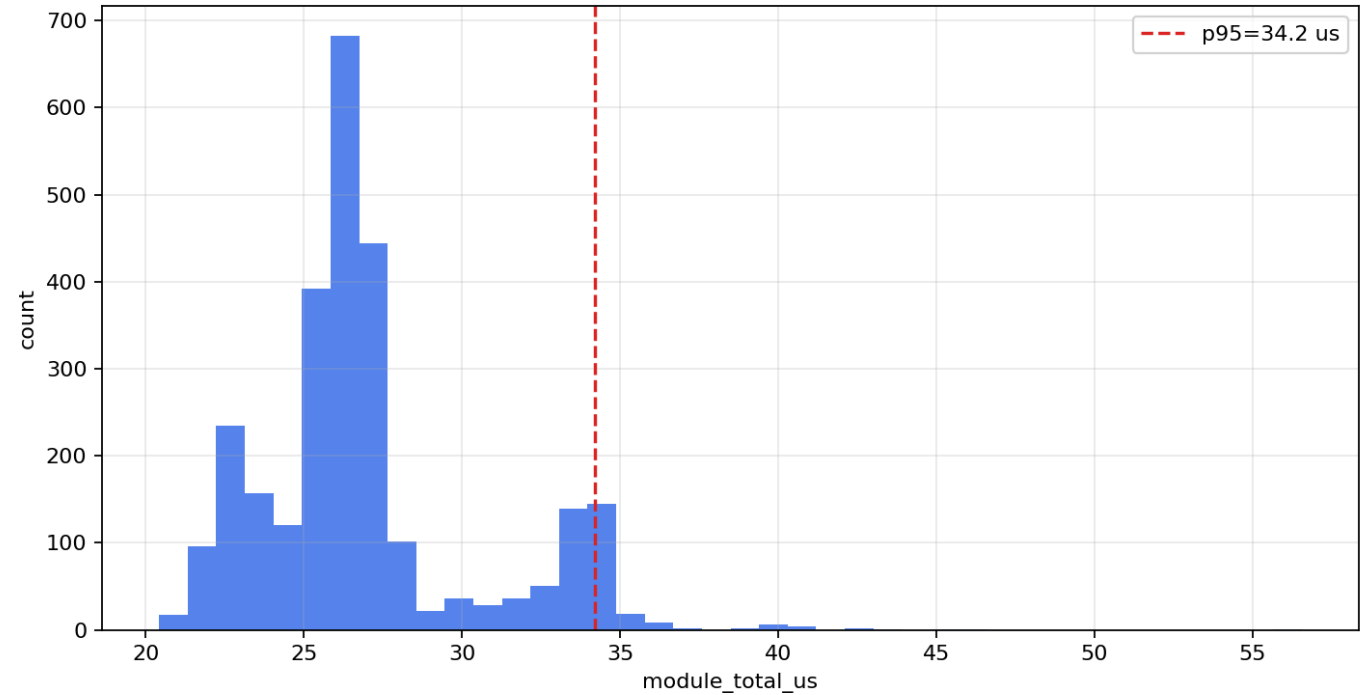
Experiment 1 current ARAIM vs advisory Predictor



current /iap/integrity monitor predictor advisory 1 advisory query current PL analyzer copied_current_flag_ratio=0.0 current advisory

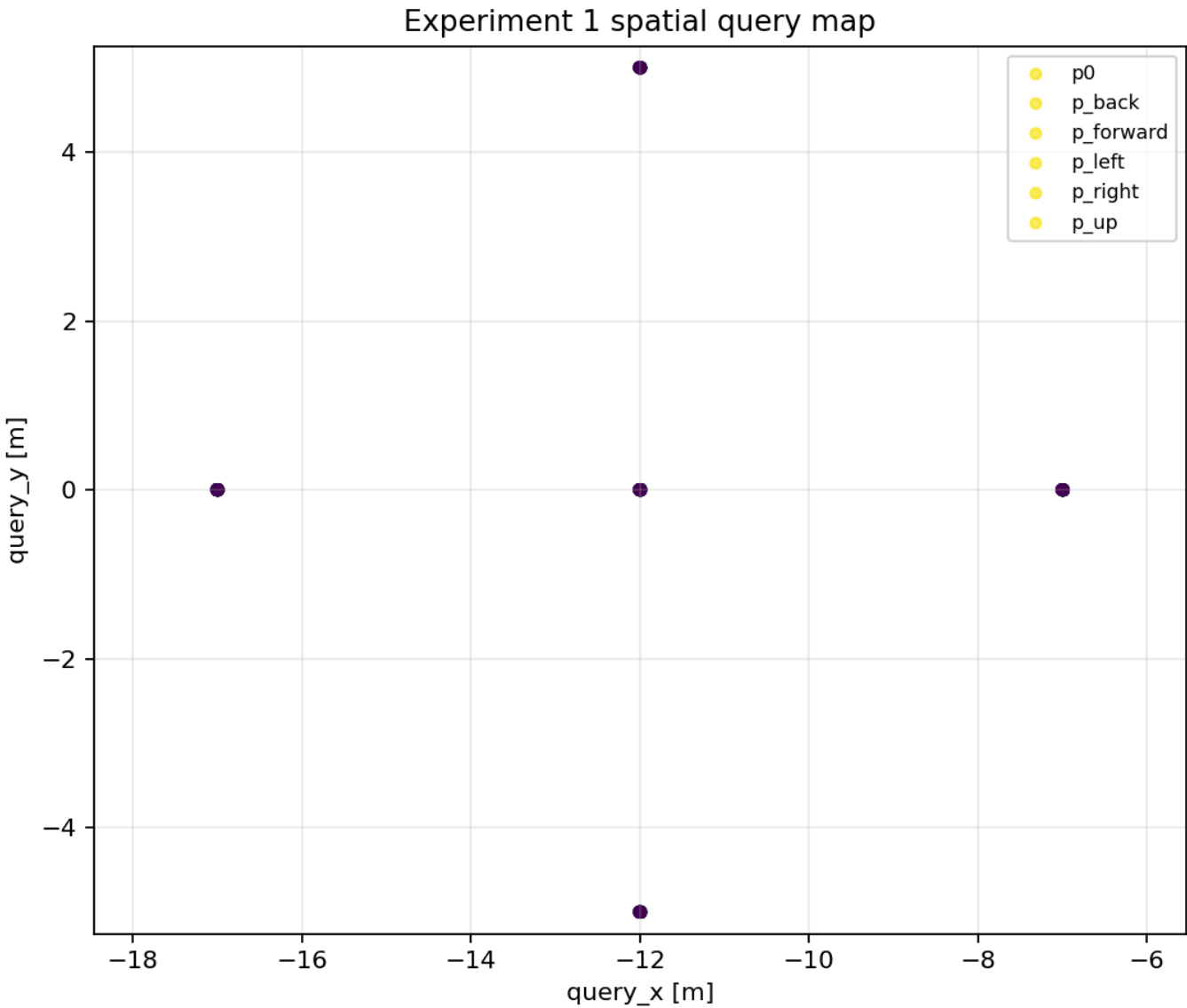
Latency distribution

Experiment 1 Predictor query latency



predictor query latency module_total_us p95 34.21 us 2000 us GNSS-only advisory query system probe ROS 2

Query spatial map



6 query offset 1 current pose p_forward/p_back/p_left/p_right/p_up odom el_fixed_neighborhood

15.6

Predictor system experiment 1 predictor_gnss_open_sky_only system launch 2748 GNSS-only advisory query selected source GNSS GNSS valid ratio 100% fallback ratio 0% IAP export/profiling/metadata/figures CSV JSON PNG open-sky GNSS predictor /iap/integrity GNSS advisory protection level planner/grid disabled predictor probe system-level evidence collection

16. System Predictor LiDAR feature-rich only

system experiment preset

predictor_lidar_feature_rich_only

src/iap/results/predictor_validation/predictor_system_predictor_lidar_feature_rich_only_bag/

16.1

predictor system launch feature-rich LiDAR LiDAR-only advisory result LiDAR FIM primitive fallback reason latency LiDAR diagnostic LiDAR-only query LiDAR fusion safety planner

16.2

run launch

experiment	predictor_lidar_feature_rich_only
araim_experiment	lidar_feature_rich
predictor_output_mode	lidar_only
validator_required_selected_source	LIDAR
predictor_enable_debug_log	true
validator_require_debug_logs	true
record_bag	true
start_planner	false predictor launch include ARAIM launch

current_pose query set /iap/integrity epoch odom LiDAR-only predictor query

```
RUN_ROOT=/home/dev/ws_iap/src/iap/results/predictor_validation/predictor_system_predictor_lidar_feature_rich_only_bag

ros2 launch iap test_predictor.launch.py  experiment:=predictor_lidar_feature_rich_only  start_rviz:=true
predictor_enable_debug_log:=true  validator_require_debug_logs:=true  run_duration_s:=90
validation_duration_s:=85  predictor_export_dir:=$RUN_ROOT/export

python3 src/iap/scripts/dev_predictor/analyze_predictor_system_e2.py  --run-dir $RUN_ROOT  --fail-on-threshold
```

16.3

2 analyzer

LIDAR selected source ratio	> 80%	100%
LiDAR valid ratio	> 80%	100%
LiDAR FIM valid ratio		100%
selected fallback ratio		0%
median lidar_n_primitives	>= 6	276
median lidar_lambda_condition	< 1e6	7.72
module_total_us p95	< 10000 us	20.46 us
valid rows NaN/Inf		0

Validator summary required debug files predictor_lidar_debug.csv predictor_lidar_primitives_debug.csv fallback_reason_by_time.csv downsampled_map.csv latency_debug.csv metadata JSON

16.4

Analyzer

```
passed: true
query_count: 308
selected_source_LIDAR_ratio: 1.0
lidar_valid_ratio: 1.0
lidar_fim_valid_ratio: 1.0
fallback_ratio: 0.0
median_lidar_n_primitives: 276
median_lidar_n_valid_normals: 276
median_lidar_lambda_trace: 284.65
median_lidar_lambda_min_eig: 28.67
median_lidar_condition: 7.72
module_total_us p50/p95/max: 11.13 / 20.46 / 32.68 us
```

CSV 308 predictor query 87.83 s 3.51 Hz selected source LIDAR LiDAR advisory query valid selected fallback count 0

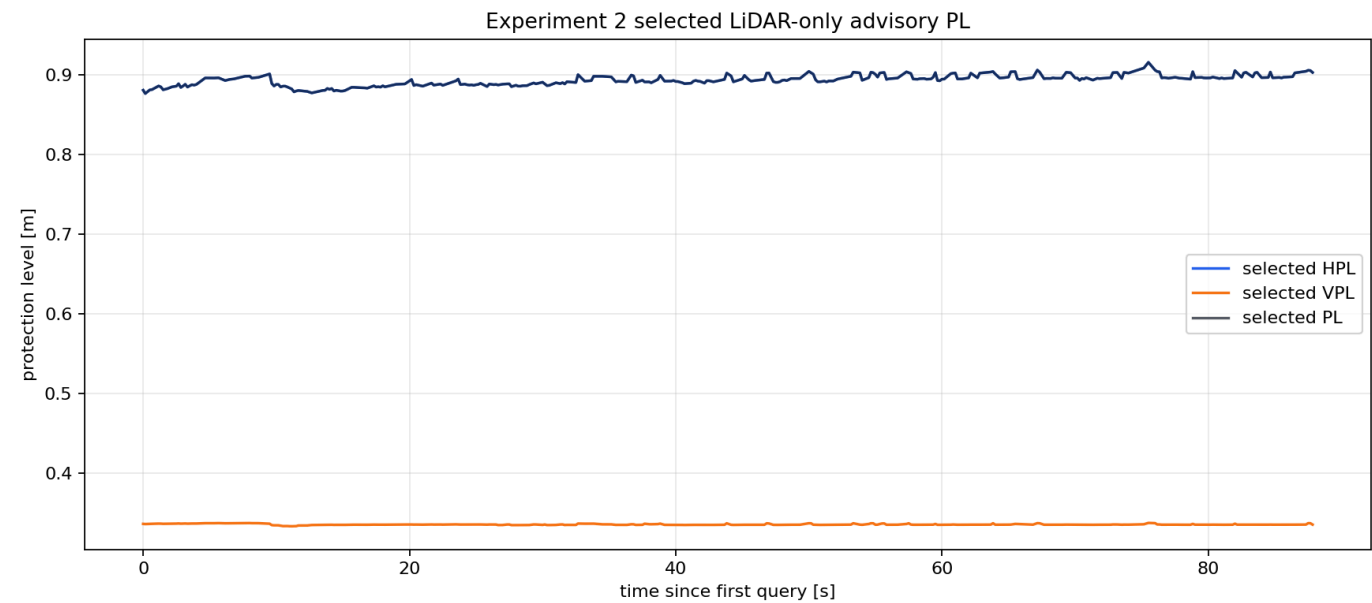
LiDAR-only advisory

PL		PL	
selected_hpl	0.8768 - 0.9161 m	selected_hpl	0.8945 m
selected_vpl	0.3329 - 0.3371 m	selected_vpl	0.3349 m
selected_pl	0.8768 - 0.9161 m	selected_pl	0.8945 m
lidar_n_primitives	267 - 278	lidar_n_primitives	276
lidar_n_valid_normals	267 - 278	lidar_n_valid_normals	276
lidar_lambda_trace	275.87 - 285.53	lidar_lambda_trace	284.65
lidar_lambda_min_eig	26.09 - 28.85	lidar_lambda_min_eig	28.67
lidar_lambda_condition	7.64 - 8.28	lidar_lambda_condition	7.72

Fallback reason gnss:no_gnss_epoch gnss:not_evaluated experiment GNSS integrity
predictor_output_mode=lidar_only GNSS selected outputLiDAR selected output fallback

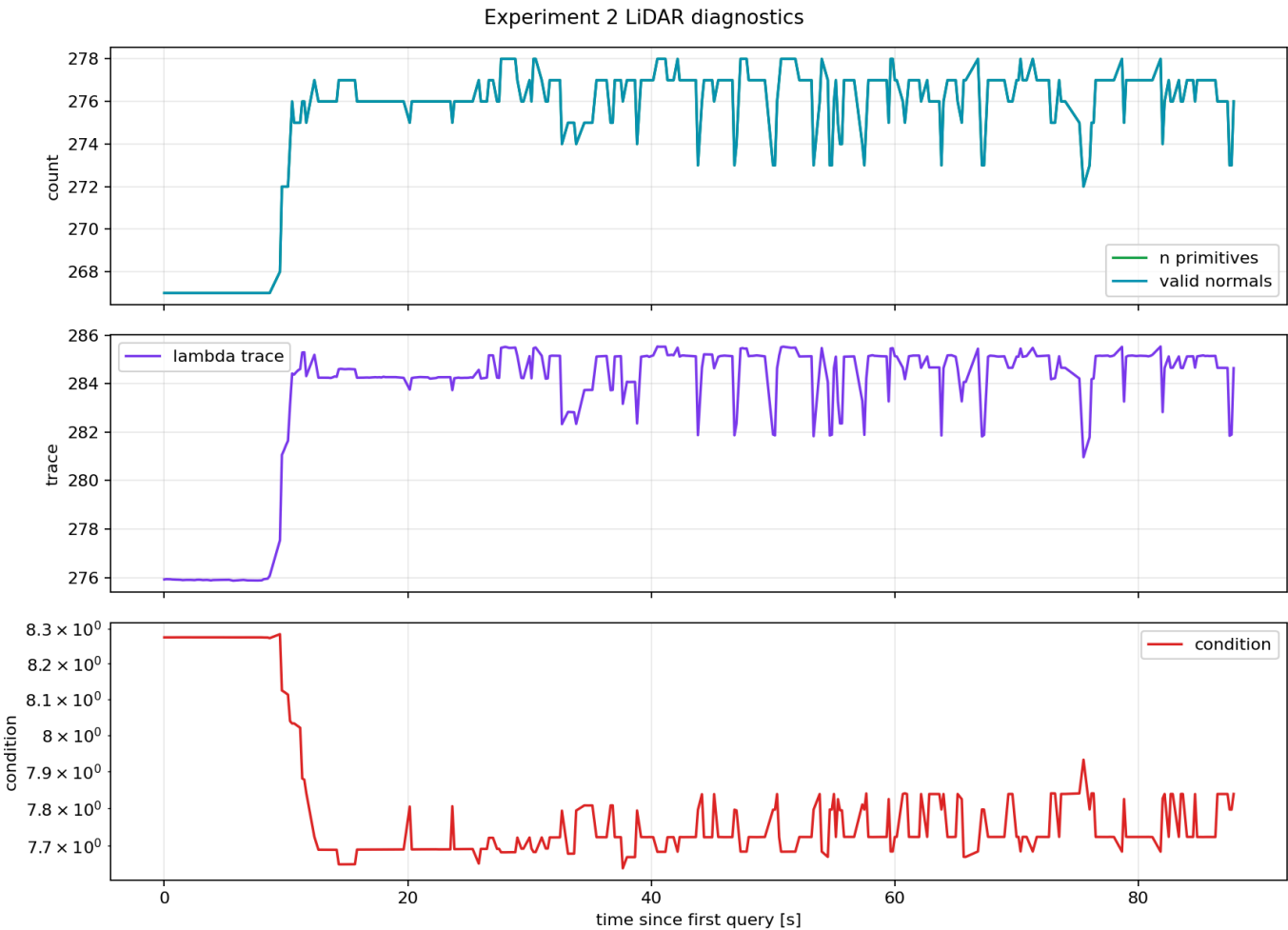
16.5

Selected PL timeline



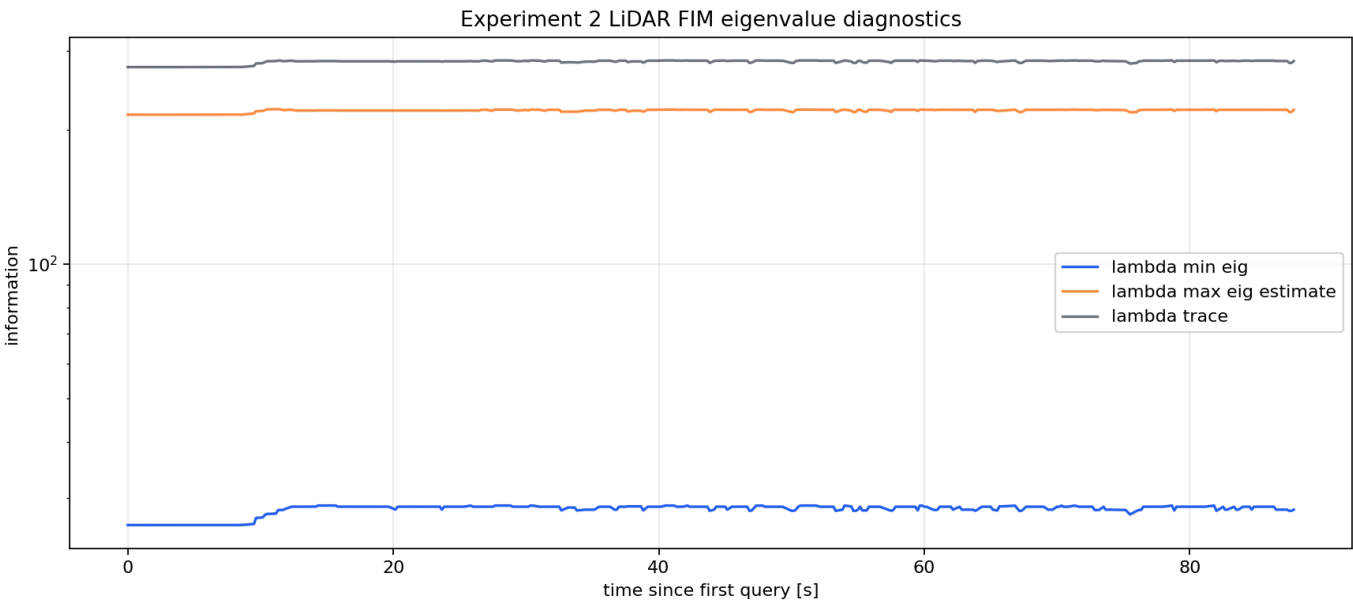
selected HPLselected VPLselected scalar PLLiDAR-only selected output run fallback
HPL 0.89 mVPL 0.335 m feature-rich LiDAR geometry selected PL

LiDAR diagnostics timeline



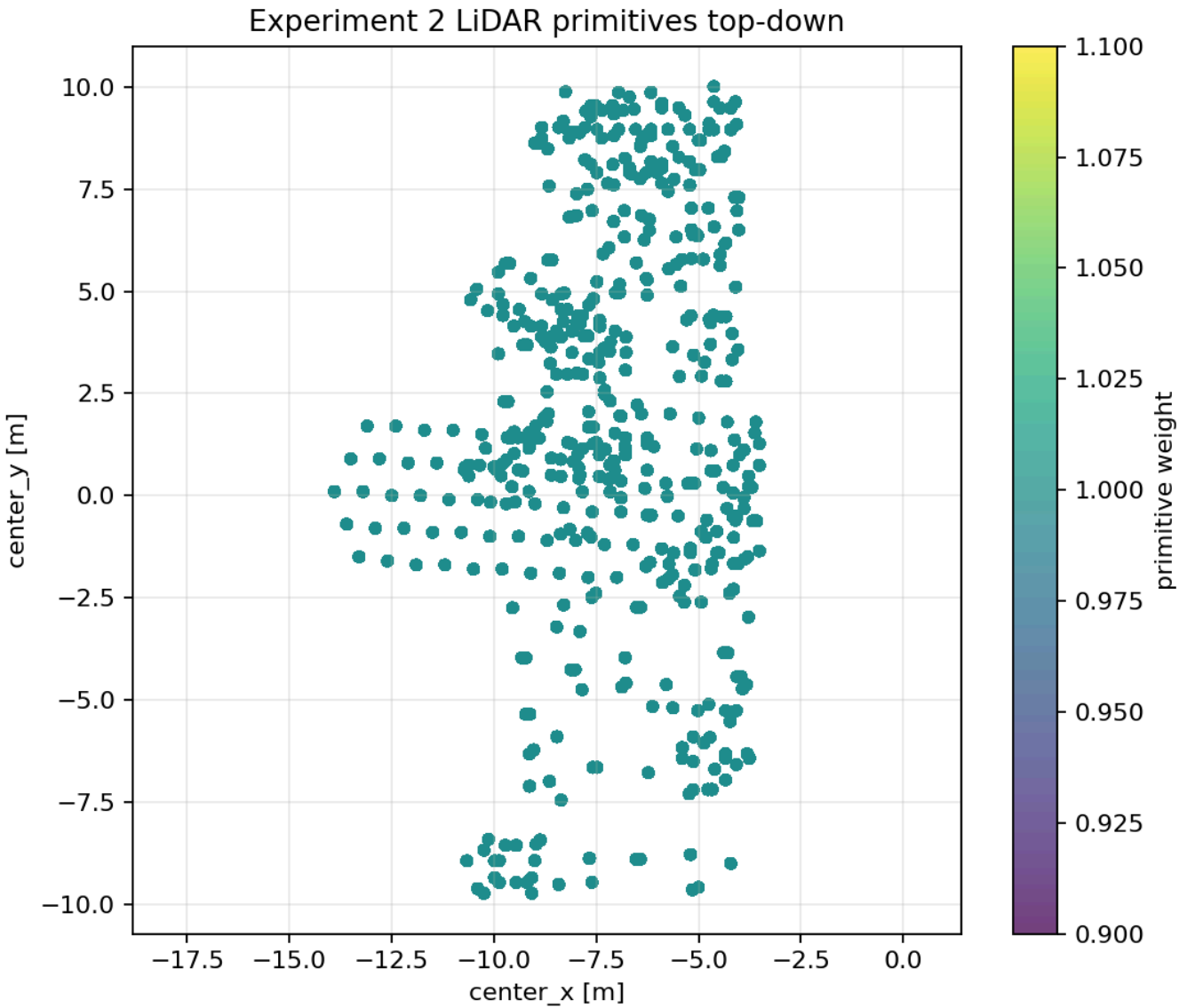
LiDAR primitives valid normals LiDAR information matrix trace condition number feature-rich map
primitive FIM primitive 276condition 7.721e6

LiDAR eigenvalues timeline



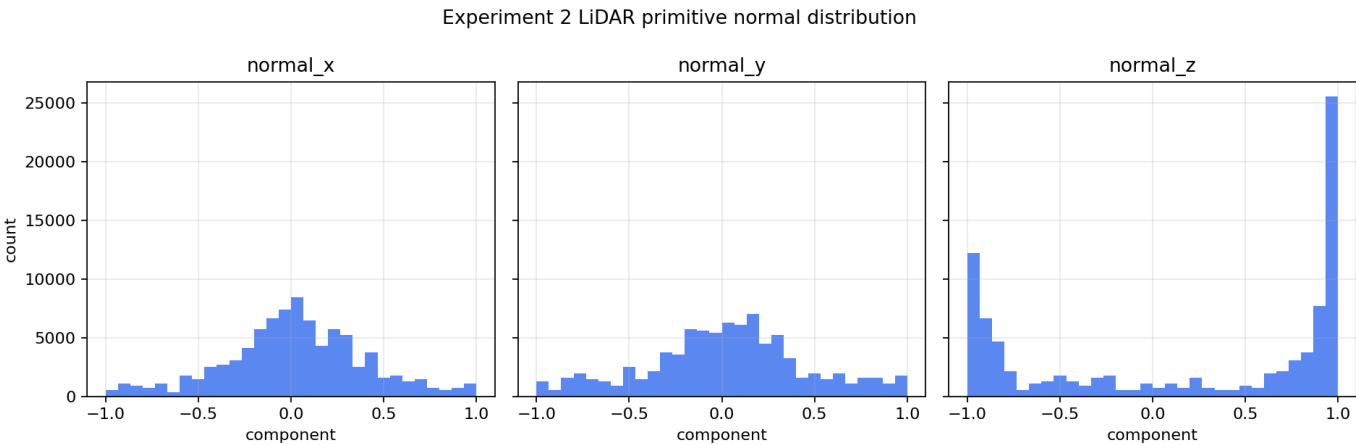
LiDAR FIM min_eig * condition FIM trace LiDAR FIM 28.67
condition 7.72

LiDAR primitives top-down



downsampled map LiDAR FIM primitives XY primitive weight feature-rich primitive

LiDAR normal distribution



primitive normal x/y/z normal LiDAR FIM

16.6

Predictor system experiment 2 predictor_lidar_feature_rich_only feature-rich LiDAR runtime 308 LiDAR-only advisory query selected source LIDAR LiDAR valid ratio FIM valid ratio 100% fallback ratio 0% IAP export/profiling/metadata/figures CSV JSON PNG

planner/grid disabled predictor probe /iap/integrity feature-rich LiDAR map LiDAR-only advisory FIM
system test plan LiDAR diagnostic LiDAR-only query availability

17. System Predictor fusion nominal

system experiment preset

```
predictor_fusion_nominal
```

```
src/iap/results/predictor_validation/predictor_system_predictor_fusion_nominal_run1/
```

17.1

predictor nominal GNSS feature-rich LiDAR current prior system launch Fusion advisory result Fusion FIM

planner success

- predictor_output_mode=fusion selected source FUSION
- GNSS LiDAR fused advisory result query valid
- Fusion FIM $\lambda_{pred} = \lambda_{prior} + \lambda_{gnss} + \lambda_{lidar}$
- Fusion debug GNSS visibility LiDAR debug fallback reason latency metadata IAP log
- planner/grid disabled predictor probe /iap/integrity system-level evidence collection

17.2

run launch

experiment	predictor_fusion_nominal
araim_experiment	fused_nominal
predictor_output_mode	fusion
validator_required_selected_source	FUSION
predictor_enable_debug_log	true
validator_require_debug_logs	true
record_bag	false
start_planner	false predictor launch include ARAIM launch

current_pose query set /iap/integrity epoch odom fusion predictor query 3 nominal fusion advisory FIM

```
RUN_ROOT=/home/dev/ws_iap/src/iap/results/predictor_validation/predictor_system_predictor_fusion_nominal_run1

ros2 launch iap test_predictor.launch.py \
  experiment:=predictor_fusion_nominal \
  start_rviz:=false \
  record_bag:=false \
  predictor_enable_debug_log:=true \
  validator_require_debug_logs:=true \
  run_duration_s:=90 \
  validation_duration_s:=85 \
  predictor_export_dir:=$RUN_ROOT/export

python3 src/iap/scripts/dev_predictor/analyze_predictor_system_e3.py \
  --run-dir $RUN_ROOT \
  --fail-on-threshold
```

17.3

3 analyzer

Item	Value	Target
FUSION selected source ratio	> 80%	100%
GNSS valid ratio	> 80%	100%
LiDAR valid ratio	> 80%	100%
Fused valid ratio	> 80%	100%
selected fallback ratio	< 10%	0%
lambda_sum_error p95	<= 1e-6	0.0
min fused_lambda_pred_min_eig	> -1e-9	27.30
module_total_us p95	< 10000 us	40.39 us
valid rows NaN/Inf	0	0

Validator summary: Required debug files: predictor_fusion_debug.csv, source_selection_debug.csv, gnss_visibility_by_query.csv, predictor_lidar_debug.csv, fallback_reason_by_time.csv, downsampled_map.csv, latency_debug.csv, metadata JSON

17.4 Results

Analyzer Results

```
passed: true
query_count: 292
selected_source_FUSION_ratio: 1.0
gnss_valid_ratio: 1.0
lidar_valid_ratio: 1.0
fused_valid_ratio: 1.0
fallback_ratio: 0.0
lambda_sum_error_p95: 0.0
fused_lambda_pred_min_eig_min: 27.2968
median_fused_lambda_condition: 7.36
module_total_us p50/p95/max: 37.78 / 40.39 / 48.14 us
```

CSV: 292 predictor query, 88.00 s, 3.32 Hz, selected source: FUSION, GNSS, LiDAR, fused advisory query: valid, selected fallback count: 0

Fusion advisory Results

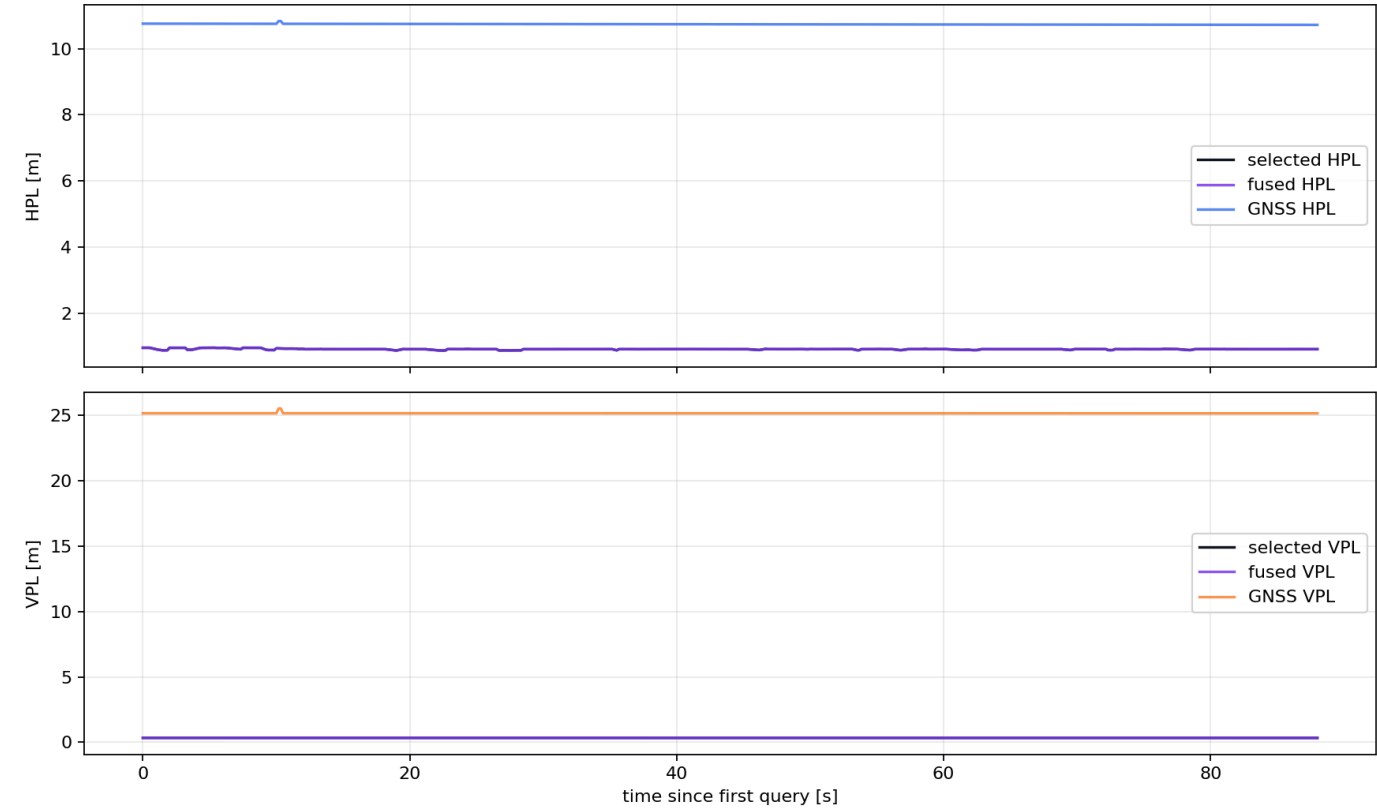
Item	Value
selected_hpl	0.8733 - 0.9570 m, 0.9129 m
selected_vpl	0.3342 - 0.3411 m, 0.3372 m
selected_pl	0.8733 - 0.9570 m, 0.9129 m
gnss_hpl	10.7199 - 10.8345 m, 10.7371 m
gnss_vpl	25.1441 - 25.5136 m, 25.1484 m
fused_lambda_prior_trace	1.7502 - 17.8633, 2.0753
fused_lambda_gnss_trace	0.8420 - 0.8550, 0.8545
fused_lambda_lidar_trace	275.2019 - 285.3188, 284.2462
fused_lambda_pred_trace	278.1229 - 296.4450, 287.1870
fused_lambda_pred_min_eig	27.2968 - 32.7786, 29.9944
fused_lambda_pred_condition	6.8478 - 7.9147, 7.3602
lambda_sum_error	0.0 - 0.0, p95 0.0

lambda_sum_error=0.0, run predictor_fusion_debug.csv, lambda_pred = lambda_prior + lambda_gnss + lambda_lidar, system experiment 3, FIM

17.5 Source

Source HPL/VPL timeline

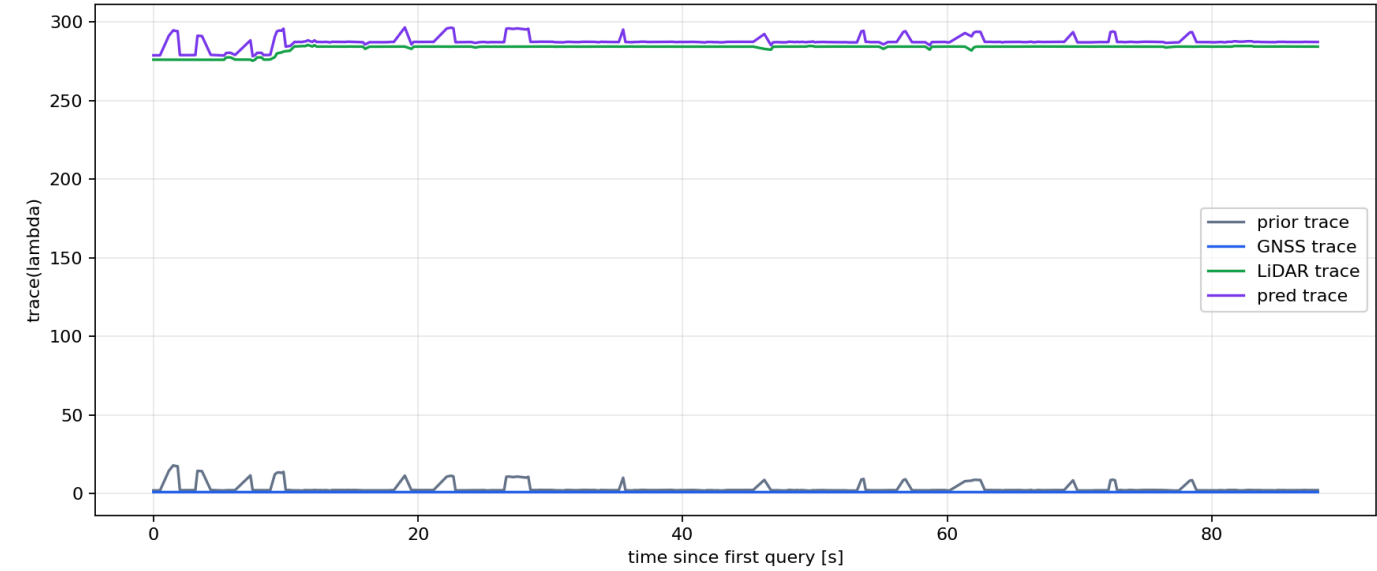
Experiment 3 source HPL/VPL timeline



selected HPL/VPL fused HPL/VPL GNSS HPL/VPL reference selected output fused output fusion advisory GNSS-only reference PL selected/fused HPL 0.91 m GNSS HPL 10.74 m feature-rich LiDAR fusion

Lambda contribution timeline

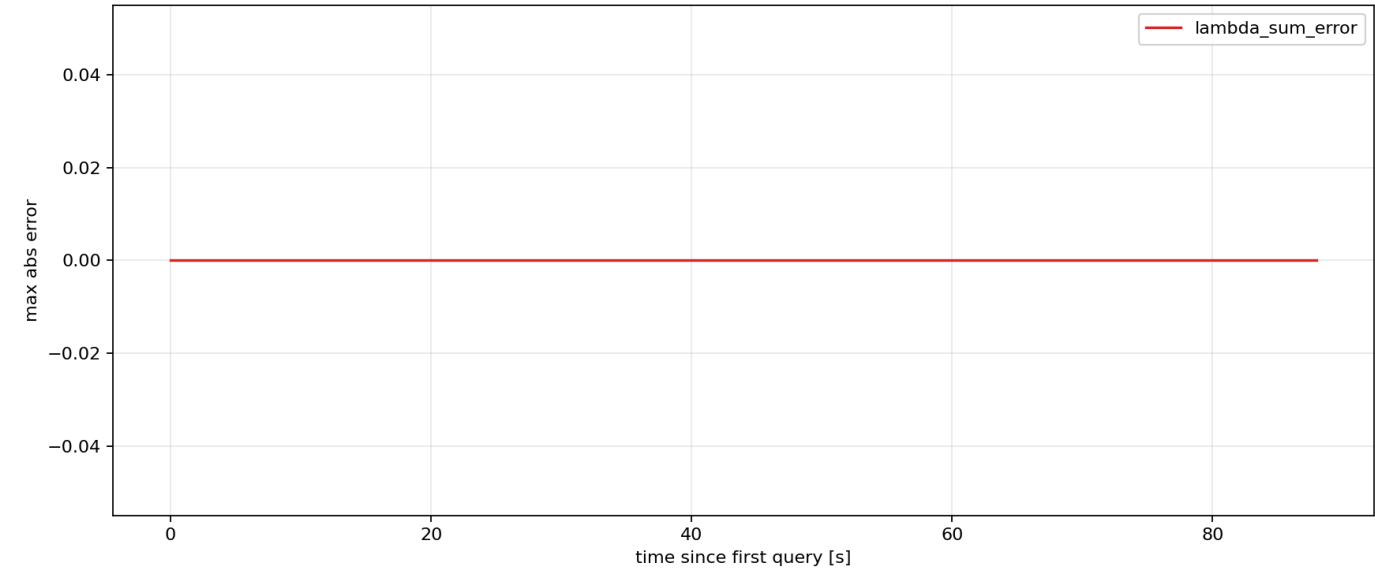
Experiment 3 lambda contribution timeline



prior trace GNSS trace LiDAR trace fused predicted trace lambda_pred LiDAR trace 284.25 GNSS trace 0.85 prior trace 2.08 fused trace feature-rich LiDAR FIM

Lambda sum error timeline

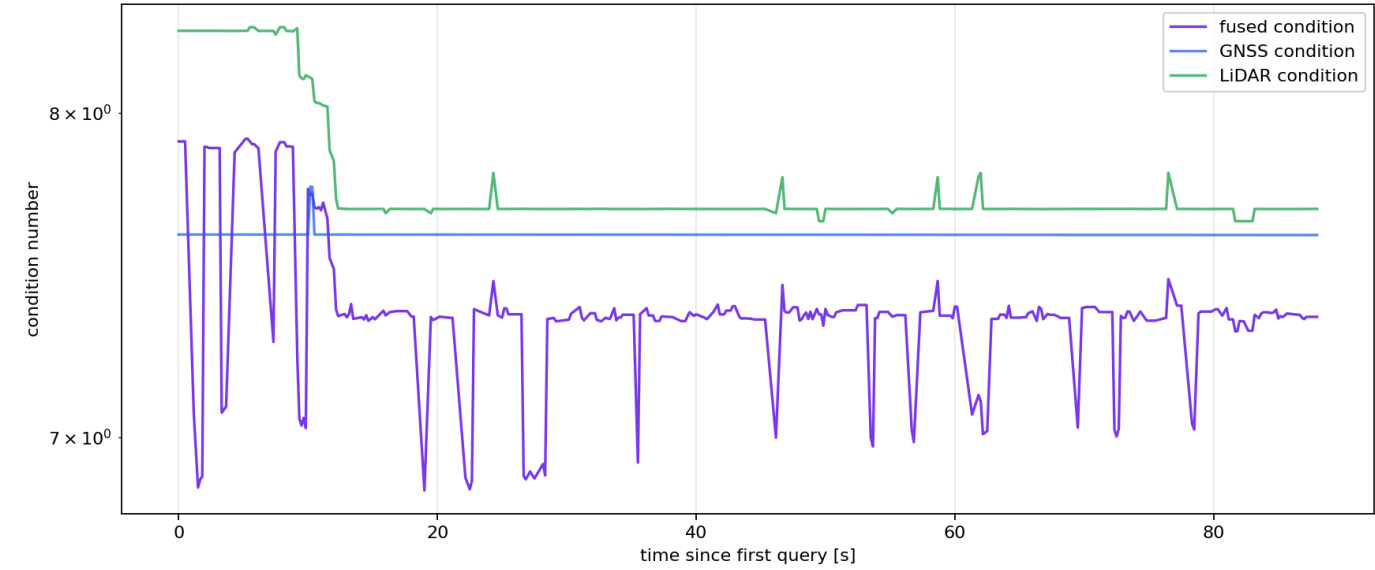
Experiment 3 lambda sum error



max_abs(lambda_pred - lambda_prior - lambda_gnss - lambda_lidar) Fusion FIM query 0.0p95 1e-6

Lambda condition timeline

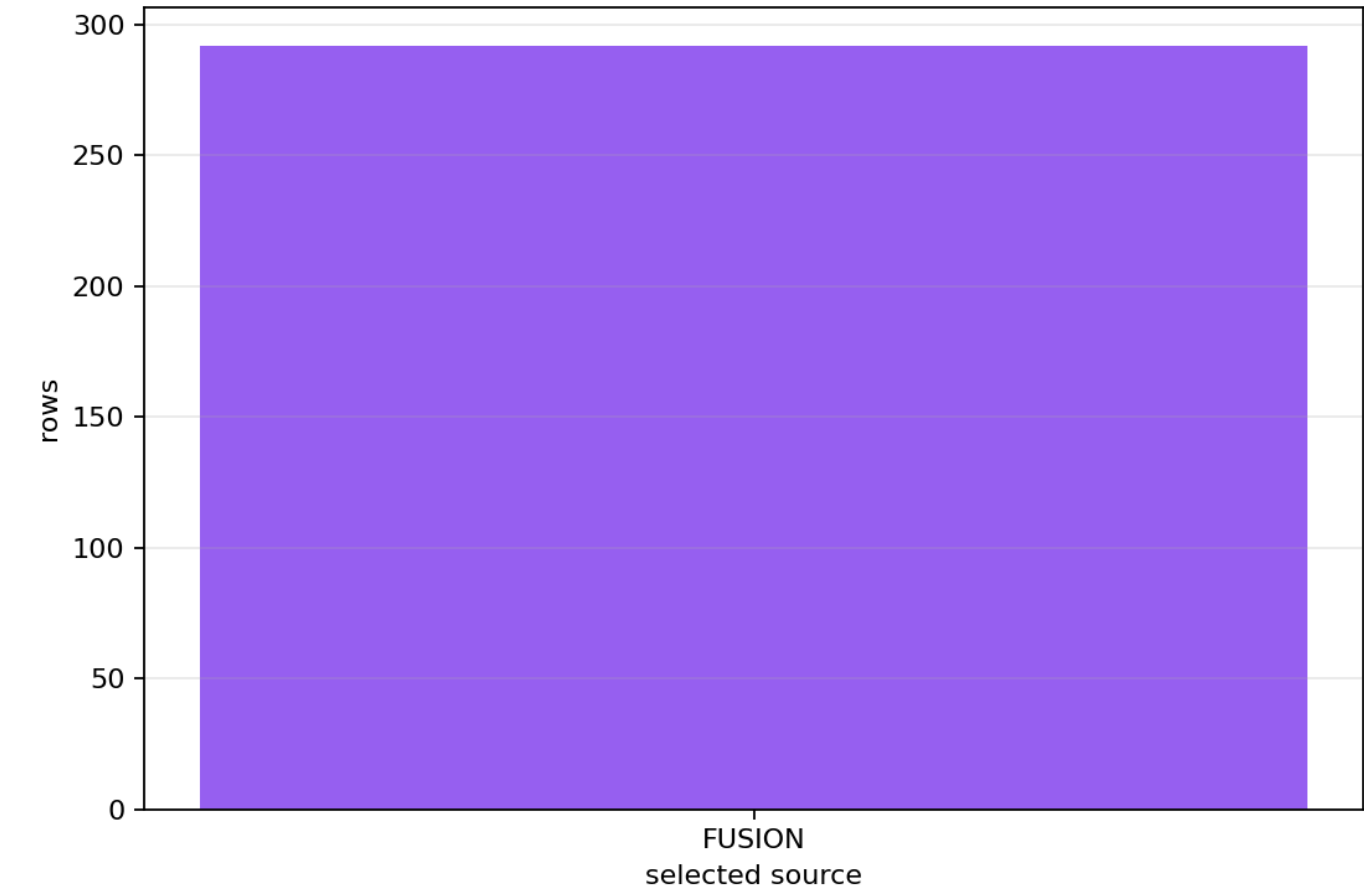
Experiment 3 lambda condition timeline



fused lambda_pred condition number GNSS condition number LiDAR condition number fusion FIM fused condition 7.36 27.29 fusion

Source selection histogram

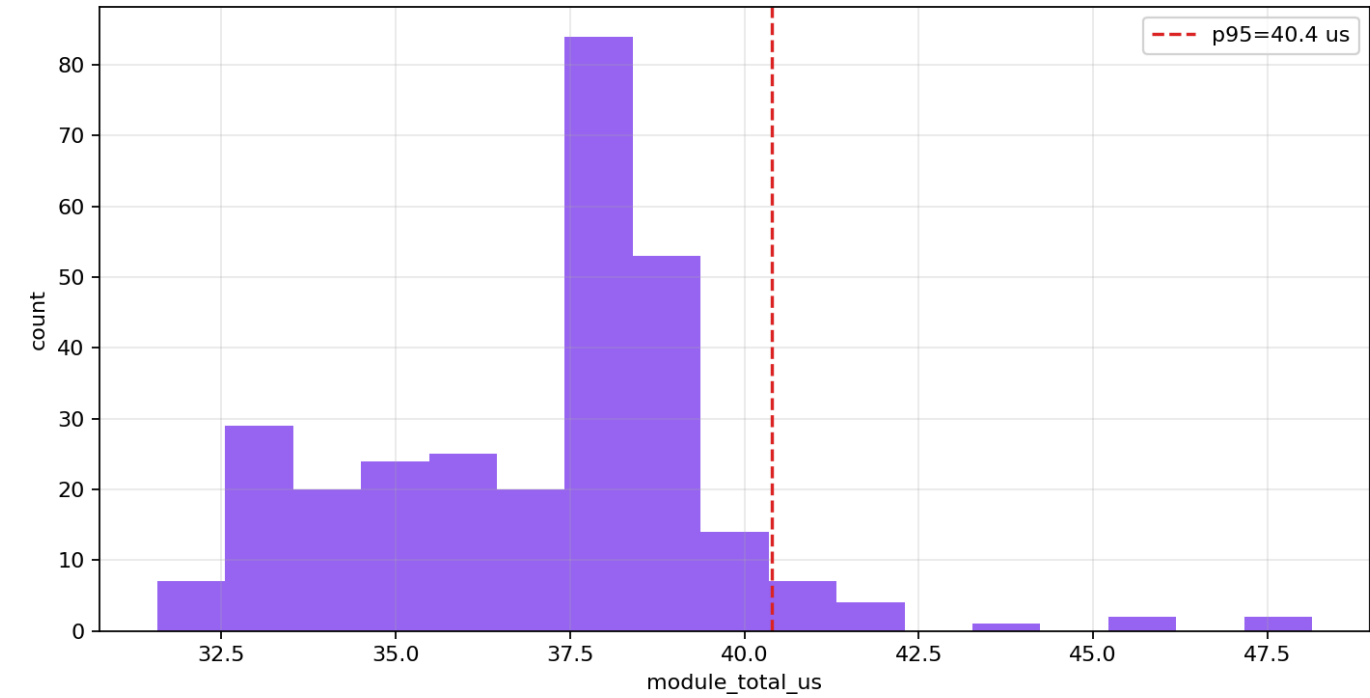
Experiment 3 source selection histogram



selected source query experiment preset source selection 292 query FUSION GNSS/LiDAR fallback selected output

Latency distribution

Experiment 3 Predictor fusion query latency



module_total_us p95 latency system probe fusion predictor query module_total_us p95 40.39 us 10000 us ROS 2

17.6

Predictor system experiment 3 `predictor_fusion_nominal` nominal GNSS + feature-rich LiDAR + current prior 292 Fusion advisory query selected source `FUSION` GNSS/LiDAR/Fused valid ratio 100% fallback ratio 0% `lambda_sum_error_p95=0.0`

planner/grid disabled predictor probe `/iap/integrity` GNSS advisory FIM LiDAR advisory FIM current prior Fusion selected output system test plan nominal fusion advisory availability FIM

18. System Predictor GNSS degraded + LiDAR good fusion

system experiment preset

```
predictor_gnss_degraded_lidar_good
```

```
src/iap/results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/
```

18.1

predictor GNSS degraded feature-rich LiDAR current prior system launch planner Fusion advisory result GNSS degradation LiDAR support Fusion FIM

- `predictor_output_mode=fusion` selected source `FUSION`
- GNSS degraded open-sky baseline HPL/PDOP
- LiDAR feature-rich diagnostics GNSS degraded
- Fusion FIM `lambda_pred = lambda_prior + lambda_gnss + lambda_lidar`
- Planner predictor risk overlay
- GNSS LiDAR Fusion fallback latency metadata analyzer CSV/PNG IAP `export/profiling/metadata/figures`

18.2

run launch

experiment	predictor_gnss_degraded_lidar_good
araim_experiment	gnss_degraded_lidar_good
predictor_output_mode	fusion
validator_required_selected_source	FUSION
predictor_enable_debug_log	true
validator_require_debug_logs	true
start_planner	true
planner odom topic	/sim/drone_0/truth_odom
planner uses predictor	false
planner risk overlay	false
predictor probe query set	current_pose

planner run1 query x=-12.03 - 11.73 m y=-0.28 - 2.86 m z=0.00 - 2.36 m Predictor probe odom planner truth-control topic predictor evidence

```
RUN_ROOT=/home/dev/ws_iap/src/iap/results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1
OPEN_SKY_RUN=/home/dev/ws_iap/src/iap/results/predictor_validation/predictor_system_predictor_gnss_open_sky_only_run1

ros2 launch iap test_predictor.launch.py \
  experiment:=predictor_gnss_degraded_lidar_good \
  start_planner:=true \
  start_rviz:=false \
  record_bag:=false \
  predictor_enable_debug_log:=true \
  validator_require_debug_logs:=true \
  run_duration_s:=120 \
  validation_duration_s:=115 \
  predictor_export_dir:=$RUN_ROOT/export
```

```
python3 src/iap/scripts/dev_predictor/analyze_predictor_system_e4.py \  
  --run-dir $RUN_ROOT \  
  --open-sky-run-dir $OPEN_SKY_RUN \  
  --fail-on-threshold
```

18.3

4 analyzer

FUSION selected source ratio	> 70%	100%
GNSS valid ratio	> 70%	100%
LiDAR valid ratio	> 70%	100%
Fused valid ratio	> 70%	100%
selected fallback ratio	< 10%	0%
median gnss_hpl vs E1 baseline	E4 > E1	10.7543 > 10.7406 m
median gnss_pdop vs E1 baseline	E4 > E1	4.6636 > 4.6619
median lidar_lambda_condition	< 1e6	7.72
lambda_sum_error p95	<= 1e-6	0.0
module_total_us p95	< 10000 us	46.49 us
valid rows NaN/Inf		0
fallback reason	fallback rows	fallback rows

predictor_e4_analysis_summary.json passed=true analyzer run1
export/test_predictor_validation_summary.json E4 analyzer summaryprobe summary CSV predictor
validator validator summary

18.4

Analyzer

```
passed: true  
query_count: 269  
selected_source_FUSION_ratio: 1.0  
gnss_valid_ratio: 1.0  
lidar_valid_ratio: 1.0  
fused_valid_ratio: 1.0  
fallback_ratio: 0.0  
median_gnss_hpl: 10.7543 m  
median_open_sky_gnss_hpl: 10.7406 m  
median_gnss_pdop: 4.6636  
median_open_sky_gnss_pdop: 4.6619  
median_lidar_lambda_condition: 7.72  
lambda_sum_error_p95: 0.0  
module_total_us p50/p95/max: 38.37 / 46.49 / 49.93 us
```

CSV 269 predictor query 67.83 s 3.95 Hz selected source FUSION GNSSLiDAR fused advisory query valid
selected fallback count 0

Fusion advisory

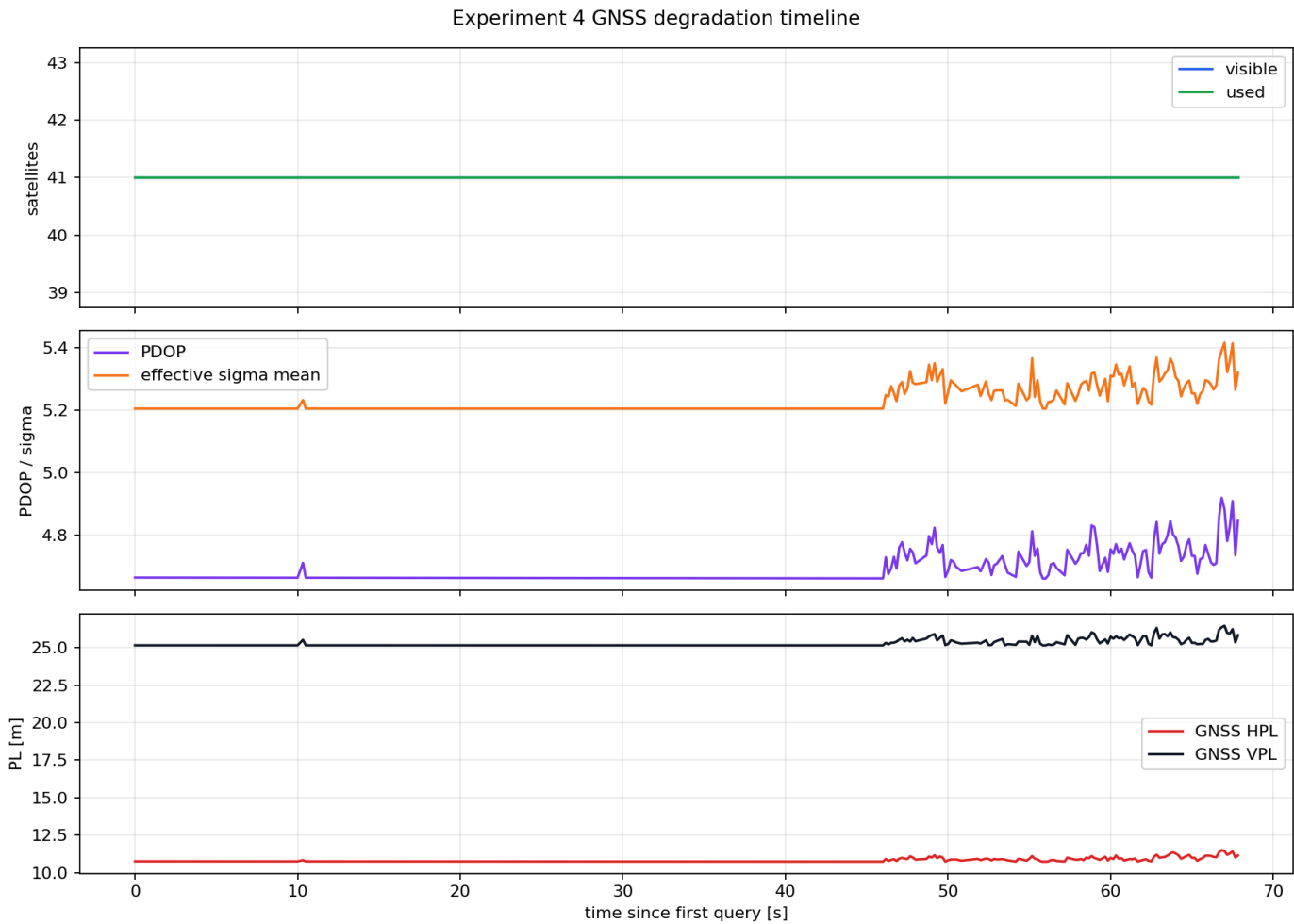
selected_hpl	0.5034 - 0.9722 m 0.9274 m
selected_vpl	0.2050 - 0.3407 m 0.3366 m
gnss_hpl	10.7330 - 11.5068 m 10.7543 m
gnss_vpl	25.1475 - 26.4521 m 25.1514 m
gnss_pdop	4.6609 - 4.9196 4.6636
GNSS visible/used satellites	visible 41used 41excluded 0
effective_sigma_mean	5.2058 - 5.4172 5.2058
lidar_n_primitives	267 - 806 277

lidar_lambda_trace	275.9161 - 888.7942	285.1882
lidar_lambda_min_eig	26.0925 - 98.2111	28.6881
lidar_lambda_condition	4.9181 - 11.9988	7.7210
fused_lambda_prior_trace	0.0341 - 13.4113	0.0746
fused_lambda_gnss_trace	0.7797 - 0.8546	0.8540
fused_lambda_lidar_trace	275.9161 - 888.7942	285.1882
fused_lambda_pred_trace	276.8233 - 889.6918	286.1689
fused_lambda_pred_min_eig	26.4495 - 98.6700	29.0671
fused_lambda_pred_condition	4.8955 - 11.8627	7.6240
lambda_sum_error	0.0 - 0.0	p95 0.0

GNSS degraded E1 baseline HPL/PDOP run1 gnss_n_visible=41 gnss_n_used=41 E4 median gnss_hpl gnss_pdop E1 open-sky baseline analyzer degradation

18.5

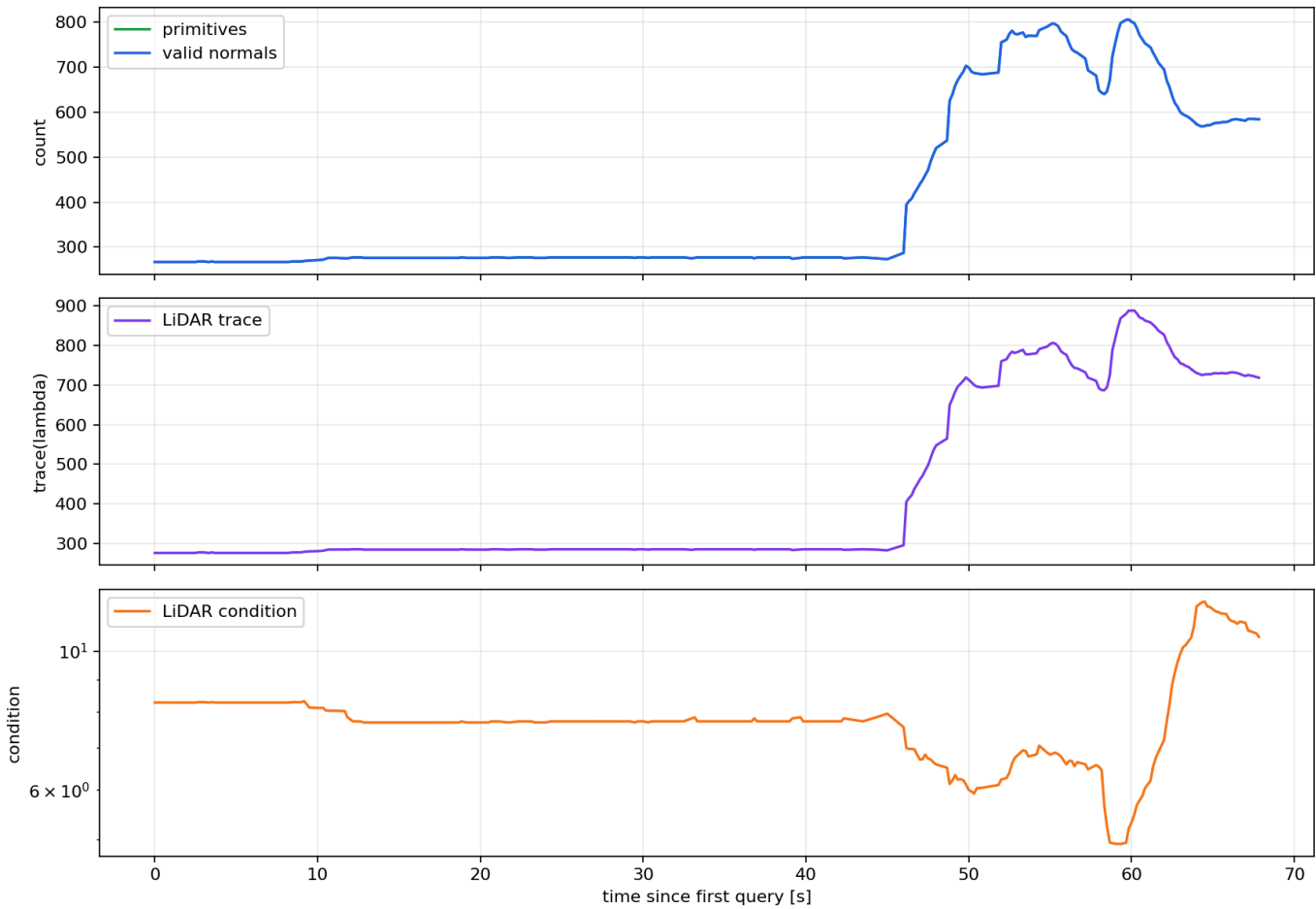
GNSS degradation timeline



GNSS HPL/VPL PDOP effective sigma visible/used satellite count degraded GNSS visible/used satellite count 41 HPL PDOP effective sigma

LiDAR support timeline

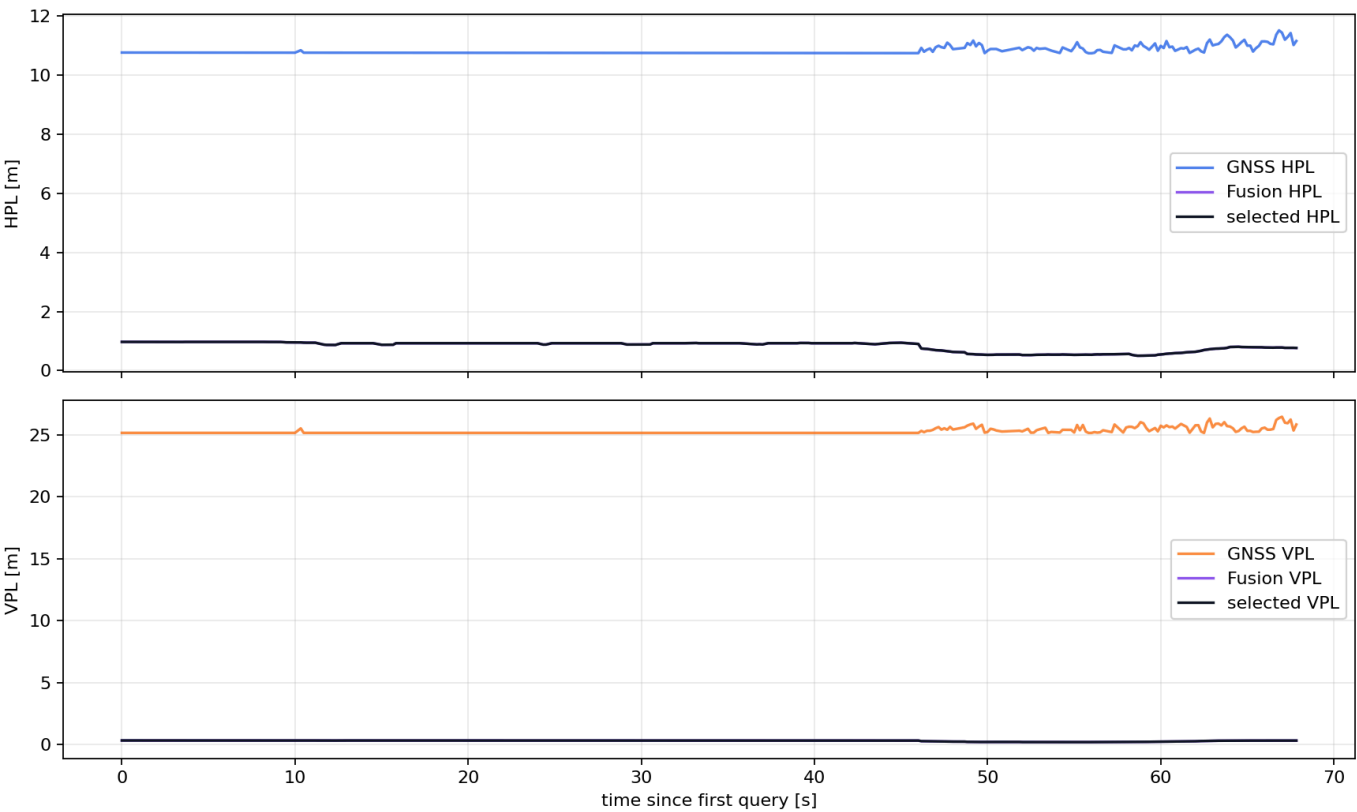
Experiment 4 LiDAR support timeline



LiDAR primitive valid normal LiDAR information trace condition number GNSS degraded LiDAR feature-rich
primitive 277 LiDAR condition 7.72 1e6

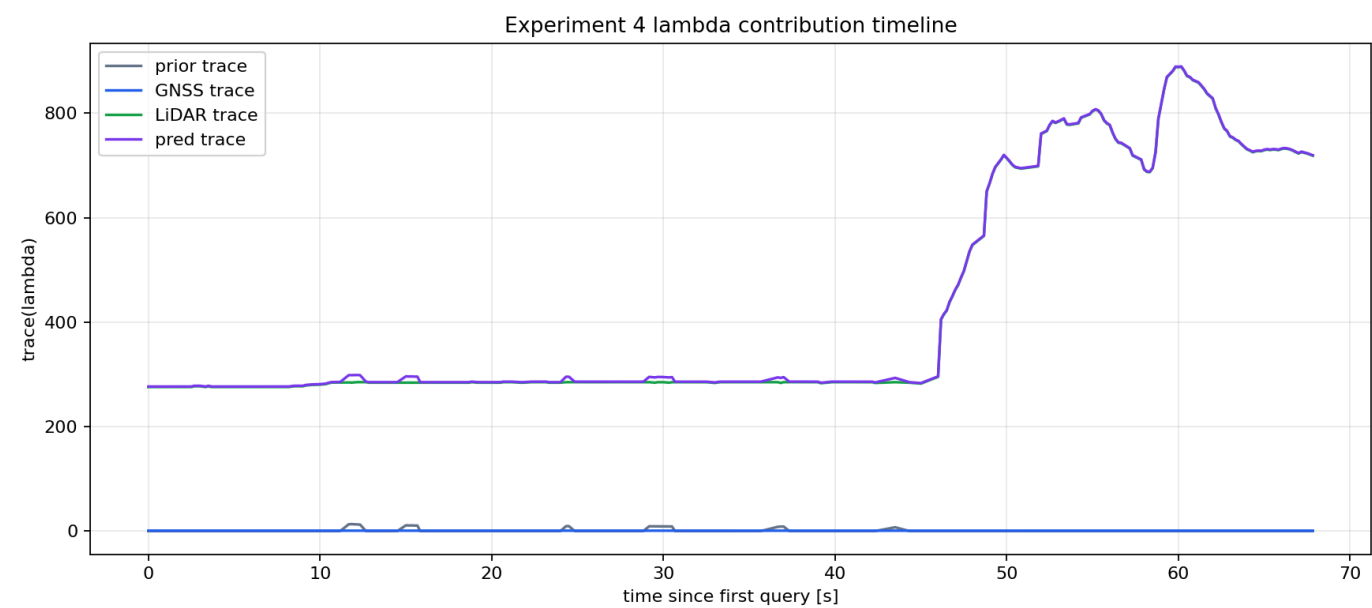
GNSS vs Fusion HPL/VPL

Experiment 4 GNSS vs Fusion HPL/VPL



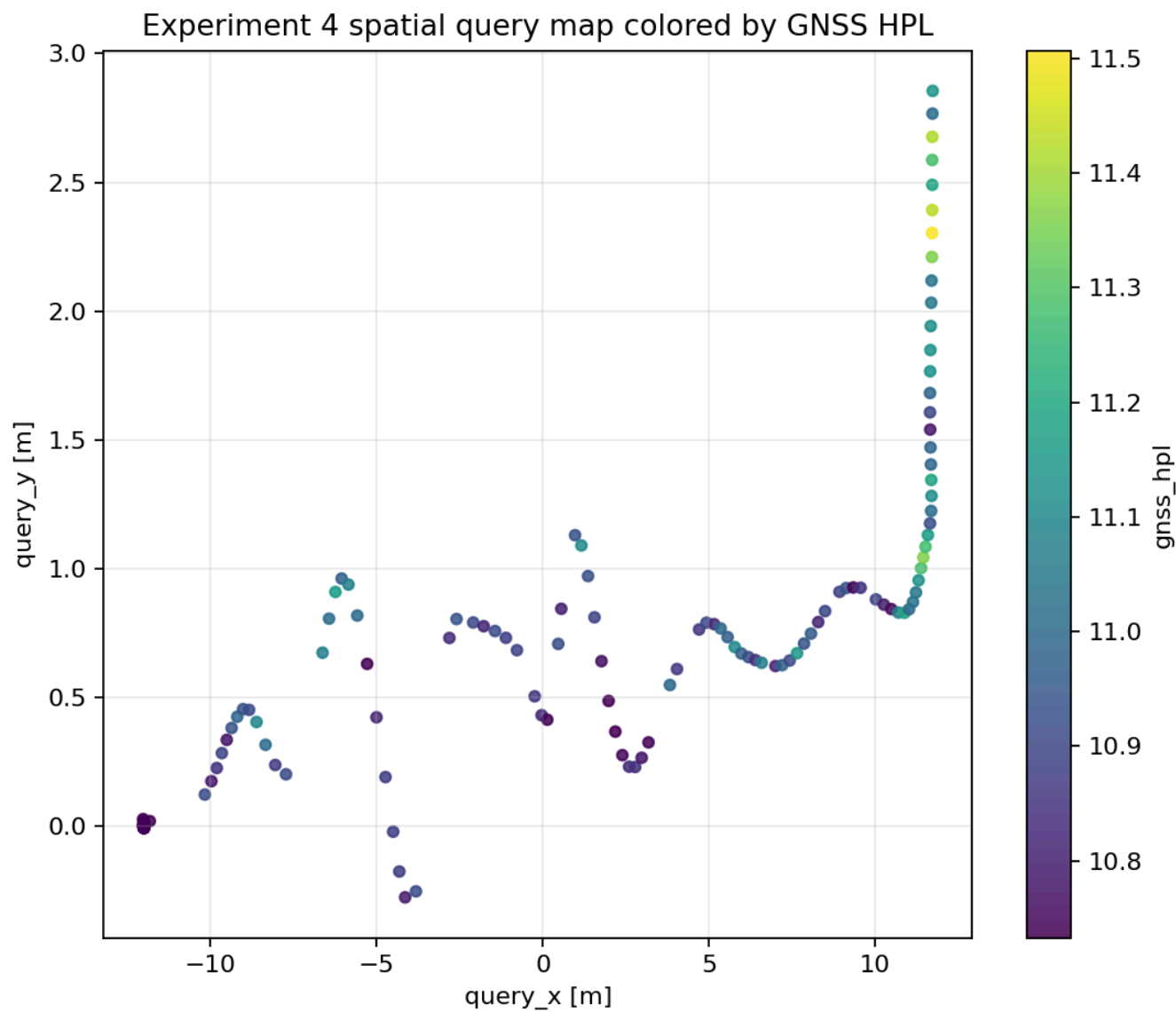
GNSS-only HPL/VPL fused/selected HPL/VPL selected output fusion fusion LiDAR/prior advisory PL
selected HPL 0.927 m GNSS HPL 10.754 m

Lambda contribution timeline



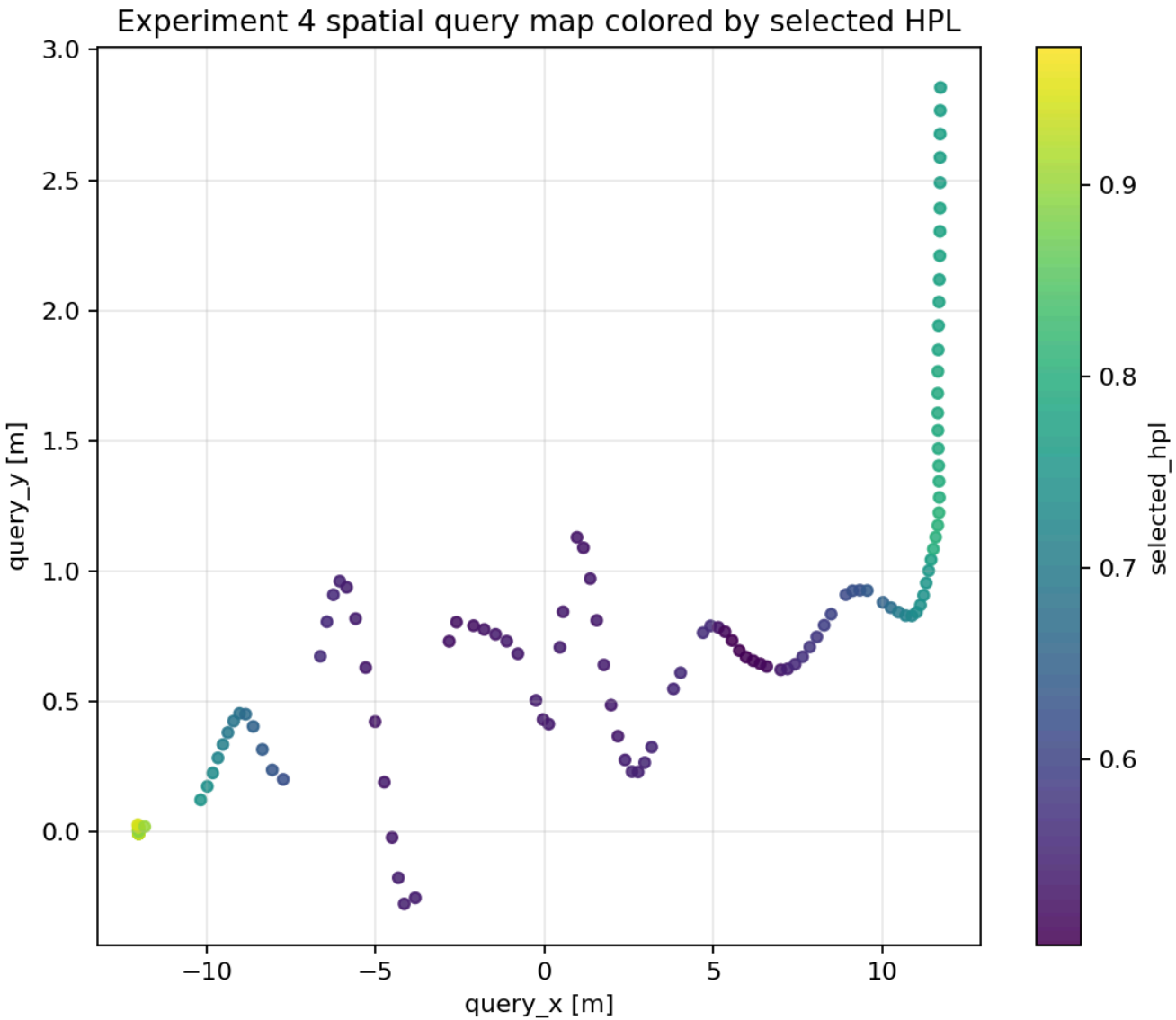
prior trace GNSS trace LiDAR trace fused predicted trace fusion information LiDAR trace
285.19 GNSS trace 0.85 feature-rich LiDAR fused information lambda_sum_error_p95=0.0

Spatial query map colored by GNSS HPL



Experiment 4 spatial query map colored by GNSS HPL. The plot shows the spatial distribution of query points colored by GNSS HPL. The x-axis represents query_x [m] and the y-axis represents query_y [m]. The color bar indicates the GNSS HPL values, ranging from 10.8 to 11.5. The query points are clustered in a path, with the highest HPL values (yellow) at the end of the path and lower values (purple) at the beginning.

Spatial query map colored by selected HPL



query selected HPL fusion selected output selected HPL 0.5034 - 0.9722 m
Fusion selected result

18.6

Predictor system experiment 4 predictor_gnss_degraded_lidar_good planner GNSS degraded feature-rich LiDAR Fusion output mode 269 Fusion advisory query selected source FUSION GNSS/LiDAR/Fused valid ratio 100% fallback ratio 0% lambda_sum_error_p95=0.0

predictor GNSS LiDAR feature-rich system Fusion selected result system test plan GNSS degradation evidence LiDAR diagnostic health Fusion source selection FIM IAP log

19. System Predictor LiDAR sparse + GNSS good fusion

system experiment preset

predictor_lidar_sparse_gnss_good

src/iap/results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/

19.1

predictor GNSS good LiDAR sparse/degraded Fusion output mode LiDAR GNSS-good advisory E5 selected source FUSION conservative gate selected/fused PL GNSS-only reference

00000000

- predictor_output_mode=fusion selected source 000 FUSION
- GNSS open-sky reference 00000GNSS HPL/PDOP 000000 baseline 00000
- LiDAR sparse/degraded indicator 00000 LiDAR diagnostics 0000
- conservative max with GNSS 000selected/fused HPL/VPL 000 GNSS HPL/VPL
- Fusion FIM 00000000lambda_pred = lambda_prior + lambda_gnss + lambda_lidar
- lidar_gate_debug.csvGNSS/LiDAR/Fusion debugfallbacklatencymetadata PNG 00 IAP export/profiling/metadata/figures 00000

19.2 0000

00 run 0000 launch 000

000	0
experiment	predictor_lidar_sparse_gnss_good
araim_experiment	lidar_degraded_gnss_good
predictor_output_mode	fusion
validator_required_selected_source	FUSION
predictor_enable_debug_log	true
validator_require_debug_logs	true
predictor_fusion_conservative_max_with_gnss	true
probe_lidar_map_max_points	500
start_planner	false
predictor probe query set	current_pose

E5 00000 planner000000 sparse/degraded LiDAR 0 fusion advisory 0000000000000000probe_lidar_map_max_points=500 0000 probe 000 LiDAR map 0000000000000000 sparse LiDAR diagnostics

00000

```
RUN_ROOT=/home/dev/ws_iap/src/iap/results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1
OPEN_SKY_RUN=/home/dev/ws_iap/src/iap/results/predictor_validation/predictor_system_predictor_gnss_open_sky_only_run1

ros2 launch iap test_predictor.launch.py \
  experiment:=predictor_lidar_sparse_gnss_good \
  start_rviz:=false \
  record_bag:=false \
  predictor_enable_debug_log:=true \
  validator_require_debug_logs:=true \
  run_duration_s:=90 \
  validation_duration_s:=85 \
  predictor_export_dir:=$RUN_ROOT/export

python3 src/iap/scripts/dev_predictor/analyze_predictor_system_e5.py \
  --run-dir $RUN_ROOT \
  --open-sky-run-dir $OPEN_SKY_RUN \
  --fail-on-threshold
```

19.3 000

00 5 analyzer 0000000000000000

000	00	0000
FUSION selected source ratio	> 80%	100%
GNSS valid ratio	> 95%	100%
Fused valid ratio	> 80%	100%
selected fallback ratio	< 10%	0%
median gnss_hpl vs E1 baseline	<= 1.10x E1 10.7380	<= 11.8147 m
median gnss_pdop vs E1 baseline	<= 1.10x E1 4.6615	<= 5.1280
LiDAR degraded indicator	0000	0000median primitives 28
conservative gate violation	0	0
lambda_sum_error p95	<= 1e-6	0.0

模块	指标	结果
module_total_us p95	< 10000 us	38.51 us
valid rows NaN/Inf	0	0
fallback reason	fallback rows 0	0 fallback rows

predictor_e5_analysis_summary.json 通过passed=true验证器summary 通过passed=trueAnalyzer/probe 运行 529 查询验证器 验证验证_duration_s=85 498 查询selected source FUSION

19.4 性能

Analyzer 性能

```
passed: true
query_count: 529
selected_source_FUSION_ratio: 1.0
gnss_valid_ratio: 1.0
lidar_valid_ratio: 1.0
fused_valid_ratio: 1.0
fallback_ratio: 0.0
median_gnss_hpl: 10.7380 m
median_open_sky_gnss_hpl: 10.7406 m
median_gnss_pdop: 4.6615
median_open_sky_gnss_pdop: 4.6619
median_lidar_n_primitives: 28
median_lidar_lambda_condition: 4.75
conservative_gate_violation_count: 0
lambda_sum_error_p95: 0.0
module_total_us p50/p95/max: 36.67 / 38.51 / 47.04 us
```

CSV 529 预测器查询88.00 s6.00 Hzselected source FUSIONselected/fused/GNSS advisory query validselected fallback count 0

Fusion advisory 性能

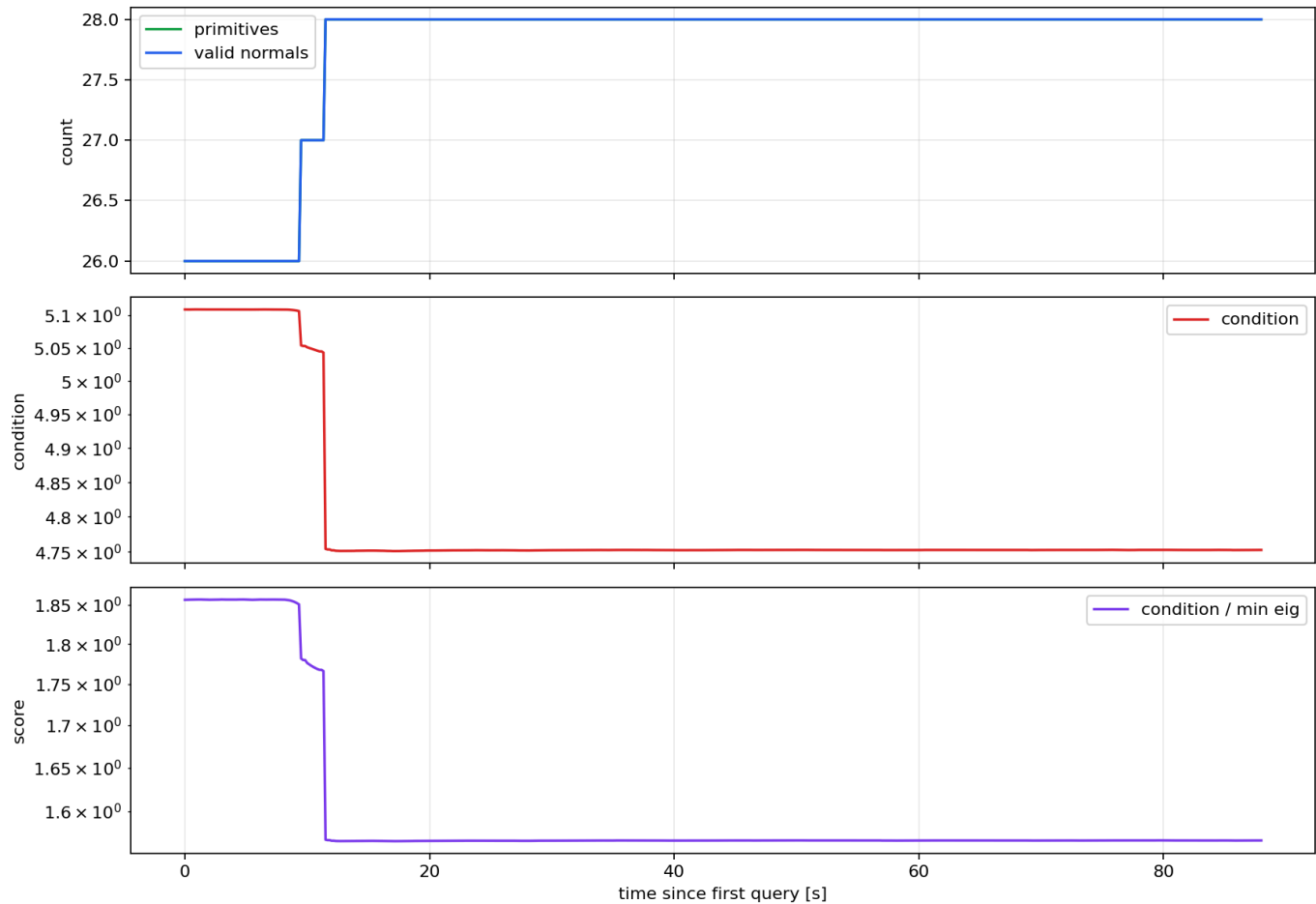
指标	结果
selected_hpl	10.7196 - 10.7569 m10.7380 m
selected_vpl	25.1441 - 25.1517 m25.1486 m
gnss_hpl	10.7196 - 10.7569 m10.7380 m
gnss_vpl	25.1441 - 25.1517 m25.1486 m
gnss_pdop	4.6591 - 4.66394.6615
GNSS visible/used satellites	visible 41used 41
lidar_n_primitives	26 - 2828
lidar_lambda_trace	22.0507 - 23.266623.2660
lidar_lambda_min_eig	2.7509 - 3.03183.0314
lidar_lambda_condition	4.7509 - 5.10884.7525
fused_lambda_prior_trace	1.7509 - 2.22842.0743
fused_lambda_gnss_trace	0.8539 - 0.85500.8544
fused_lambda_lidar_trace	22.0507 - 23.266623.2660
fused_lambda_pred_trace	24.6569 - 26.348826.1733
fused_lambda_pred_condition	3.3735 - 3.72873.4359
selected_hpl - gnss_hpl	0.0 - 0.0 m
selected_vpl - gnss_vpl	0.0 - 0.0 m
lambda_sum_error	0.0 - 0.0p95 0.0

E5 LiDAR sparse/degraded primitive feature-rich runE5 median lidar_n_primitives=28E4 feature-rich run 277fusion_conservative_max_with_gnss=truesource flags conservative_max 529 查询selected/fused PL GNSS reference GNSS-only selected output

19.5 性能

LiDAR degradation timeline

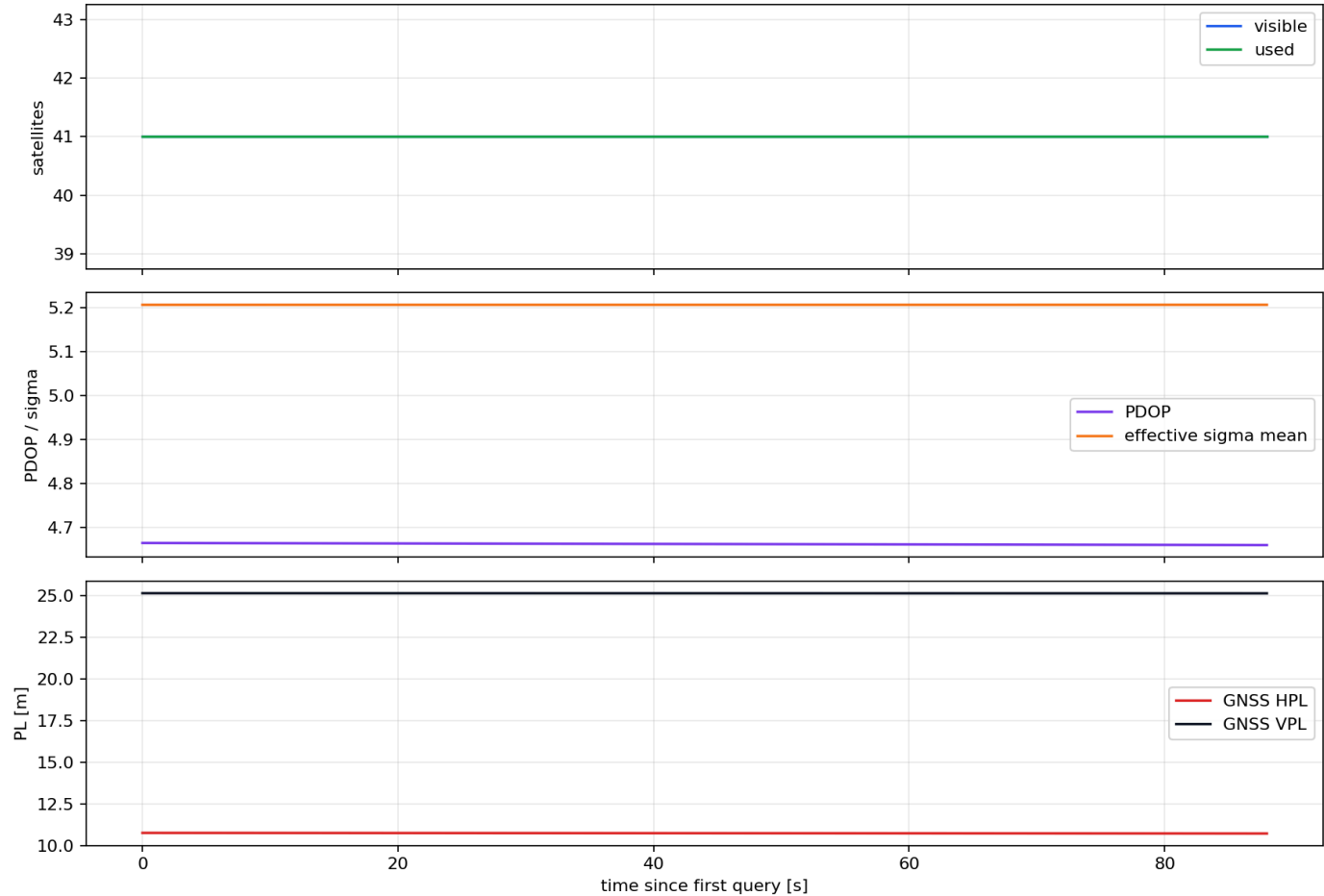
Experiment 5 LiDAR degradation timeline



LiDAR primitive valid normal LiDAR condition number degradation score E5 LiDAR sparse/degraded primitive 26 - 28 feature-rich sparse indicator

GNSS stability timeline

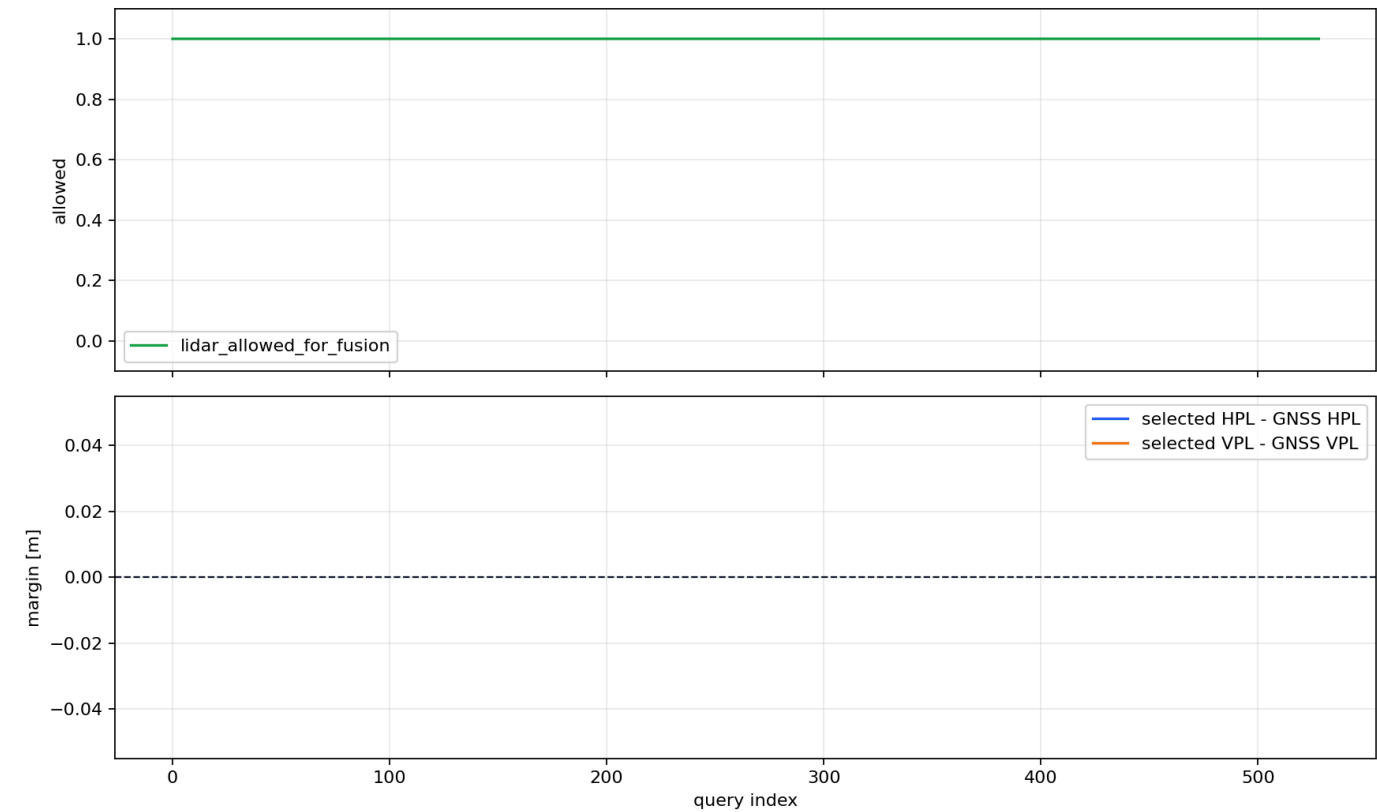
Experiment 5 GNSS stability timeline



GNSS visible/used satellite count PDOP effective sigma GNSS HPL/VPL E5 GNSS-good visible/used satellite count
41 median GNSS HPL/PDOP E1 open-sky baseline

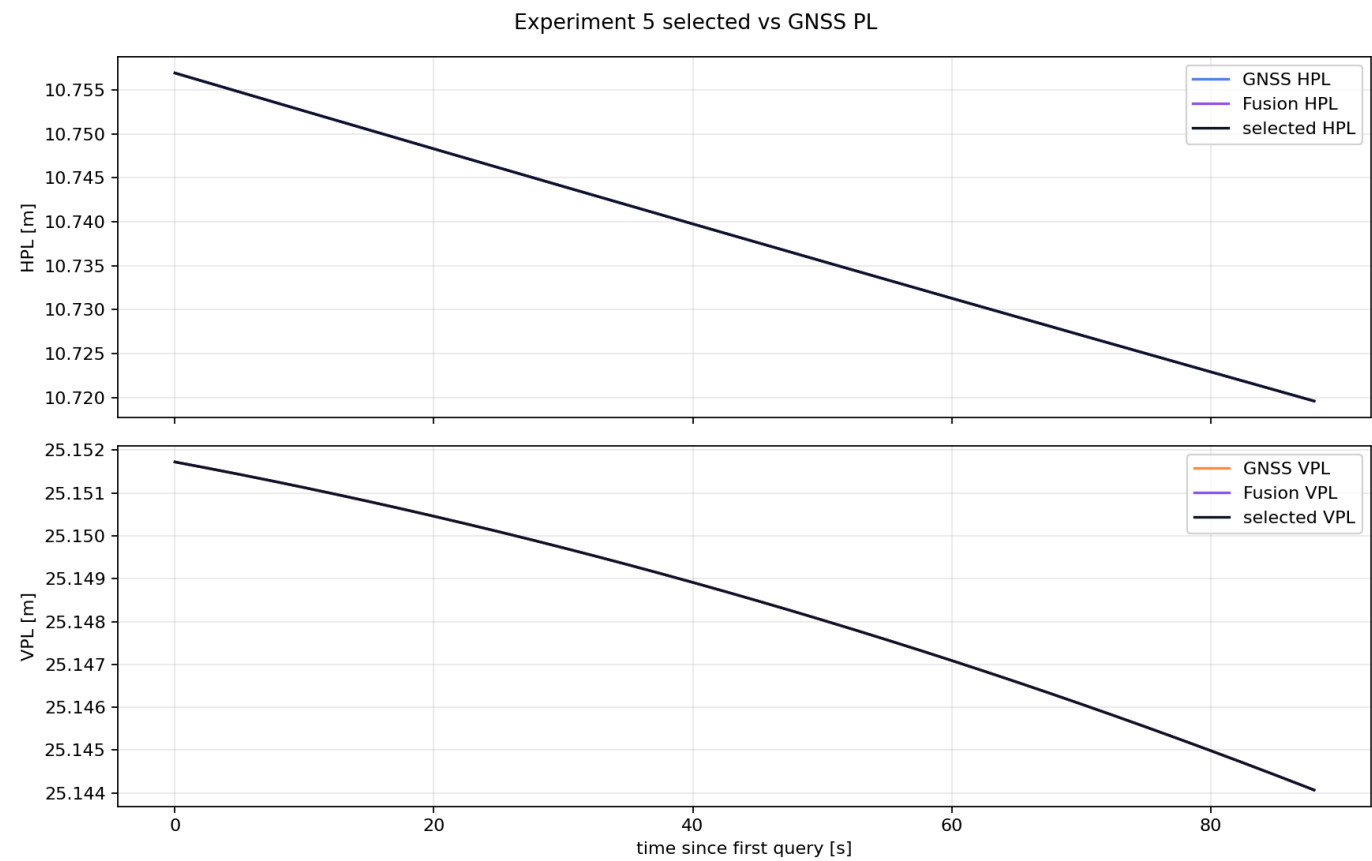
Fusion gate timeline

Experiment 5 fusion gate timeline



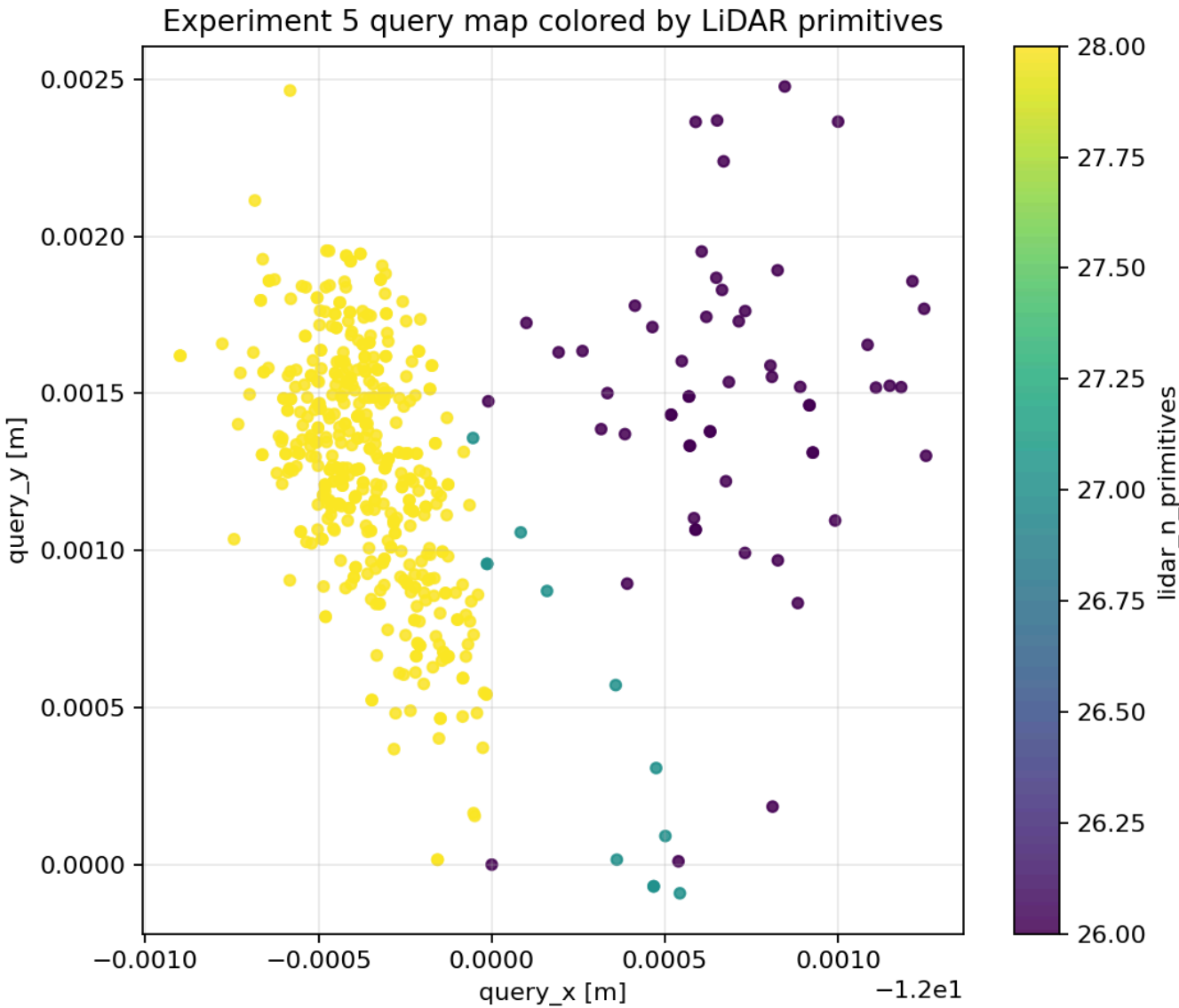
analyzer
 lidar_allowed_for_fusion
 selected_hpl - gnss_hpl
 selected_vpl - gnss_vpl
 LiDAR
 selected output
 GNSS reference
 margin 0.0
 conservative gate violation count 0

Selected vs GNSS PL



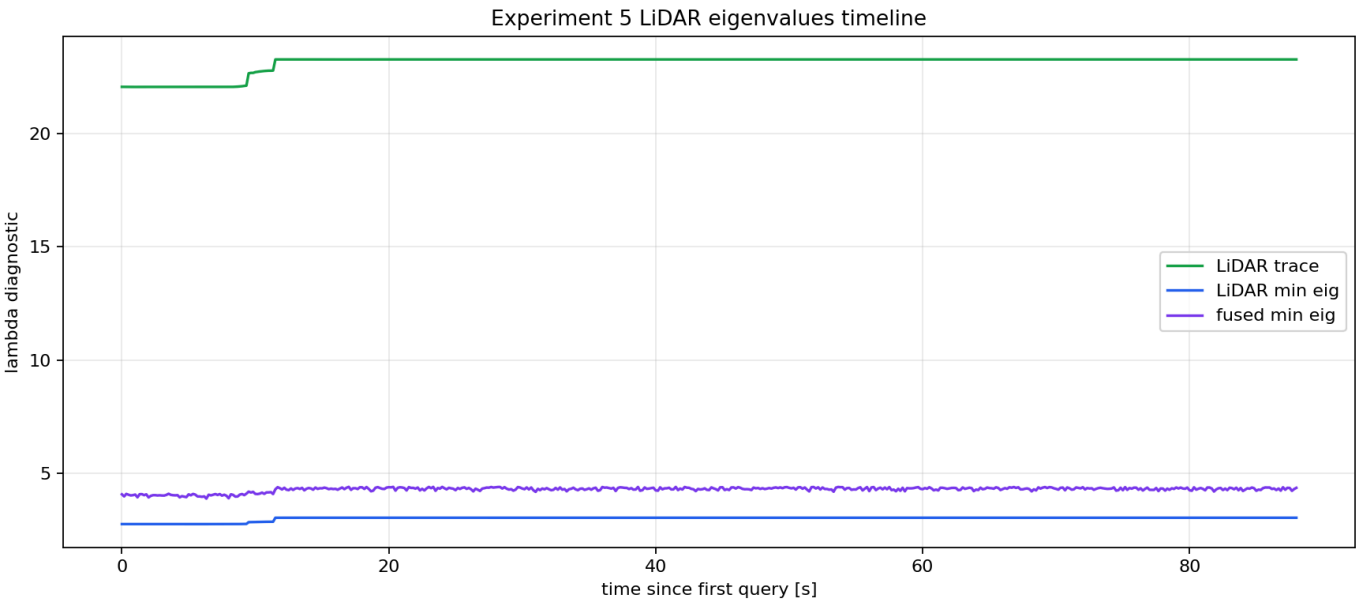
selected/fused HPL/VPL
 GNSS HPL/VPL
 selected source
 FUSION
 selected PL
 GNSS-only reference
 selected HPL/VPL
 GNSS HPL/VPL
 conservative max with GNSS

LiDAR primitives top-down



query lidar_n_primitives sparse LiDAR primitive planner query set current_pose
sparse primitive count

LiDAR eigenvalues timeline



LiDAR trace LiDAR min eigenvalue fused min eigenvalue sparse LiDAR FIM fused information LiDAR
trace 23.27 E4 feature-rich 285.19 sparse/degraded

19.6

Predictor system experiment 5 predictor_lidar_sparse_gnss_good GNSS goodLiDAR sparse/degraded Fusion output mode 529 Fusion advisory queryselected source FUSIONGNSS/Fused valid ratio 100%fallback ratio 0% conservative_gate_violation_count=0lambda_sum_error_p95=0.0

sparse LiDAR predictor Fusion selected result conservative max with GNSS LiDAR selected/fused PL GNSS-only reference system test plan LiDAR sparse/degraded evidenceGNSS stabilityFusion source selectionconservative gate IAP log

20. System Predictor GNSS outage + LiDAR recovery

system experiment preset

predictor_gnss_outage_lidar_recovery

src/iap/results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/

20.1

predictor GNSS outage window fallback GNSS-required policyLiDAR/Fusion advisory GNSS epoch GNSS advisoryselected output Fusion NONE/INVALID fallback

outage window

- gnss_valid=false GNSS advisory selected output
- selected_valid=falseselected source NONE/INVALID
- fallbackfallback reasonno_gnss_epoch too_few_sats
- GNSS recovery selected output valid
- IAP export/profiling/metadata/figures

20.2

run launch

experiment	predictor_gnss_outage_lidar_recovery
araim_experiment	manual
predictor_output_mode	fusion
validator_required_selected_source	source
predictor_enable_debug_log	true
validator_require_debug_logs	true
predictor_fusion_conservative_max_with_gnss	false
gnss_ephemeris_source	synthetic
gnss_enabled_constellations	GPS
gnss_scenario_file	generated:gnss_outage
gnss_enable_fault_injection	true
probe_gnss_epoch_max_age_s	0.5
probe_require_gnss_for_selected_output	true
start_planner	false
predictor probe query set	current_pose
analyzer policy	gnss_required
nominal outage window	35.0 - 60.0 s
analyzer observed outage window	35.0 - 58.8 s

GNSS simulator nominal outage window 35.0 - 60.0 s predictor query 58.83 s GNSS epoch query outageE6 analyzer run4 35.0 - 58.8 s outage window

```
RUN_ROOT=/home/dev/ws_iap/src/iap/results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_
run4

ros2 launch iap test_predictor.launch.py \
  experiment:=predictor_gnss_outage_lidar_recovery \
  start_rviz:=false \
  start_planner:=false \
  record_bag:=false \
  run_duration_s:=90 \
  validation_duration_s:=85 \
  predictor_enable_debug_log:=true \
  validator_require_debug_logs:=true \
  predictor_export_dir:=$RUN_ROOT/export

ros2 run iap analyze_predictor_system_e6.py \
  --run-dir $RUN_ROOT \
  --policy gnss_required \
  --outage-start-s 35.0 \
  --outage-end-s 58.8
```

20.3

6 analyzer

pre-outage selected valid ratio	>= 50%	96.4%
outage gnss_valid ratio	<= 20%	0.0%
outage invalid or explained fallback ratio	>= 95%	100%
post-recovery selected valid ratio	>= 50%	100%
module_total_us p95	< 15000 us	19.95 us
valid rows NaN/Inf		0
required debug files		

predictor_e6_analysis_summary.json passed=truefailures=[]test_predictor_validation_summary.json passed=true
required debug logs

20.4

Analyzer

```
passed: true
query_count: 326
window_counts: before=139, outage=70, after=117
before_selected_valid_ratio: 0.9640
outage_gnss_valid_ratio: 0.0
outage_valid_ratio: 0.0
outage_invalid_or_explained_fallback_ratio: 1.0
after_selected_valid_ratio: 1.0
source_histogram: FUSION=251, NONE=75
outage_source_histogram: NONE=70
fallback_reason_histogram: selected/module/gnss/fused no_gnss_epoch = 75
module_total_us p50/p95/max: 16.05 / 19.95 / 34.60 us
predictor_invalid_start_s: 35.17 s
predictor_valid_recovery_s: 58.83 s
recovery_delay_s: 0.03 s
```

CSV 326 predictor query analyzer 35.0 - 58.8 s outage window outage 70 query

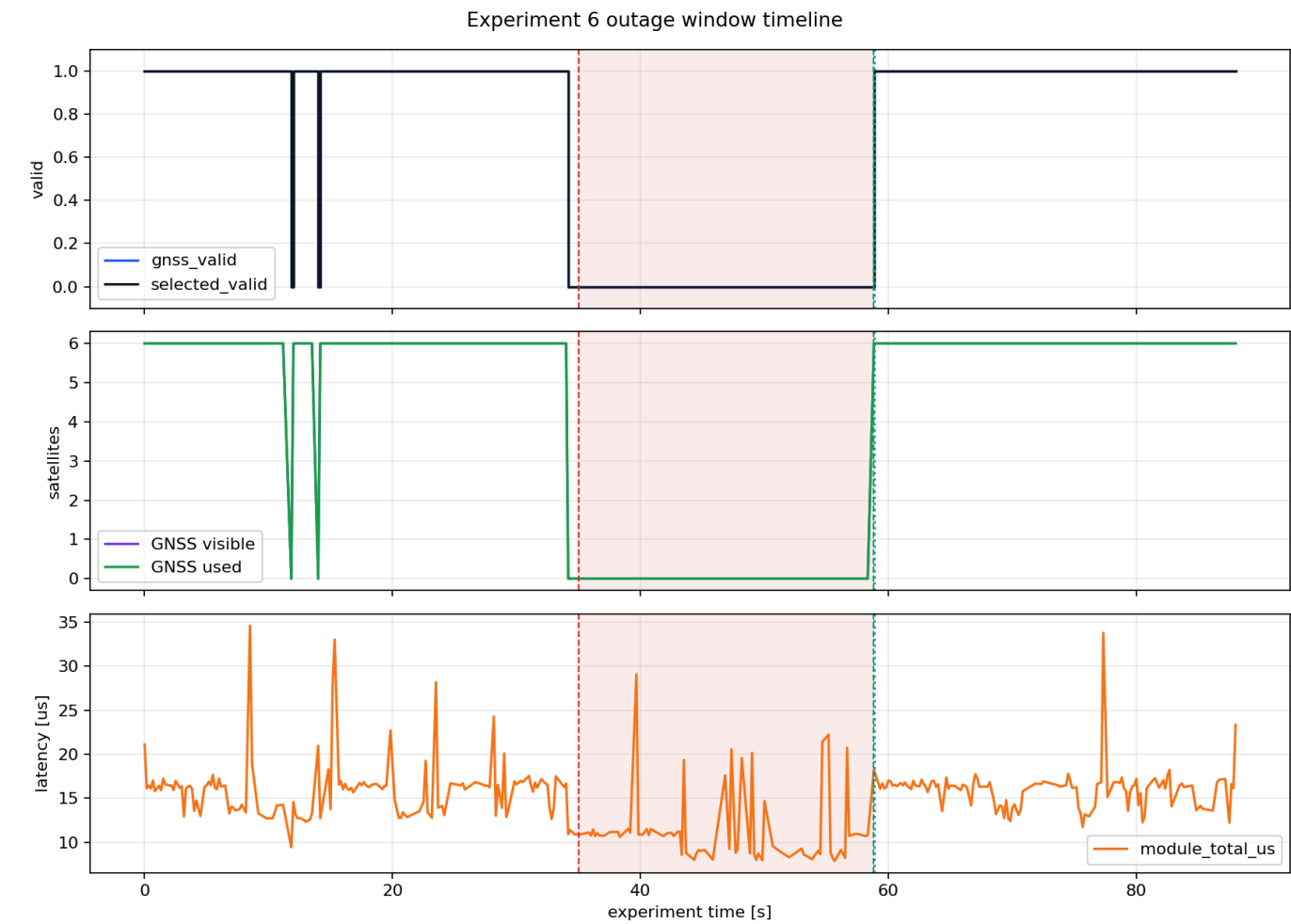
	before outage	outage window	after recovery
query count	139	70	117
selected source	FUSION	NONE 70/70	FUSION 117/117
selected valid ratio	96.4%	0.0%	100%
GNSS valid ratio	96.4%	0.0%	100%
selected fallback ratio	3.6%	100%	0%

	before outage	outage window	after recovery
median gnss_n_visible	6	0	6
outage fallback reason	-	gnss:no_gnss_epoch	-
recovery delay	-	-	0.03 s

run4 GNSS outage outage window GNSS gnss_valid_ratio=0.0selected source NONEselected output invalid fallback reason gnss:no_gnss_epochGNSSselected output 0.03 s valid FUSION

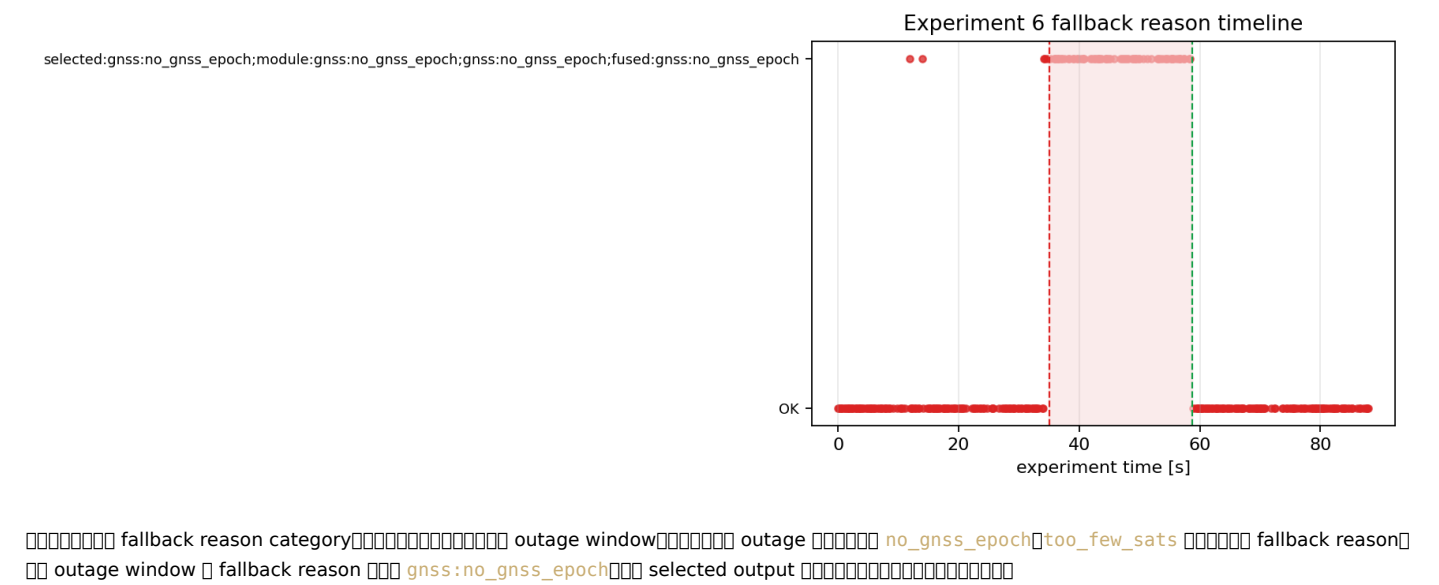
20.5

Outage window timeline

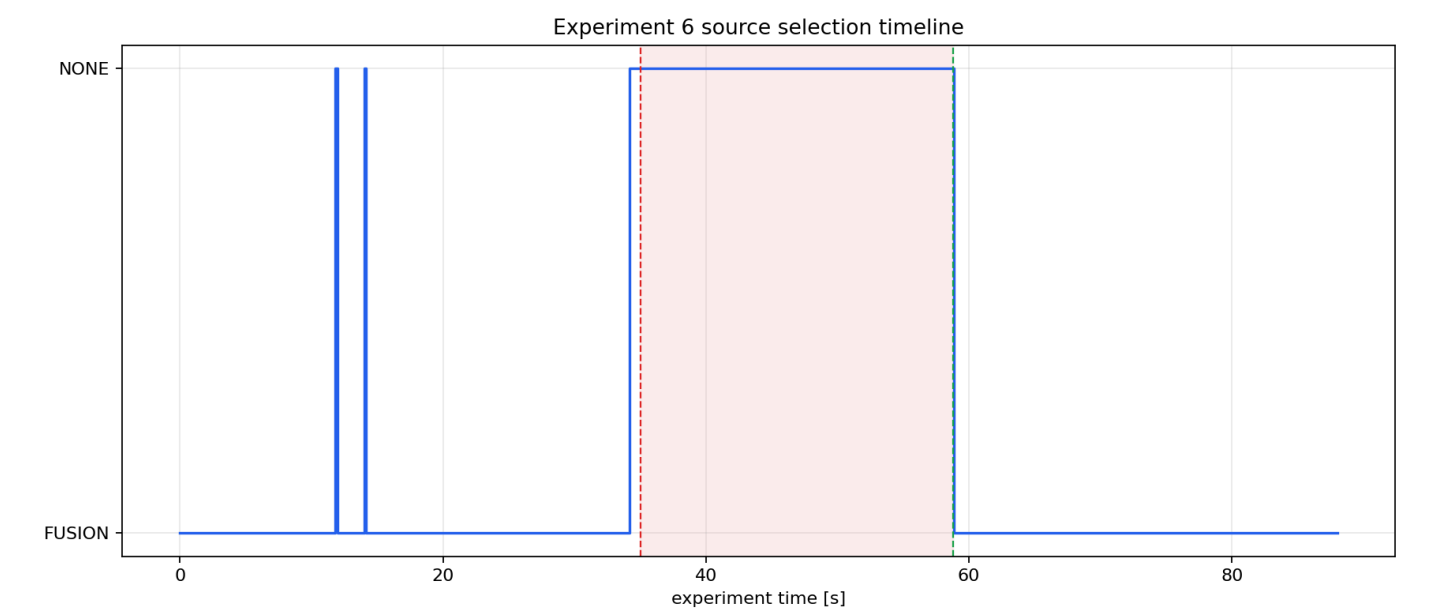


analyzer GNSS outage window 35.0 - 58.8 sgnss_validselected_validGNSS visible/used satellite countmodule_total_usoutage selected output latency gnss_valid selected_valid 0 1 GNSS-required policy

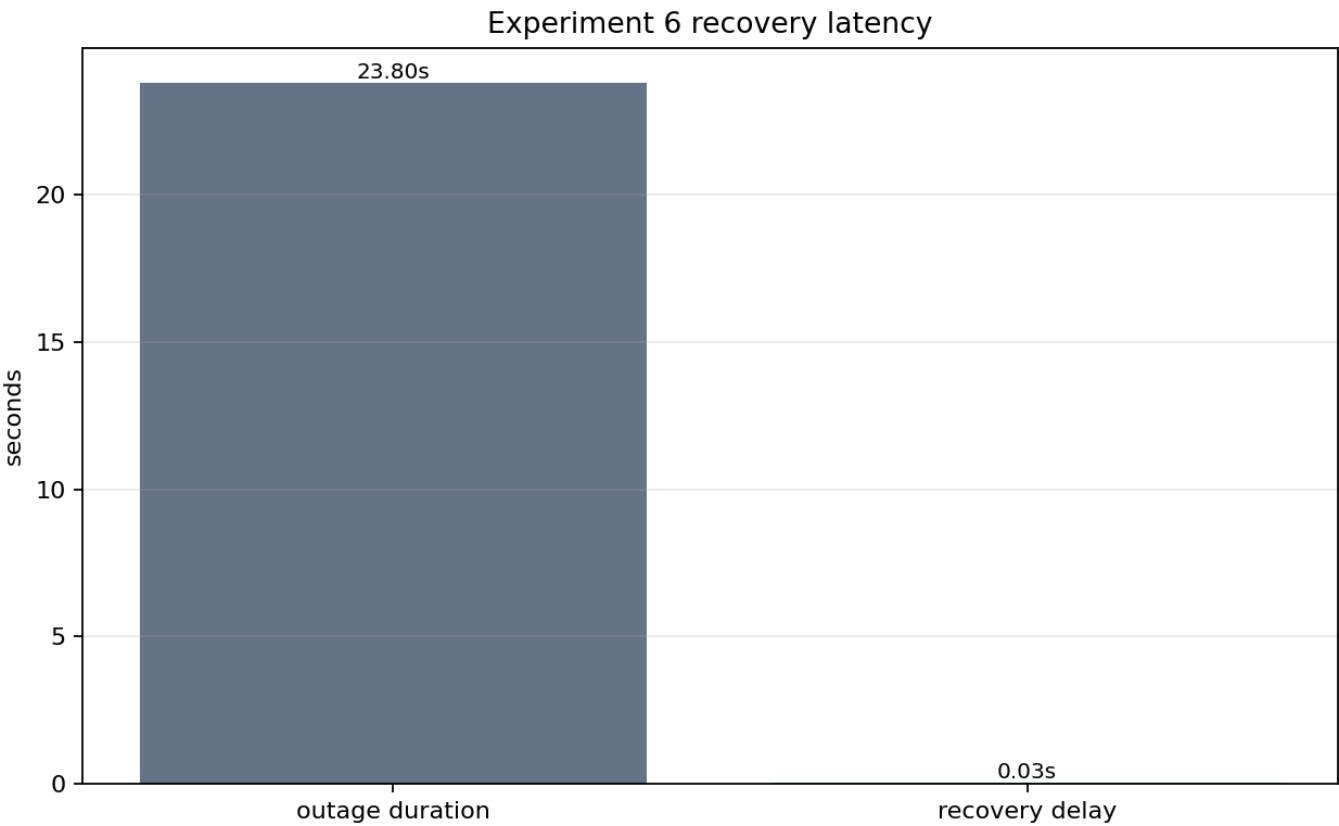
Fallback reason timeline



Source selection timeline

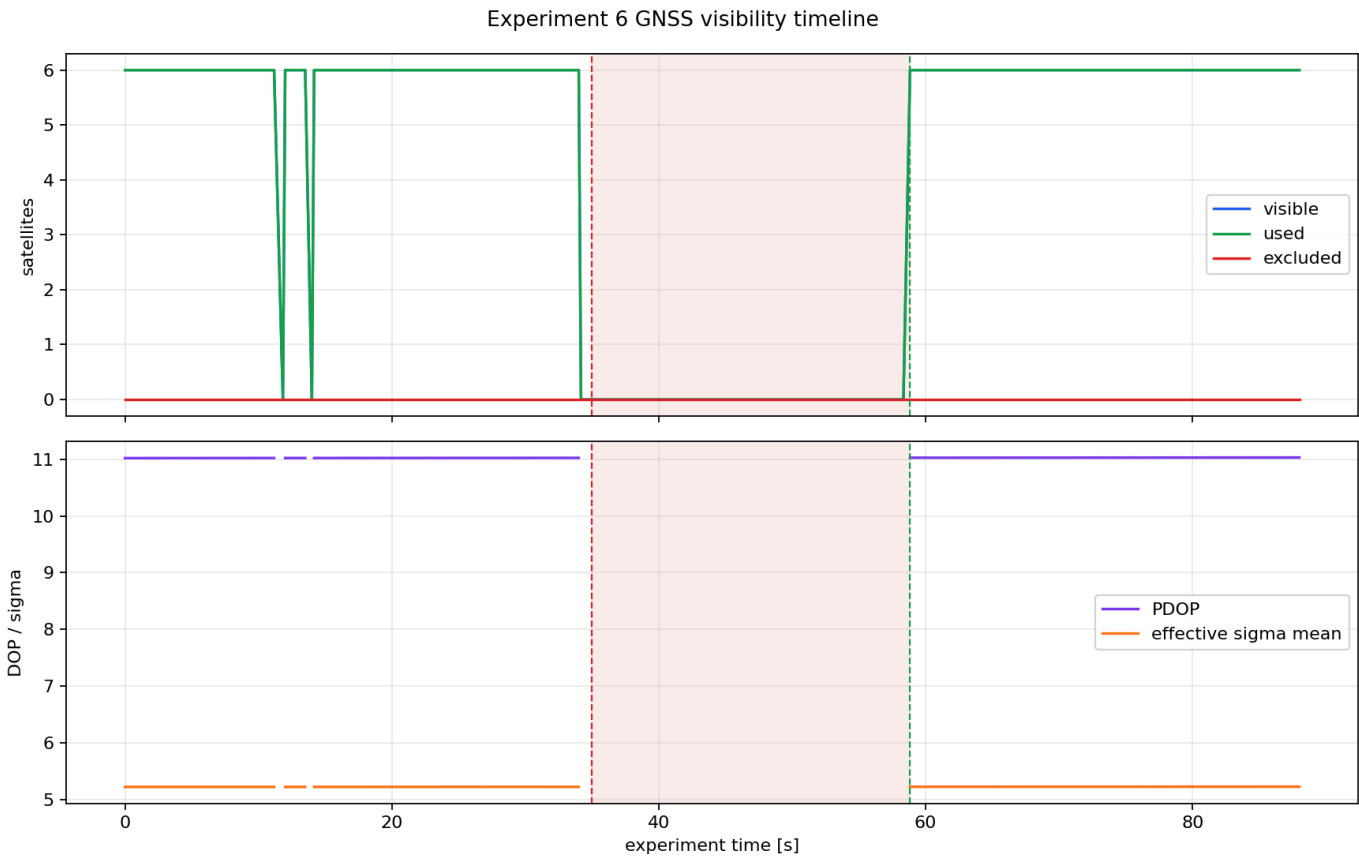


Recovery latency



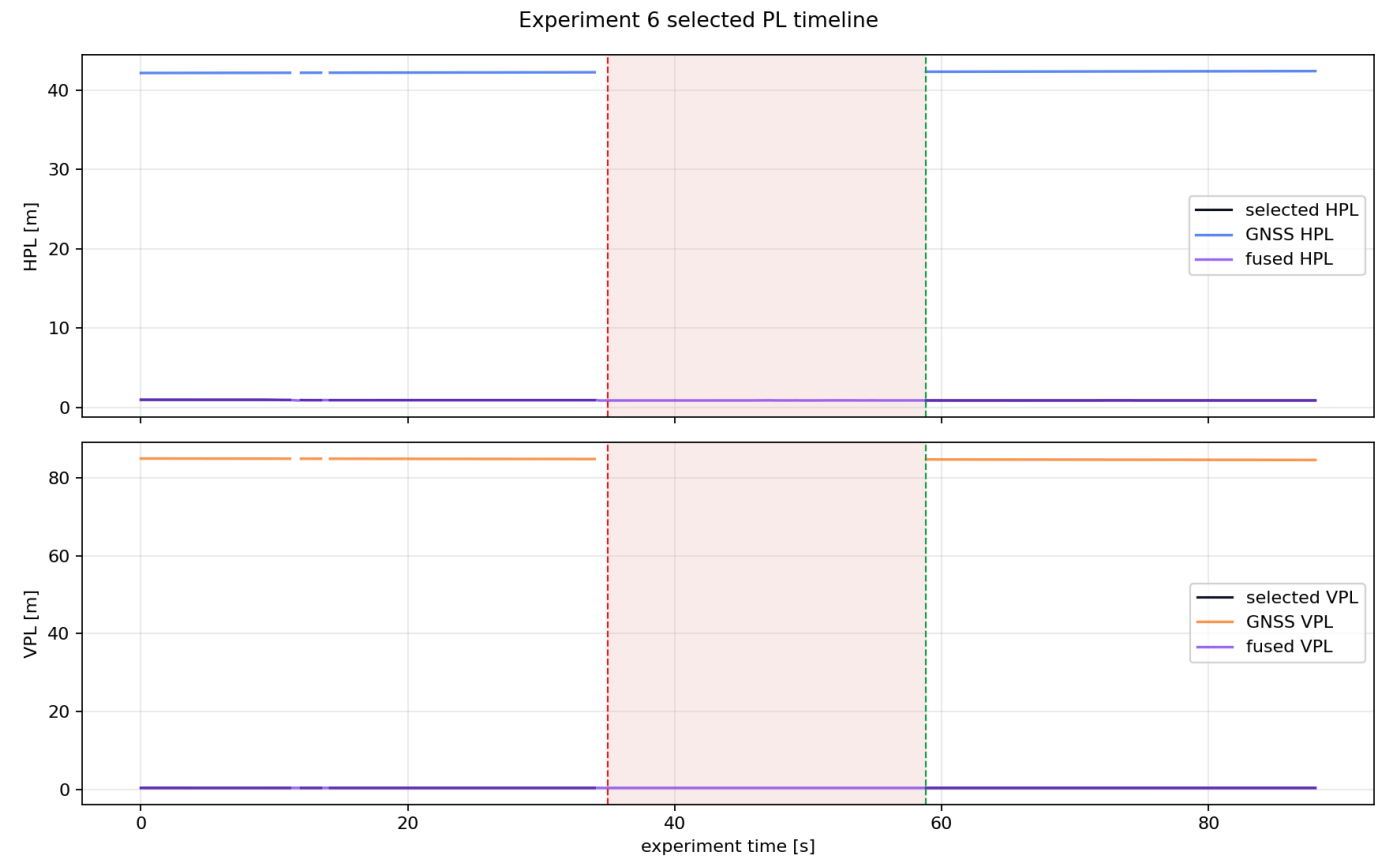
Analyzer: outage duration recovery delay predictor_invalid_start_s=35.17 s predictor_valid_recovery_s=58.83 s recovery_delay_s=0.03 s
selected output invalid/NONE valid

GNSS visible timeline



visible used excluded satellite count PDOP effective sigma mean outage window GNSS fault injection drop visible/used satellite count outage window 0 6 run4 GNSS outage

Selected PL timeline



selected HPL/VPL GNSS HPL/VPL fused HPL/VPL outage window outage selected PL selected output outage window invalid/NONE selected PL selected PL FUSION

20.6

Predictor system experiment 6 predictor_gnss_outage_lidar_recovery GNSS outage 326 predictor query outage window GNSS 0 gnss_valid_ratio=0.0 selected output invalid NONE gnss:no_gnss_epoch fallback reason GNSS selected output valid FUSION 0.03 s

GNSS-required conservative policy predictor GNSS outage Fusion selected result CSV SON latency profiling metadata figure IAP log export/profiling/metadata/ figures/

21. System Predictor no-source negative

system experiment preset

```
predictor_no_source_negative
```

```
src/iap/results/predictor_validation/predictor_system_predictor_no_source_negative_run1/
```

21.1

predictor advisory source selected PL current integrity HPL/VPL advisory result negative case GNSS LiDAR primitives Predictor output mode fusion

- gnss_valid=false
- lidar_valid=false
- fused_valid=false
- selected_valid=false
- selected_fallback=true
- selected source NONE
- fallback reason GNSS/LiDAR source
- invalid selected rows selected_hpl/selected_vpl/selected_pl finite
- selected PL current integrity HPL/VPL

21.2 配置

在 run 目录下 launch 配置

配置	值
experiment	predictor_no_source_negative
araim_experiment	fallback_only
predictor_output_mode	fusion
predictor_enable_debug_log	true
validator_require_debug_logs	true
validator_required_selected_source	至少一个 valid source
validator_require_selected_valid	false
validator_require_gnss_valid	false
validator_require_lidar_valid	false
validator_require_fusion_valid	false
probe_use_lidar_primitives	false
probe_require_gnss_for_selected_output	false
probe_enable_current_prior	true
start_planner	false
predictor probe query set	current_pose

在 run 目录下

```
RUN_ROOT=/home/dev/ws_iap/src/iap/results/predictor_validation/predictor_system_predictor_no_source_negative_run1

ros2 launch iap test_predictor.launch.py \
  experiment:=predictor_no_source_negative \
  start_rviz:=true \
  record_bag:=false \
  predictor_enable_debug_log:=true \
  validator_require_debug_logs:=true \
  run_duration_s:=90 \
  validation_duration_s:=85 \
  predictor_export_dir:=$RUN_ROOT/export

python3 src/iap/scripts/dev_predictor/analyze_predictor_system_e7.py \
  --run-dir $RUN_ROOT \
  --fail-on-threshold
```

在 test_predictor_validator.py 中 summary 运行结果 passed=false，predictor_lidar_primitives_debug.csv 中 required debug log failure 为 negative case，probe_use_lidar_primitives=false，LiDAR primitive debug 中 E7 的 analyzer 中 predictor_e7_analysis_summary.json 中 invalid_output_audit.json 中

21.3 结果

7 analyzer 结果

配置	期望	实际
query_count	> 0	530
selected_valid_ratio	== 0	0.0%
selected_fallback_ratio	== 100%	100%
gnss_valid_ratio	== 0	0.0%
lidar_valid_ratio	== 0	0.0%
fused_valid_ratio	== 0	0.0%
finite selected PL count	0	0
empty fallback reason count	0	0
copied current flag count	0	0
module_total_us p95	< 10000 us	4.39 us

predictor_e7_analysis_summary.json passed=truefailures=[]invalid_output_audit.json passed=trueinvalid selected rows
finite selected PLcurrent/advisory copy

21.4

Analyzer

```
passed: true
query_count: 530
selected_valid_ratio: 0.0
selected_fallback_ratio: 1.0
gnss_valid_ratio: 0.0
lidar_valid_ratio: 0.0
fused_valid_ratio: 0.0
module_valid_ratio: 0.0
module_fallback_ratio: 1.0
source_histogram: NONE=530
finite_selected_pl_count: 0
fallback_reason_empty_count: 0
copied_current_flag_count: 0
module_total_us p50/p95/max: 3.06 / 4.39 / 7.89 us
```

CSV 530 predictor query 88.17 s query selected invalid + selected fallbackselected source NONEGNSS source
no_gnss_epochLiDAR source missing_lidar_normalsfallback reason query

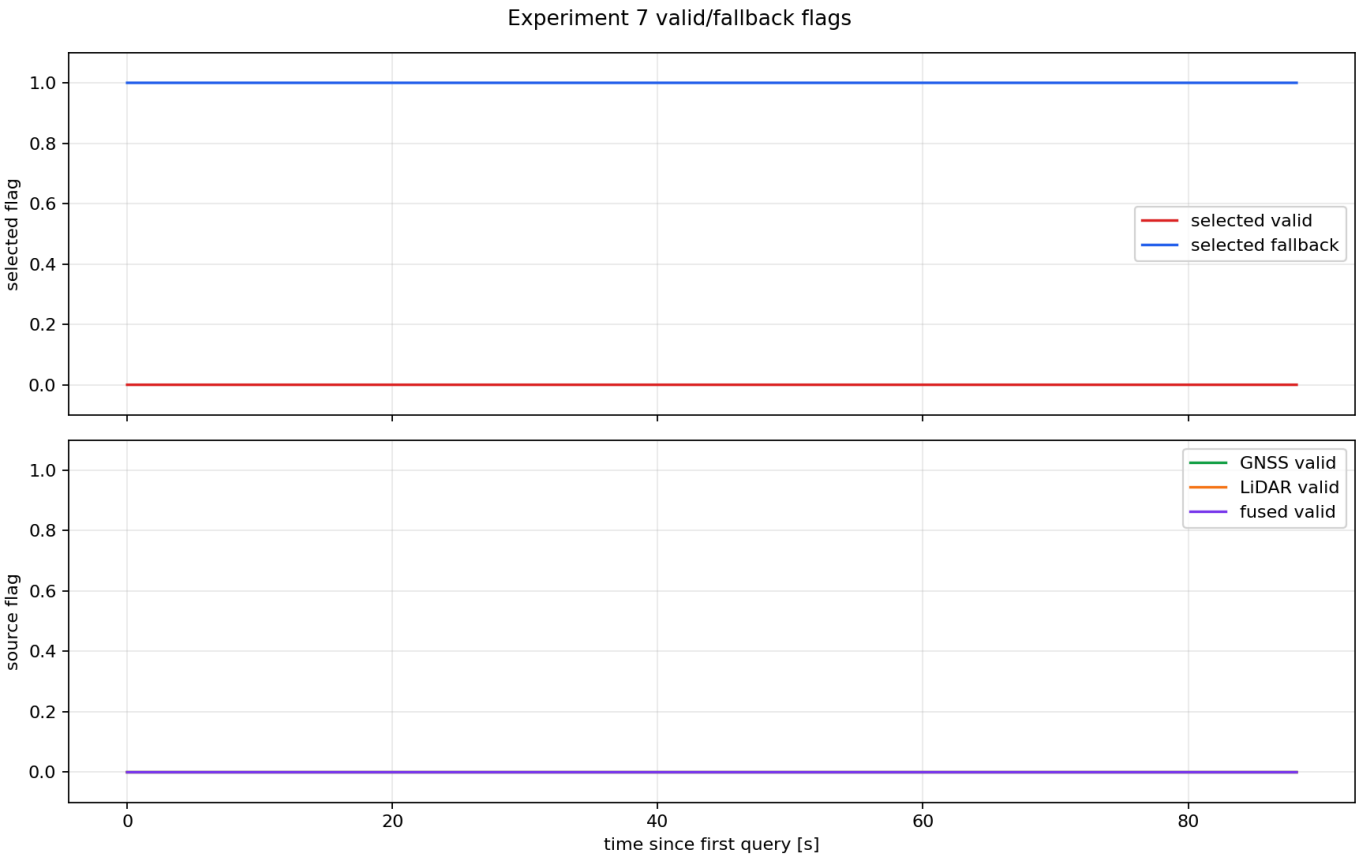
```
gnss:no_gnss_epoch;lidar:missing_lidar_normals
```

Current integrity run current HPL/VPL 0.150 m selected HPL/VPL/PL NaNcopied_current_flag_count=0 predictor
current integrity finite PL advisory selected output

query count	530
selected source	NONE 530/530
selected valid count	0
selected fallback count	530
GNSS valid count	0
LiDAR valid count	0
Fusion valid count	0
finite selected PL count	0
copied current flag count	0
empty fallback reason count	0
current HPL/VPL median	0.150 m / 0.150 m
module latency p95	4.39 us

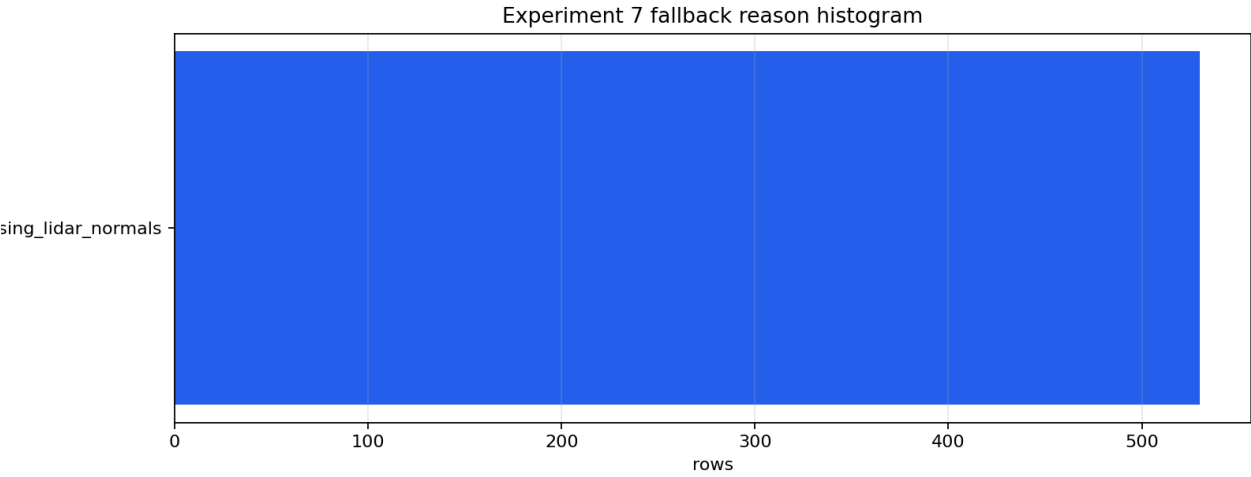
21.5

Valid/fallback flags



selected_valid selected_fallbackgnss_validlidar_valid fused_validno-source negative selected output invalid fallbackselected_valid=0selected_fallback=1GNSS/LiDAR/Fusion valid flag 0

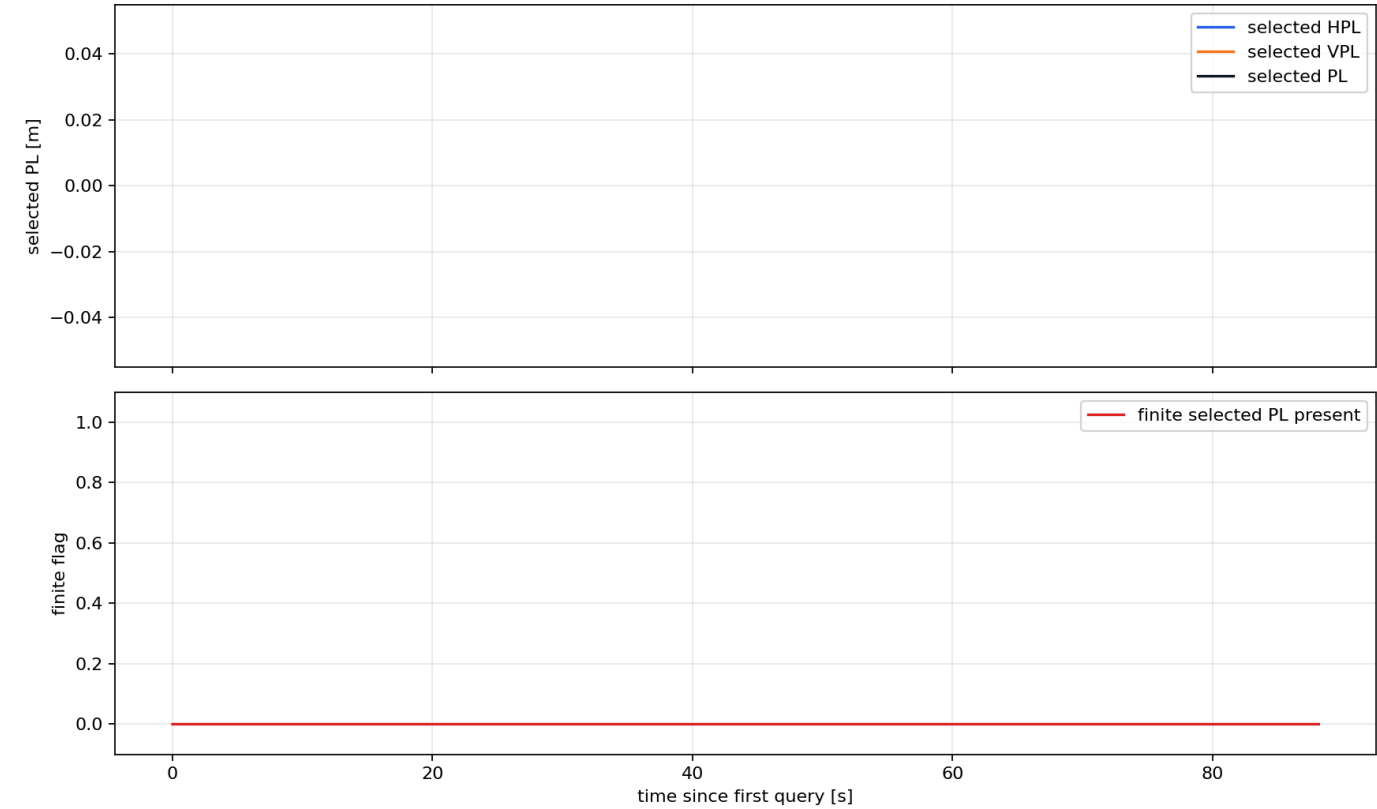
Fallback reason histogram



fallback reason categorynegative case query 530 query reason gnss:no_gnss_epochlidar:missing_lidar_normalsGNSS source LiDAR source

Selected PL timeline

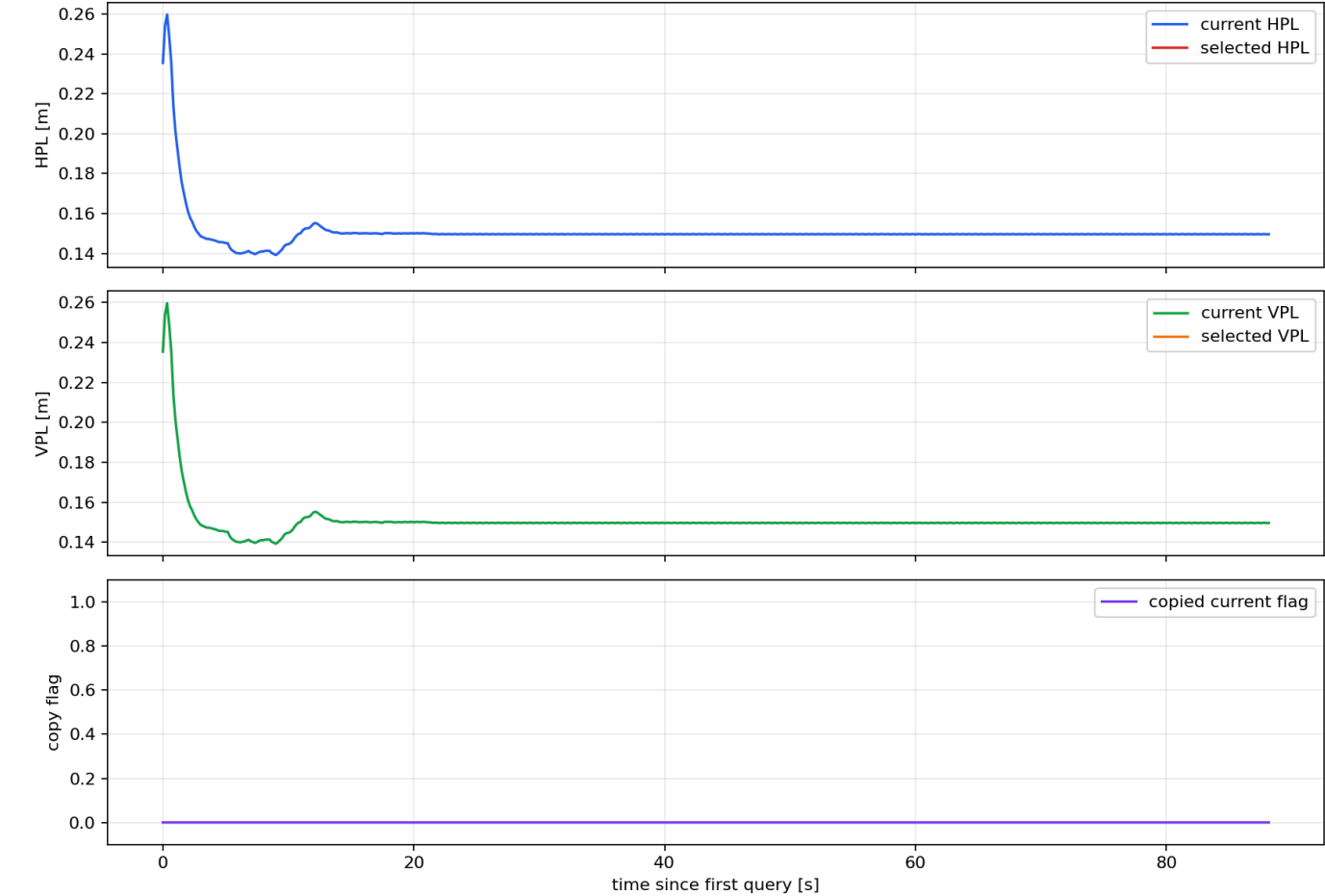
Experiment 7 selected PL timeline



selected HPL/VPL/PL finite selected PL flag invalid selected rows PL selected HPL/VPL/PL NaN finite selected PL flag 0 predictor finite PL

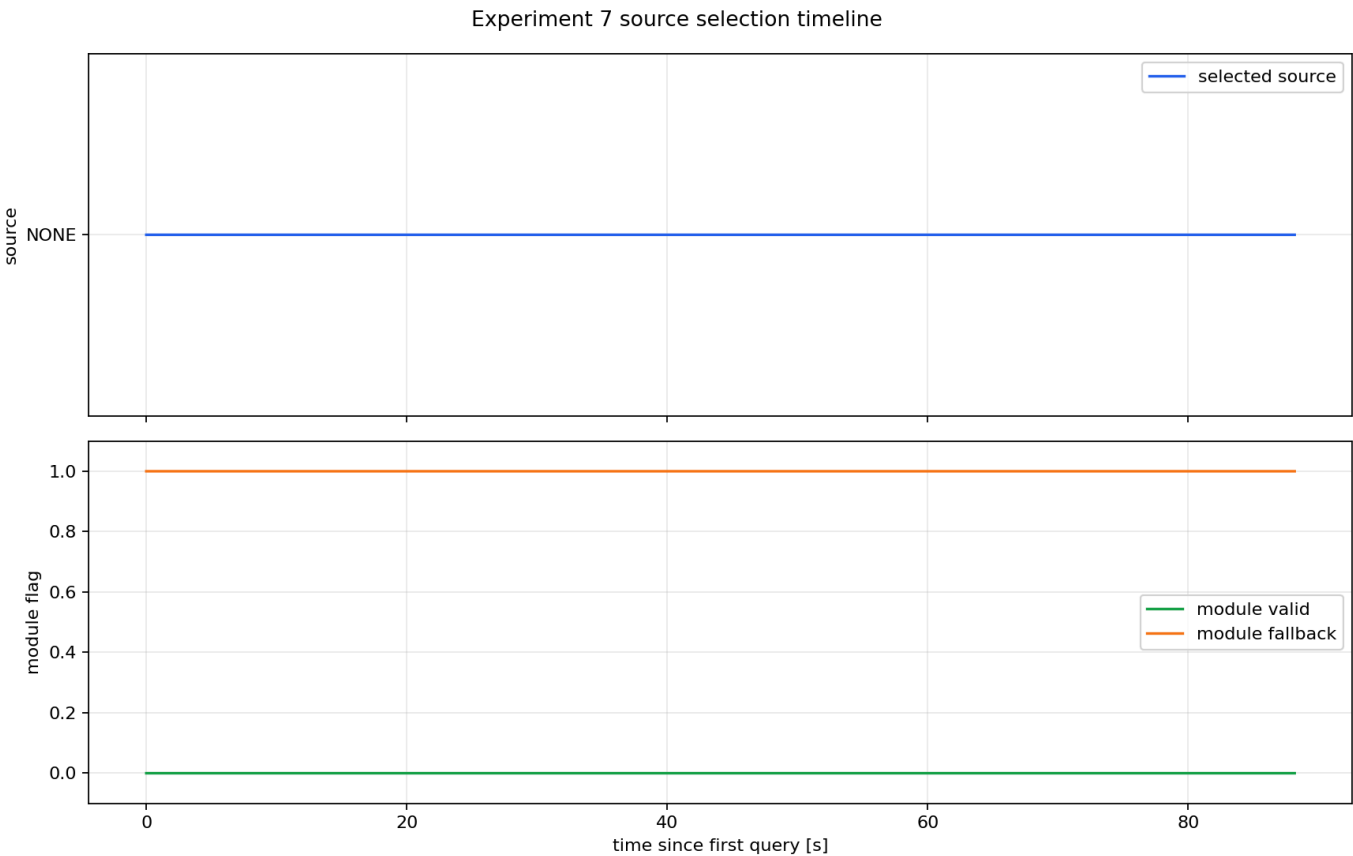
Current vs selected PL

Experiment 7 current vs selected PL



current HPL/VPL selected HPL/VPL copied current flag current integrity finite PL selected advisory current PL current HPL/VPL selected HPL/VPL NaN copy flag 0

Source selection timeline



selected source category module valid/fallback flag no-source selected source NONE module fallback selected source NONE module valid 0 module fallback 1

21.6

Predictor system experiment 7 predictor_no_source_negative GNSS disabled LiDAR primitives disabled Fusion output mode 530 predictor query query selected invalid selected source NONE fallback ratio 100% fallback reason gnss:no_gnss_epoch;lidar:missing_lidar_normals

advisory source negative case predictor finite selected PL current integrity finite HPL/VPL selected advisory CSV CSVJSON latency profiling metadata figure IAP log export/profiling/metadata/ figures/

22. System Predictor current/advisory separation

system experiment

```
predictor_current_advisory_separation
```

```
src/iap/results/predictor_validation/predictor_system_predictor_current_advisory_separation_run1
```

22.1

predictor GNSS open-sky GNSS-only selected output current ARAIM advisory predictor probe /iap/integrity epoch current variant

Variant	Current HPL/VPL	Current state	selected output
normal	current integrity	0	GNSS advisory
current_high	1000 / 1000 m	0	GNSS advisory high current
current_unsafe	500 / 600 m	2	GNSS advisory unsafe current

planner selected HPL/VPL GNSS advisory HPL/VPL artificial current variant `copied_current_flag=0`

22.2

run

```
RUN_ROOT=/home/dev/ws_iap/src/iap/results/predictor_validation/predictor_system_predictor_current_advisory_separation_run1

ros2 launch iap test_predictor.launch.py \
  experiment:=predictor_current_advisory_separation \
  start_rviz:=false \
  record_bag:=false \
  predictor_enable_debug_log:=true \
  validator_require_debug_logs:=true \
  run_duration_s:=90 \
  validation_duration_s:=85 \
  predictor_export_dir:=$RUN_ROOT/export

python3 src/iap/scripts/dev_predictor/analyze_predictor_system_e8.py \
  --run-dir $RUN_ROOT \
  --fail-on-threshold
```

run_manifest.json

araim_experiment	gnss_open_sky
predictor_output_mode	gnss_only
start_planner	false
planner_uses_predictor	false
probe_current_variant_set	e8_current_advisory_separation
probe_current_high_hpl_m / vpl_m	1000.0 / 1000.0
probe_current_unsafe_hpl_m / vpl_m	500.0 / 600.0
probe_current_unsafe_state	2

22.3

run IAP log

Category	
export/	test_predictor_query_probe.csvtest_predictor_query_probe_summary.json test_predictor_validation_summary.json
export/	current_advisory_separation.csvcurrent_advisory_variant_summary.jsonpredictor_e8_analysis_summary.json
export/	source_selection_debug.csvgnss_epoch_debug.csvgnss_visibility_by_query.csvfallback_reason_by_time.csv
export/	predictor_lidar_debug.csvpredictor_lidar_primitives_debug.csvpredictor_fusion_debug.csv downsampled_map.csv
profiling/	latency_debug.csv
metadata/	run_manifest.jsonpredictor_launch_config.jsonpredictor_probe_config.json
figures/	E8_current_vs_advisory_bar.pngE8_copied_current_flag_timeline.pngE8_selected_vs_gnss_timeline.png E8_variant_latency_distribution.png

Runtime validator summary `passed=true` 85 validation window 1494 query selected source GNSS E8 analyzer run CSV 1587 query

22.4

predictor_e8_analysis_summary.json `passed=true` failure reason

		run1
query count	> 0	1587
required variants	normal/current_high/current_unsafe	
variant count	529 / 529 / 529	
selected source GNSS ratio	> 0.95	1.000

Item	Value	run1 Value
selected valid ratio	> 0.95	1.000
GNSS valid ratio	> 0.95	1.000
selected GNSS HPL/VPL mismatch count	0	0
artificial current copied count	0	0
current_high median HPL/VPL	1000 / 1000 m	1000.0 / 1000.0 m
current_unsafe state != 2 count	0	0
non-finite valid fields	0	0
module latency p95	< 2000 us	33.77 us

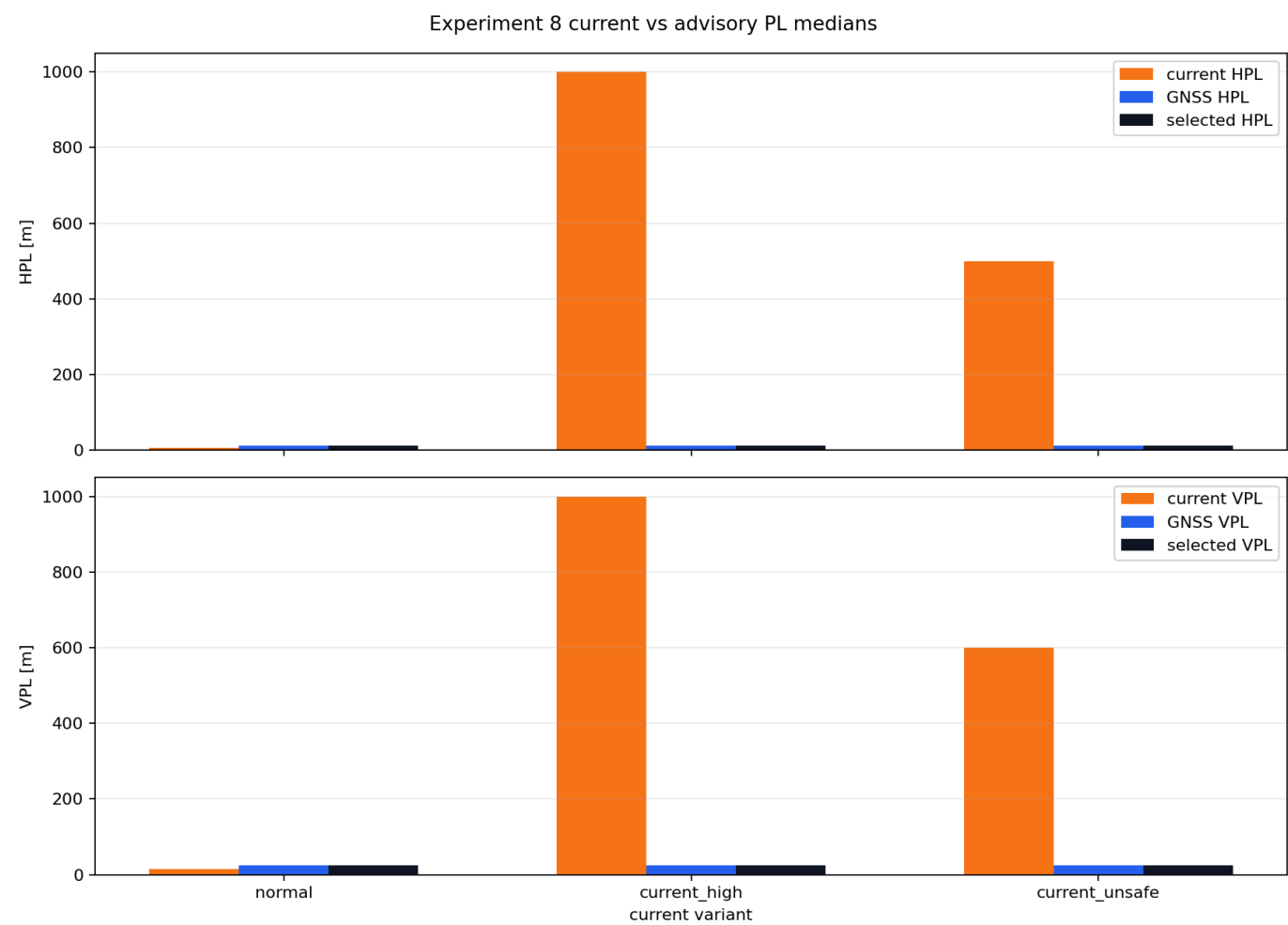
variant

Variant	Median current HPL/VPL	Median GNSS HPL/VPL	Median selected HPL/VPL
normal	5.0496 / 14.4060 m	10.7380 / 25.1486 m	10.7380 / 25.1486 m
current_high	1000.0 / 1000.0 m	10.7380 / 25.1486 m	10.7380 / 25.1486 m
current_unsafe	500.0 / 600.0 m	10.7380 / 25.1486 m	10.7380 / 25.1486 m

selected HPL/VPL GNSS advisory current ARAIM input current 1000/1000 m unsafe 500/600 m, state=2selected output GNSS-only predictor

22.5

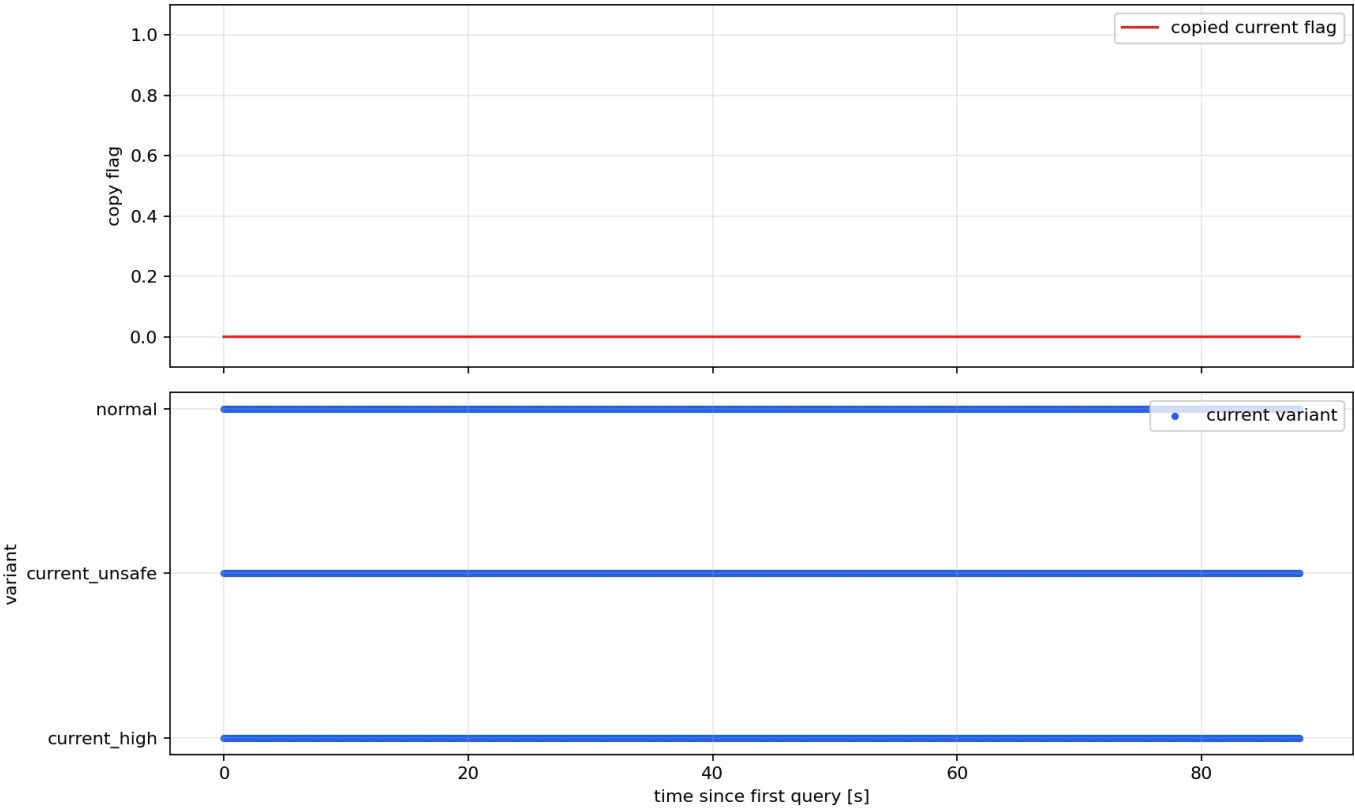
Current vs advisory bar



variant median current HPL/VPLGNSS HPL/VPL selected HPL/VPLcurrent unsafe selected output GNSS advisory current_high current_unsafe current PL GNSS/selected PL selected GNSS

Copied current flag timeline

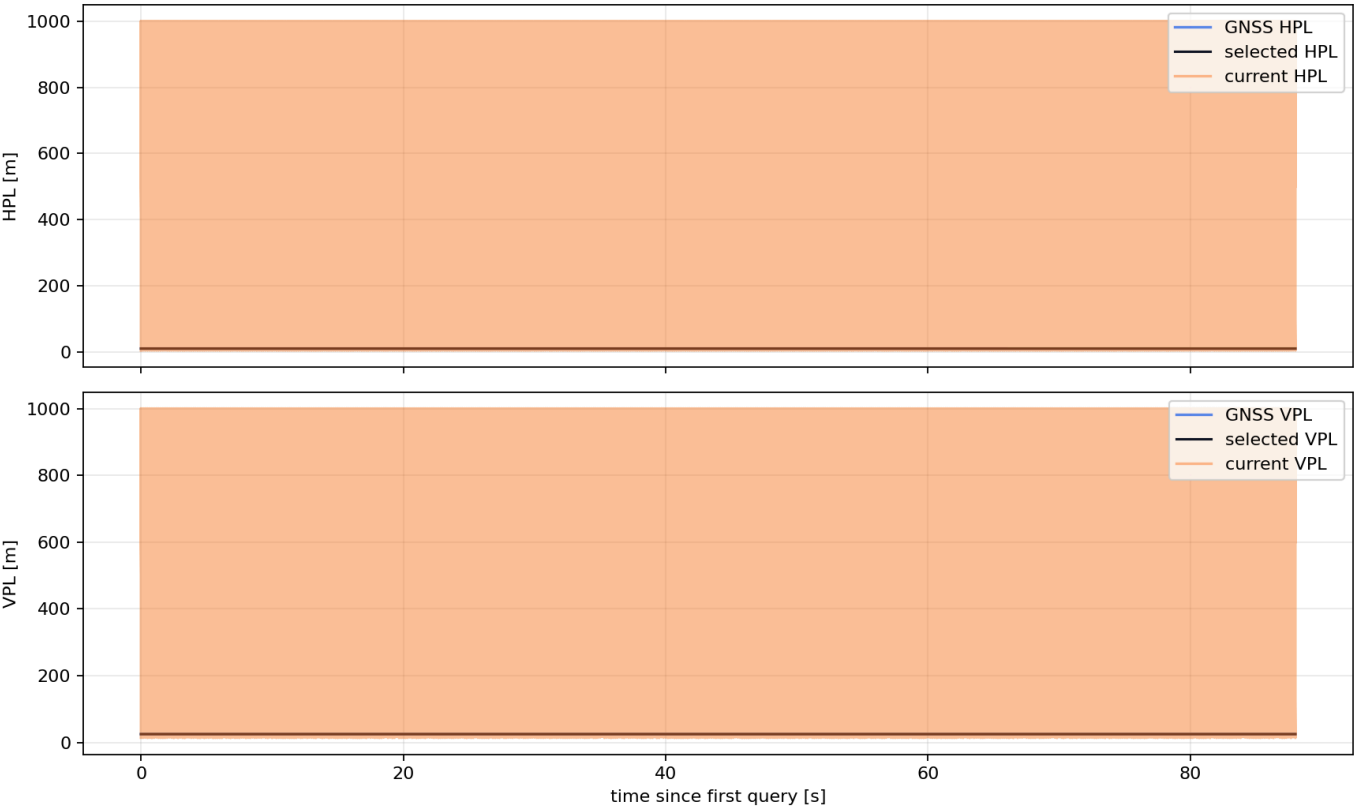
Experiment 8 copied current flag timeline



query copied_current_flag current variant current variant selected advisory variant copy flag
0 current-to-advisory

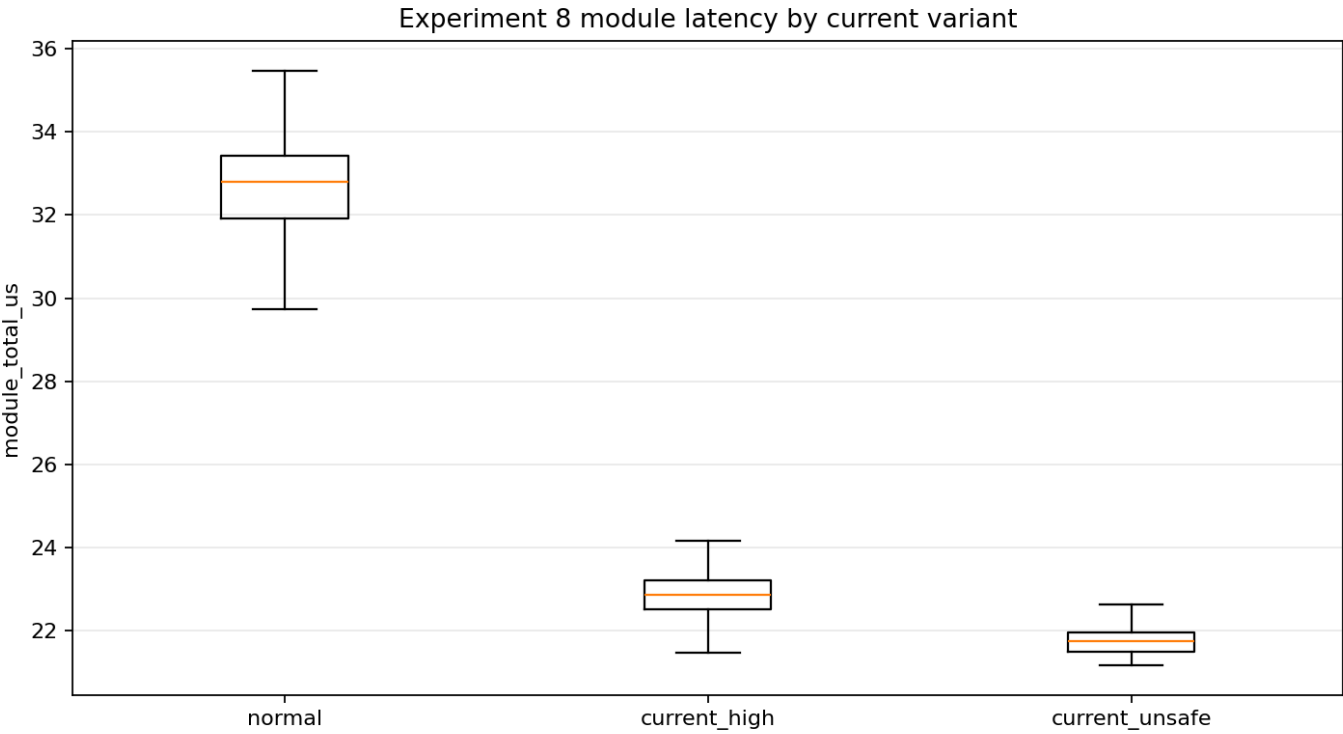
Selected vs GNSS timeline

Experiment 8 selected vs GNSS timeline



GNSS HPL/VPL selected HPL/VPL selected output run GNSS advisory selected HPL/VPL GNSS
HPL/VPL selected_minus_gnss_hpl/vpl 0

Variant latency distribution



current variant `normal` `module_total_us` current `normal` module latency p95 `33.77 us` `2000 us` variant `normal`

22.6

Predictor system experiment 8 `predictor_current_advisory_separation` GNSS open-sky GNSS-only selected output `1587` analyzer query `529` selected source `GNSS` selected valid ratio `100%` GNSS valid ratio `100%`

`current_high=1000/1000 m` `current_unsafe=500/600 m`, state=2 selected advisory selected HPL/VPL GNSS HPL/VPL `copied_current_flag_count=0` selected/GNSS mismatch count `0` predictor advisory output current integrity input system launch

23. System Predictor GNSS sigma degradation

system experiment

```
predictor_gnss_sigma_degradation
```

```
src/iap/results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1
```

23.1

predictor GNSS open-sky GNSS-only selected output GNSS pseudorange sigma inflation launch preset `1x` `2x` `4x` run E9 analyzer

- `effective_sigma_mean/max` sigma scale `1x` -> `2x` -> `4x`
- `gnss_hpl/gnss_vpl` `selected_hpl/selected_vpl` sigma scale
- `gnss_lambda_trace` sigma scale
- selected source `GNSS` GNSS valid ratio `100%` fallback ratio `0%`

23.2

sweep

Scale	Run dir	gnss_pr_noise_base	gnss_dop_noise_base	probe_gnss_sigma_scale
1x	scale_1x	1.0	0.03	1.0
2x	scale_2x	2.0	0.06	2.0

Scale	Run dir	gnss_pr_noise_base	gnss_dop_noise_base	probe_gnss_sigma_scale
4x	scale_4x	4.0	0.12	4.0

probe_gnss_sigma_scale predictor advisory sigma sweep GNSS epoch pr_sigma visibility predictor effective canopy
sigma gnss_pr_noise_base gnss_dop_noise_base simulator/current ARAIM

```
SWEEP_ROOT=/home/dev/ws_iap/src/iap/results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_ru
n1
RUN_1X=$SWEEP_ROOT/scale_1x
RUN_2X=$SWEEP_ROOT/scale_2x
RUN_4X=$SWEEP_ROOT/scale_4x

ros2 launch iap test_predictor.launch.py \
  experiment:=predictor_gnss_sigma_degradation \
  start_rviz:=false \
  record_bag:=false \
  predictor_enable_debug_log:=true \
  validator_require_debug_logs:=true \
  run_duration_s:=90 \
  validation_duration_s:=85 \
  gnss_pr_noise_base:=1.0 \
  gnss_dop_noise_base:=0.03 \
  probe_gnss_sigma_scale:=1.0 \
  predictor_export_dir:=$RUN_1X/export

ros2 launch iap test_predictor.launch.py \
  experiment:=predictor_gnss_sigma_degradation \
  start_rviz:=false \
  record_bag:=false \
  predictor_enable_debug_log:=true \
  validator_require_debug_logs:=true \
  run_duration_s:=90 \
  validation_duration_s:=85 \
  gnss_pr_noise_base:=2.0 \
  gnss_dop_noise_base:=0.06 \
  probe_gnss_sigma_scale:=2.0 \
  predictor_export_dir:=$RUN_2X/export

ros2 launch iap test_predictor.launch.py \
  experiment:=predictor_gnss_sigma_degradation \
  start_rviz:=false \
  record_bag:=false \
  predictor_enable_debug_log:=true \
  validator_require_debug_logs:=true \
  run_duration_s:=90 \
  validation_duration_s:=85 \
  gnss_pr_noise_base:=4.0 \
  gnss_dop_noise_base:=0.12 \
  probe_gnss_sigma_scale:=4.0 \
  predictor_export_dir:=$RUN_4X/export

python3 src/iap/scripts/dev_predictor/analyze_predictor_system_e9.py \
  --scale-run 1:$RUN_1X \
  --scale-run 2:$RUN_2X \
  --scale-run 4:$RUN_4X \
  --output-run-dir $SWEEP_ROOT \
  --fail-on-threshold
```

23.3

scale run IAP log

Category	
scale_*/export/	test_predictor_query_probe.csvtest_predictor_query_probe_summary.json test_predictor_validation_summary.json
scale_*/export/	source_selection_debug.csvgnss_epoch_debug.csvgnss_visibility_by_query.csv fallback_reason_by_time.csv
scale_*/export/	predictor_lidar_debug.csvpredictor_lidar_primitives_debug.csvpredictor_fusion_debug.csv downsampled_map.csv
scale_*/profiling/	latency_debug.csv
scale_*/metadata/	run_manifest.jsonpredictor_launch_config.jsonpredictor_probe_config.json

E9 analyzer sweep root

Category	
export/	gnss_sigma_sweep_system.csvpredictor_e9_analysis_summary.json
figures/	E9_sigma_vs_pl.pngE9_sigma_vs_lambda_trace.pngE9_effective_sigma_by_scale.pngE9_latency_by_scale.png

23.4

predictor_e9_analysis_summary.json passed=truefailures=[] scale selected sourceGNSS validfallback latency

Scale	Query count	GNSS selected ratio	GNSS valid ratio	Fallback ratio	Latency p95
1x	530	1.000	1.000	0.000	34.25 us
2x	530	1.000	1.000	0.000	34.16 us
4x	530	1.000	1.000	0.000	34.35 us

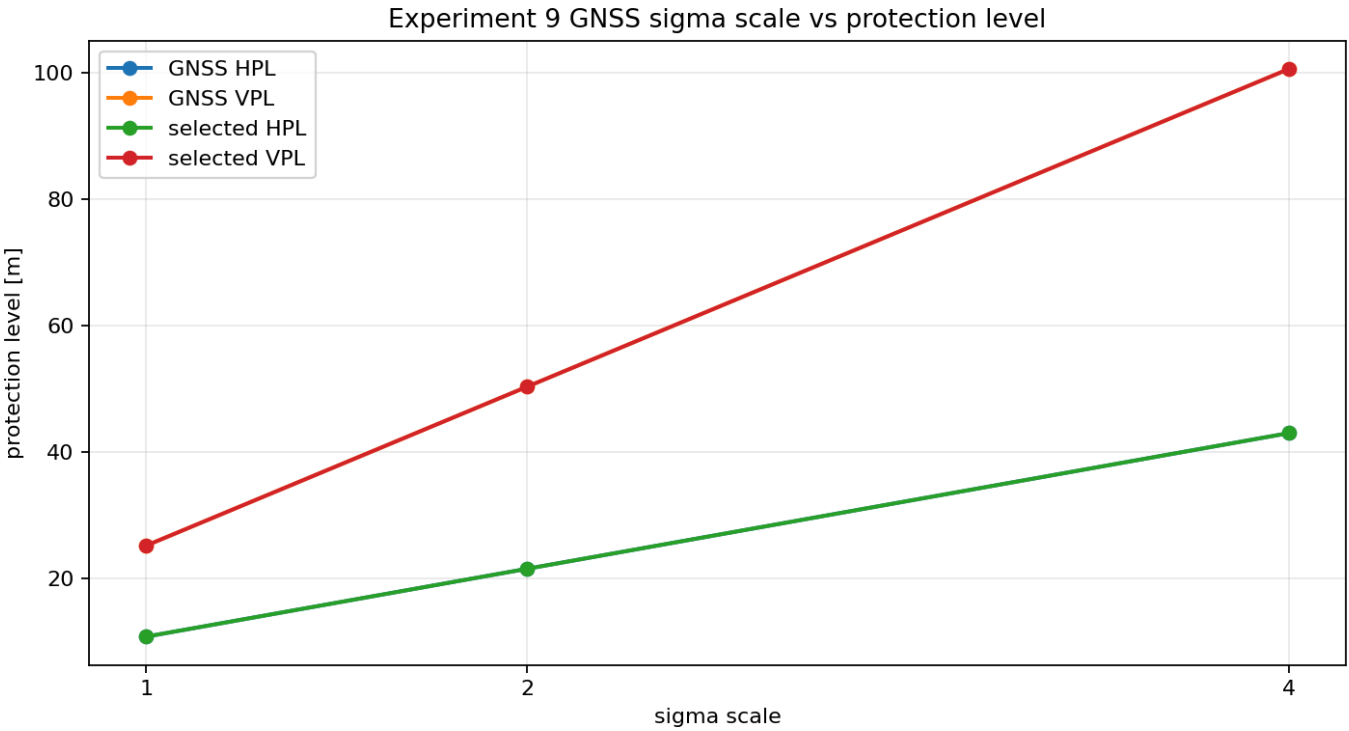
	1x	2x	4x	
median effective sigma mean	5.2058 m	10.4115 m	20.8231 m	
median effective sigma max	5.5790 m	11.1578 m	22.3157 m	
median GNSS HPL	10.7381 m	21.4761 m	42.9523 m	
median GNSS VPL	25.1486 m	50.2971 m	100.5943 m	
median selected HPL	10.7381 m	21.4761 m	42.9523 m	
median selected VPL	25.1486 m	50.2971 m	100.5943 m	
median GNSS lambda trace	0.8544	0.2136	0.0534	

Analyzer monotonic checks true

```
effective_sigma_mean_increasing: true
effective_sigma_max_increasing: true
gnss_hpl_increasing: true
gnss_vpl_increasing: true
selected_hpl_increasing: true
selected_vpl_increasing: true
gnss_lambda_trace_decreasing: true
```

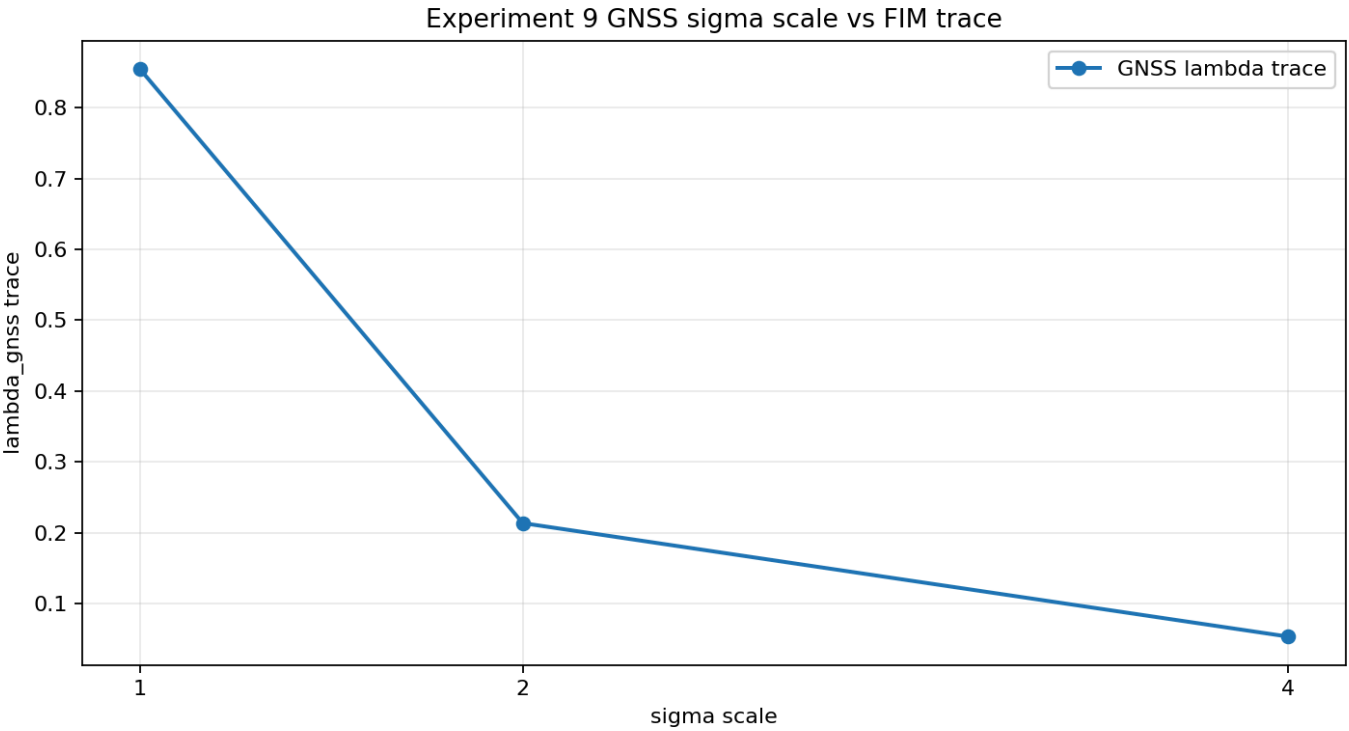
23.5

Sigma vs PL



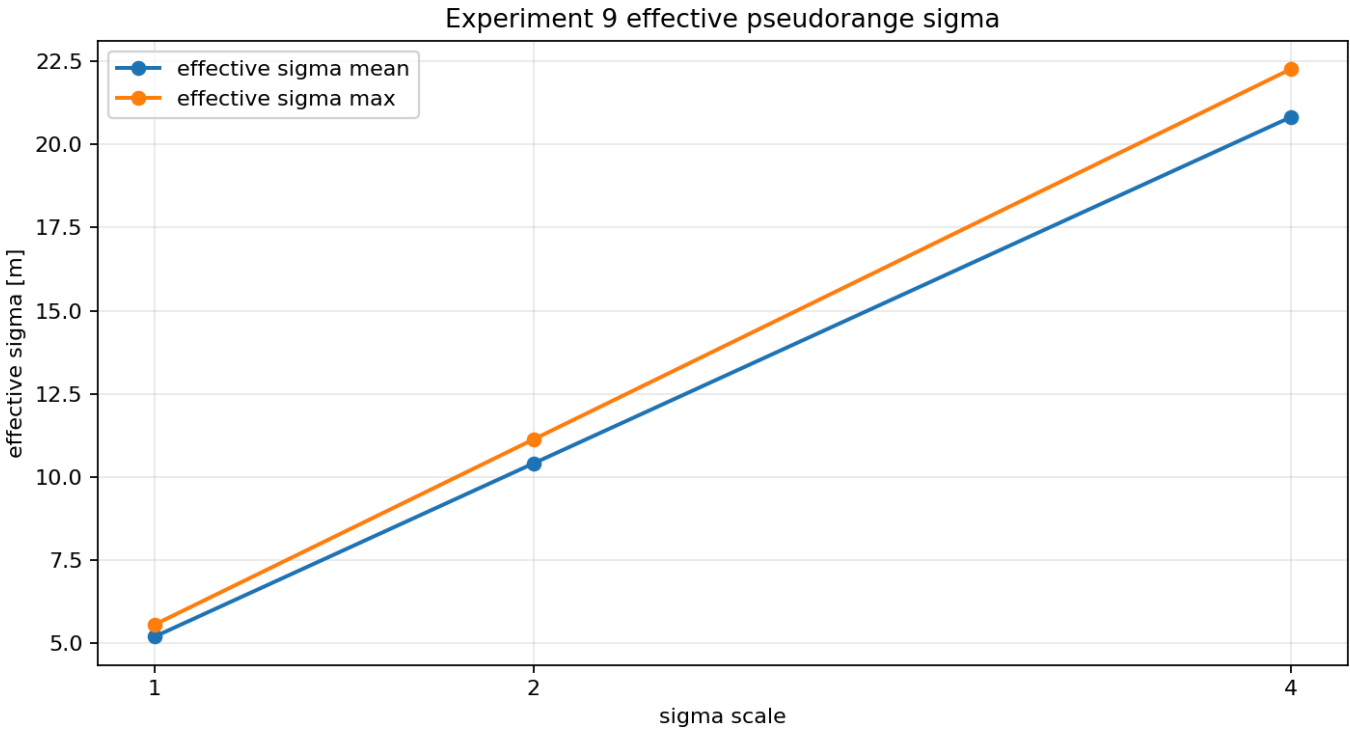
sigma scale protection level GNSS HPL GNSS VPL selected HPL selected VPL measurement sigma inflation
GNSS advisory PL selected output PL 1x -> 2x -> 4x selected HPL/VPL GNSS HPL/VPL GNSS-only selected output

Sigma vs lambda trace

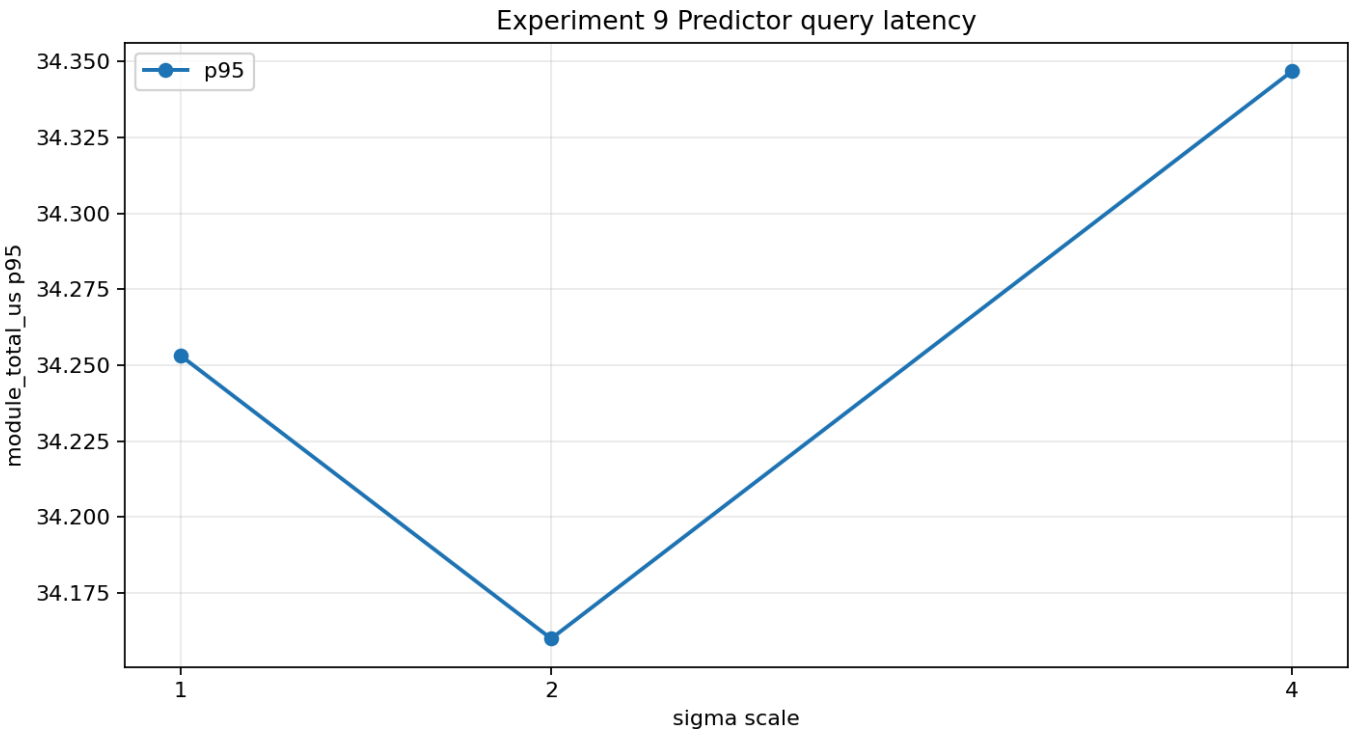


sigma scale GNSS position-only FIM trace sigma GNSS information matrix lambda_trace 0.8544 0.2136 0.0534

Effective sigma by scale



Experiment 9 scale 1 2 4 median effective_sigma_mean effective_sigma_max analyzer launch predictor GNSS geometry/FIM effective sigma mean/max 1x/2x/4x sigma sweep predictor advisory Latency by scale



Experiment 9 scale 1 2 4 module_total_us p95 sigma inflation scale p95 34 us 2000 us E9 sigma sweep runtime latency

23.6

Predictor system experiment 9 predictor_gnss_sigma_degradation GNSS open-sky GNSS-only selected output 1x/2x/4x run 530 predictor query selected source GNSS GNSS valid ratio 100% fallback ratio 0% predictor advisory pseudorange sigma inflation effective_sigma_mean/max GNSS HPL/VPL selected HPL/VPL scale GNSS FIM lambda_trace scale GNSS predictor PL FIM diagnostics measurement noise inflation sigma degradation fallback source per-scale CSV summary JSON metadata latency profiling sweep-level CSV/PNG IAP log

24. System Predictor corridor LiDAR degeneracy

system experiment

```
predictor_corridor_lidar_degeneracy
```

```
src/iap/results/predictor_validation/predictor_system_predictor_corridor_lidar_degeneracy_run1
```

24.1

predictor corridor LiDAR geometry LiDAR degeneracy diagnostics X

- along_corridor_information cross_corridor_information
- weak_axis_ratio = along / cross 0.25
- lidar_lambda_condition degeneracy_score
- LiDAR analyzer fusion source gate/fallback reason
- finite latency system analyzer

24.2

run lidar_corridor_degenerate ARAIM preset lidar_only predictor selected output

experiment	predictor_corridor_lidar_degeneracy
araim_experiment	lidar_corridor_degenerate
predictor_output_mode	lidar_only
start_planner	false
probe_use_lidar_primitives	true
probe_enable_map_snapshot	true
probe_lidar_map_max_points	2500
lidar_fim_condition_max	1000000.0
lidar_fim_radius_m	8.0

```
RUN_ROOT=/home/dev/ws_iap/src/iap/results/predictor_validation/predictor_system_predictor_corridor_lidar_degeneracy_run1

ros2 launch iap test_predictor.launch.py \
  experiment:=predictor_corridor_lidar_degeneracy \
  start_rviz:=false \
  record_bag:=false \
  predictor_enable_debug_log:=true \
  validator_require_debug_logs:=true \
  run_duration_s:=90 \
  validation_duration_s:=85 \
  predictor_export_dir:=$RUN_ROOT/export

python3 src/iap/scripts/dev_predictor/analyze_predictor_system_e10.py \
  --run-dir $RUN_ROOT \
  --fail-on-threshold
```

24.3

run IAP log

Category	
export/	test_predictor_query_probe.csvtest_predictor_query_probe_summary.json test_predictor_validation_summary.json
export/	source_selection_debug.csvgnss_epoch_debug.csvgnss_visibility_by_query.csvfallback_reason_by_time.csv

Category	
export/	predictor_lidar_debug.csv predictor_lidar_primitives_debug.csv predictor_fusion_debug.csv downsampled_map.csv
export/	lidar_corridor_degeneracy_system.csv predictor_e10_analysis_summary.json
profiling/	latency_debug.csv
metadata/	run_manifest.json predictor_launch_config.json predictor_probe_config.json
figures/	E10_corridor_map_topdown.png E10_lidar_condition_timeline.png E10_along_vs_cross_corridor_information.png E10_lidar_gate_timeline.png E10_lidar_eigenvalues_timeline.png

24.4

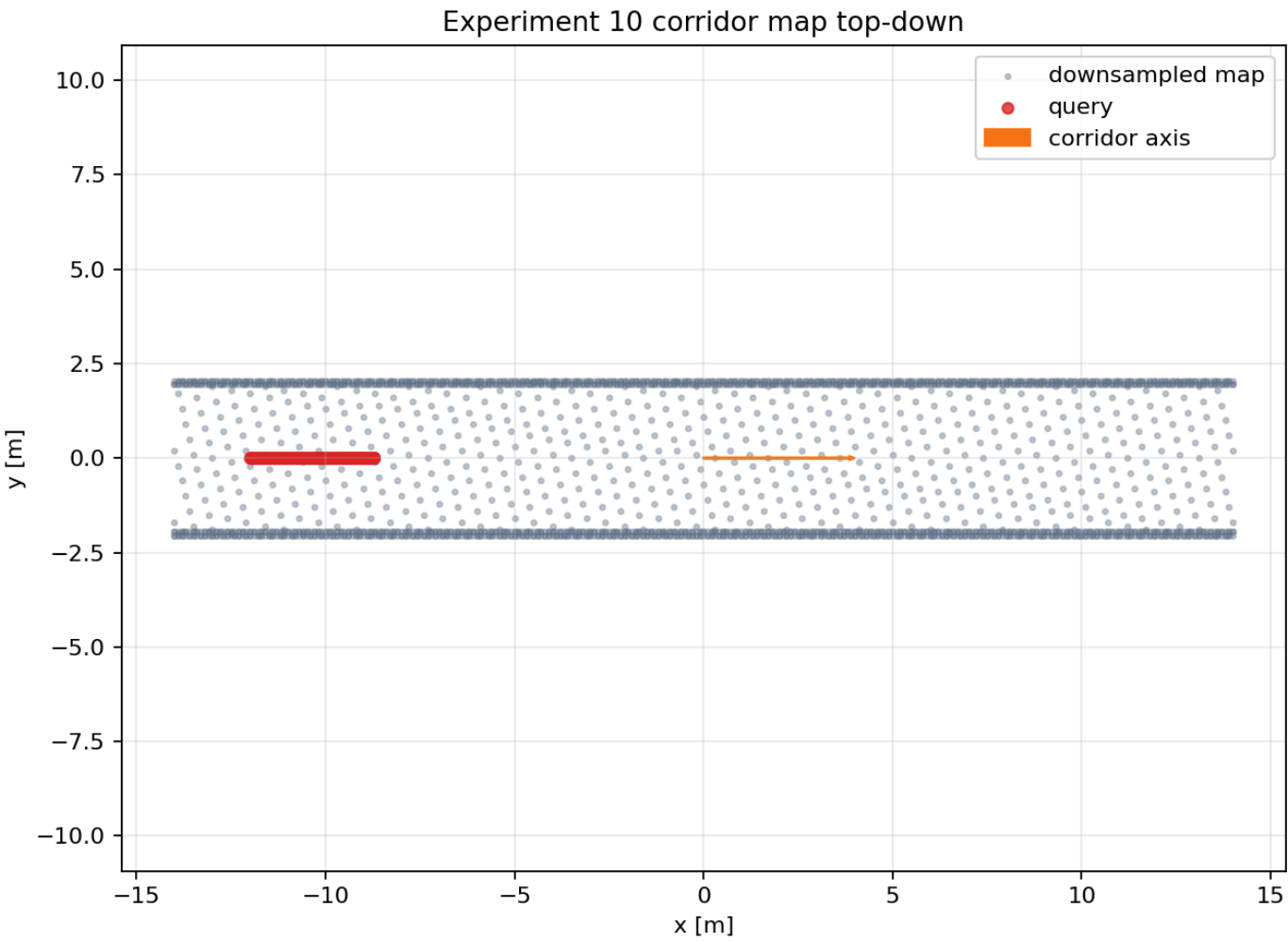
`predictor_e10_analysis_summary.json` `passed=true``failures=[]`

Query count	529	> 0
LiDAR debug rows	529	> 0
Fusion debug rows	529	> 0
Downsampled map rows	2500	> 0
median along corridor information	1.0857	cross direction
median cross corridor information	1148.4353	along direction
median weak axis ratio	0.000945	< 0.25
median LiDAR condition	1064.9819	
median degeneracy score	1057.7632	>= 1000
median LiDAR primitives	702	primitive
LiDAR allowed for fusion false ratio	1.000	> 0.80
Gate reason nonempty ratio	1.000	> 0.80
nonfinite diagnostic fields	0	0
module total latency p95	19.28 us	< 10000 us

CSV `lidar_corridor_degeneracy_system.csv` query corridor axisLiDAR primitiveFIM conditionlidar_alpha`lidar_tdop` along/cross informationweak-axis ratio`degeneracy score` gate reason analyzer query `lidar_allowed_for_fusion=false` gate reason

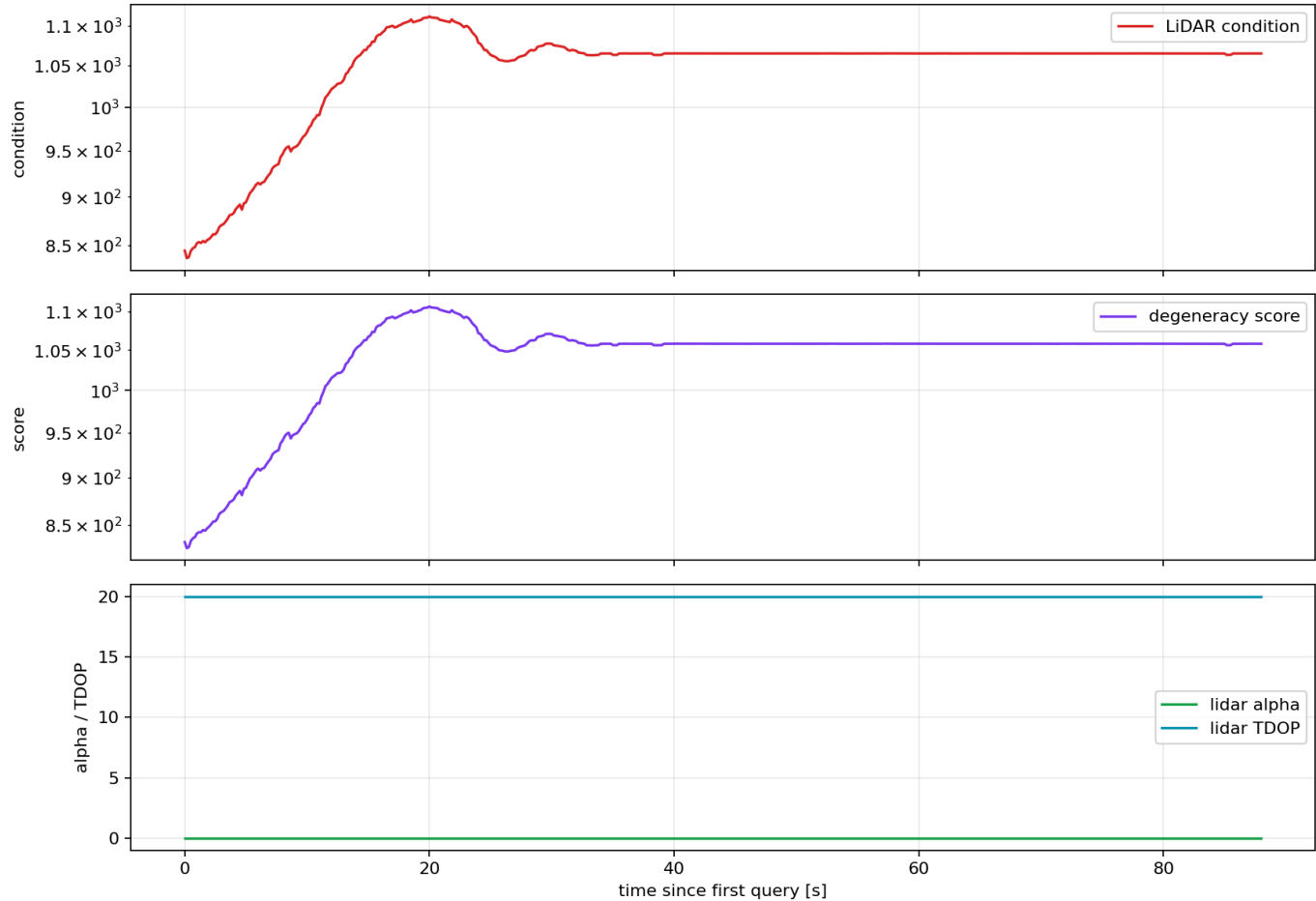
24.5

Corridor map top-down



Corridor geometry analyzer
downsampled map query corridor axis
X
corridor geometry
feature-rich
LiDAR condition timeline

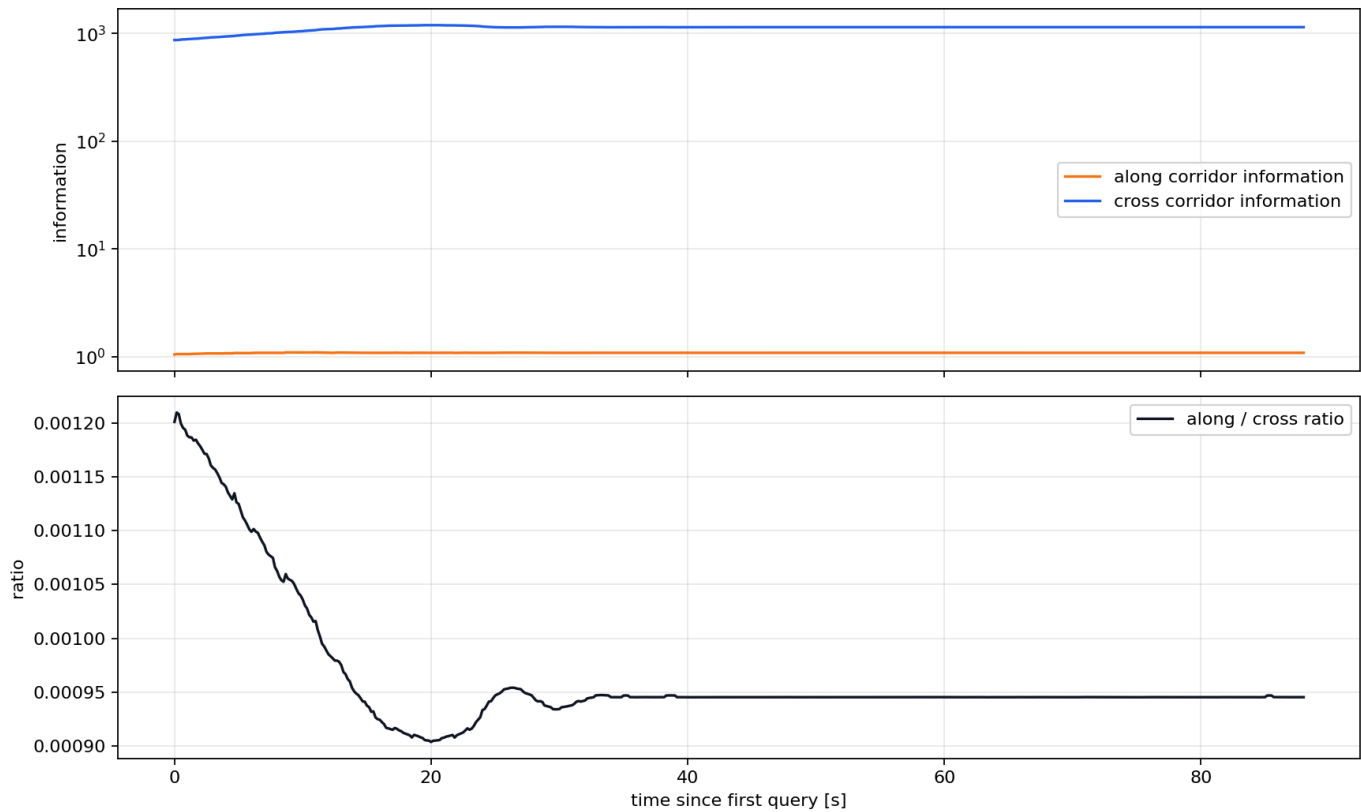
Experiment 10 LiDAR corridor degeneracy diagnostics



LiDAR FIM condition degeneracy score lidar_alpha lidar_tdop query condition degeneracy score corridor geometry LiDAR FIM lidar_alpha=0 lidar_tdop=20 LiDAR support

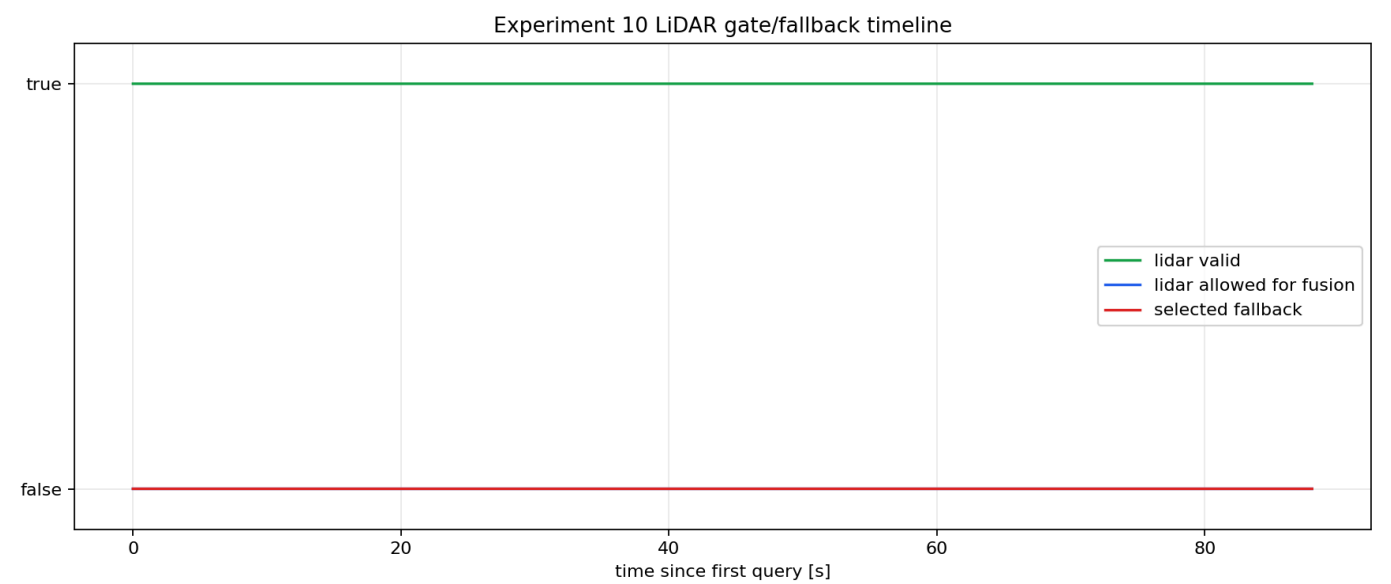
Along vs cross corridor information

Experiment 10 along-vs-cross corridor information



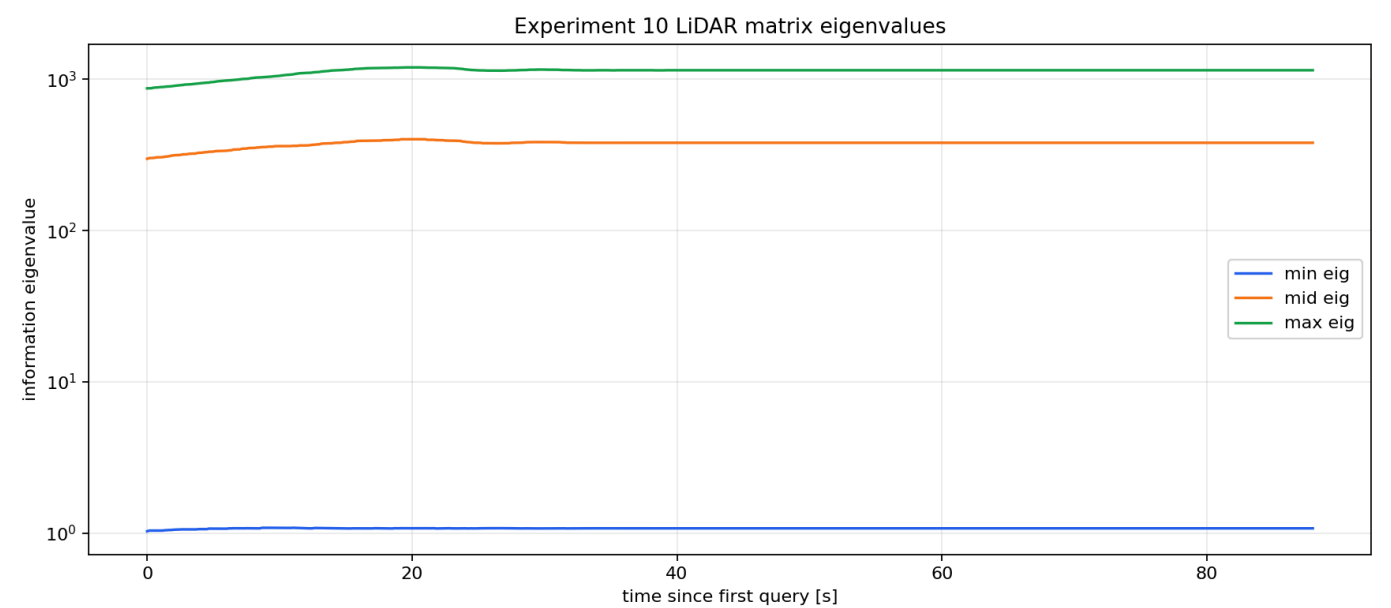
XXXXXXXXXXXX X information Y information along/cross ratio median along information 1.0857median cross information 1148.4353ratio 0.000945XXXXXXXXXXXX

LiDAR gate timeline



XXXXXXXX LiDAR validanalyzer lidar_allowed_for_fusion selected fallback flag LiDAR query valid lidar-only selected resultanalyzer query lidar_allowed_for_fusion=falsecorridor degeneracy gate XXXXXXXXXXX fusion

LiDAR eigenvalues timeline



XXXXXXXX lambda_lidar 3x3 information matrix eigenvalue eigenvalue eigenvalue LiDAR XXXXXXXXXXX information

24.6

Predictor system experiment 10 predictor_corridor_lidar_degeneracy corridor LiDAR geometryLiDAR-only selected output 529
predictor query CSVmetadatalatency profiling E10 PNG IAP log

corridor geometry LiDAR FIM median weak_axis_ratio=0.0009450.25 median
degeneracy_score=1057.7632analyzer query lidar_allowed_for_fusion=falsegate reason XXXXXXXXXXX predictor
system logging corridor LiDAR degeneracyfusion gate XXXXXXXXXXX system-level evidence

25. System Predictor stale snapshot guard

system experiment

```
predictor_stale_snapshot_guard
```

0000000000

```
src/iap/results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3
```

25.1 0000

00000 predictor 0 fusion output mode 00 input freshness guard00000 probe 00000 stale variants000000 ROS topic000000 /iap/integrity callback 0
00000000 query rows0

0000000000

- normal variant 0 GNSS0LiDAR0prior 000000 valid FUSION selected result0
- stale_odom0stale_integrity0stale_gnss0stale_snapshot 00 stale variant 00000 invalid/fallback0
- stale rows 00000 finite selected HPL/VPL/PL0
- fallback reason 0000000000 stale source 000
- latency 00 system analyzer 000

25.2 0000

00 run 00 fused_nominal ARAIM preset0fusion predictor selected output0000 freshness guard0

00	0
experiment	predictor_stale_snapshot_guard
araim_experiment	fused_nominal
predictor_output_mode	fusion
start_planner	false
probe_stale_variant_set	e11_stale_snapshot_guard
probe_stale_age_s	2.0
predictor_enable_freshness_guard	true
predictor_max_odom_age_s	0.5
predictor_max_integrity_age_s	0.5
predictor_max_gnss_age_s	0.5
predictor_max_snapshot_age_s	0.5

000000

```
RUN_ROOT=/home/dev/ws_iap/src/iap/results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3

ros2 launch iap test_predictor.launch.py \
  experiment:=predictor_stale_snapshot_guard \
  start_rviz:=false \
  record_bag:=false \
  predictor_enable_debug_log:=true \
  validator_require_debug_logs:=true \
  run_duration_s:=90 \
  validation_duration_s:=85 \
  predictor_export_dir:=$RUN_ROOT/export

python3 src/iap/scripts/dev_predictor/analyze_predictor_system_e11.py \
  --run-dir $RUN_ROOT \
  --fail-on-threshold
```

25.3 000000

00 run 0 IAP log 0000000000

Category	00
export/	test_predictor_query_probe.csv0test_predictor_query_probe_summary.json0 test_predictor_validation_summary.json
export/	source_selection_debug.csv0gnss_epoch_debug.csv0gnss_visibility_by_query.csv0fallback_reason_by_time.csv
export/	predictor_lidar_debug.csv0predictor_lidar_primitives_debug.csv0predictor_fusion_debug.csv0 downsampled_map.csv
export/	stale_snapshot_debug.csv0stale_snapshot_audit.json0predictor_e11_analysis_summary.json

Category	
profiling/	latency_debug.csv
metadata/	run_manifest.jsonpredictor_launch_config.jsonpredictor_probe_config.json
figures/	E11_age_vs_validity.pngE11_stale_reason_timeline.pngE11_stale_source_histogram.pngE11_latency_distribution.png

25.4

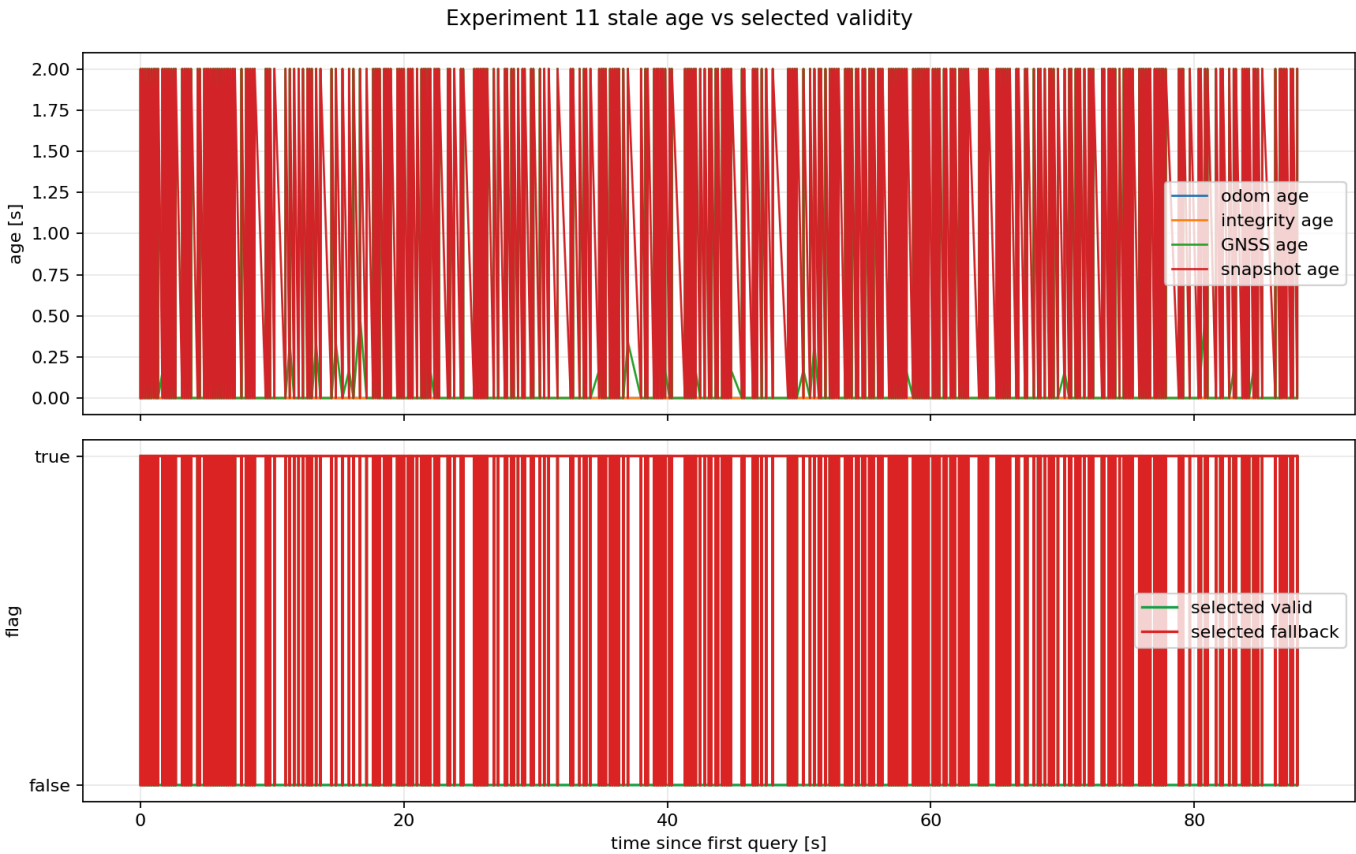
predictor_e11_analysis_summary.json passed=truefailures=[]

Query count	1590	> 0
Stale debug rows	1590	query count
Required variants	normal/stale_odom/stale_integrity/stale_gnss/stale_snapshot	
Rows per variant	318	
Normal selected valid ratio	1.000	> 0.80
Stale selected valid count	0	0
Stale selected fallback ratio	1.000	1.000
Stale reason nonempty ratio	1.000	1.000
Stale nonfinite selected PL ratio	1.000	1.000
Bad age rows	0	0
Bad reason rows	0	0
Finite PL rows in stale variants	0	0
module total latency p95	39.07 us	< 10000 us

stale variant age stale_odom median odom age 2.0 sstale_integrity median integrity age 2.0 sstale_gnss median GNSS age 2.0 sstale_snapshot median snapshot age 2.0 sfallback reason histogram stale_odom=318stale_integrity=318stale_gnss_epoch=318stale_snapshot=318

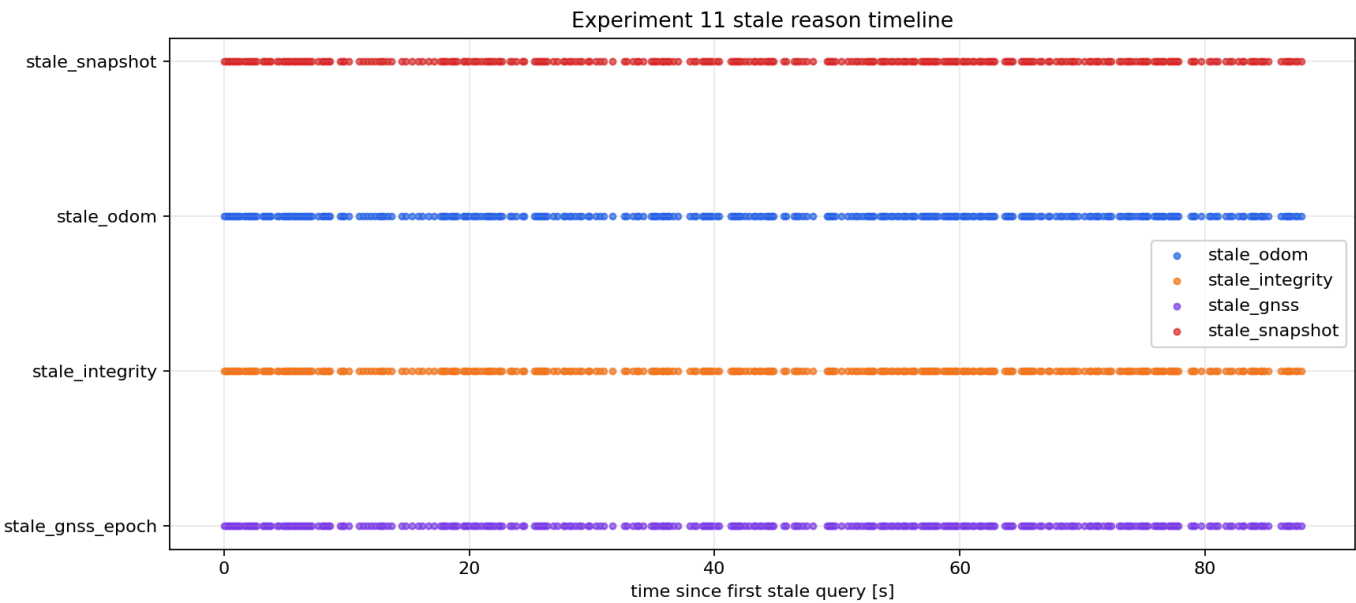
25.5

Age vs validity



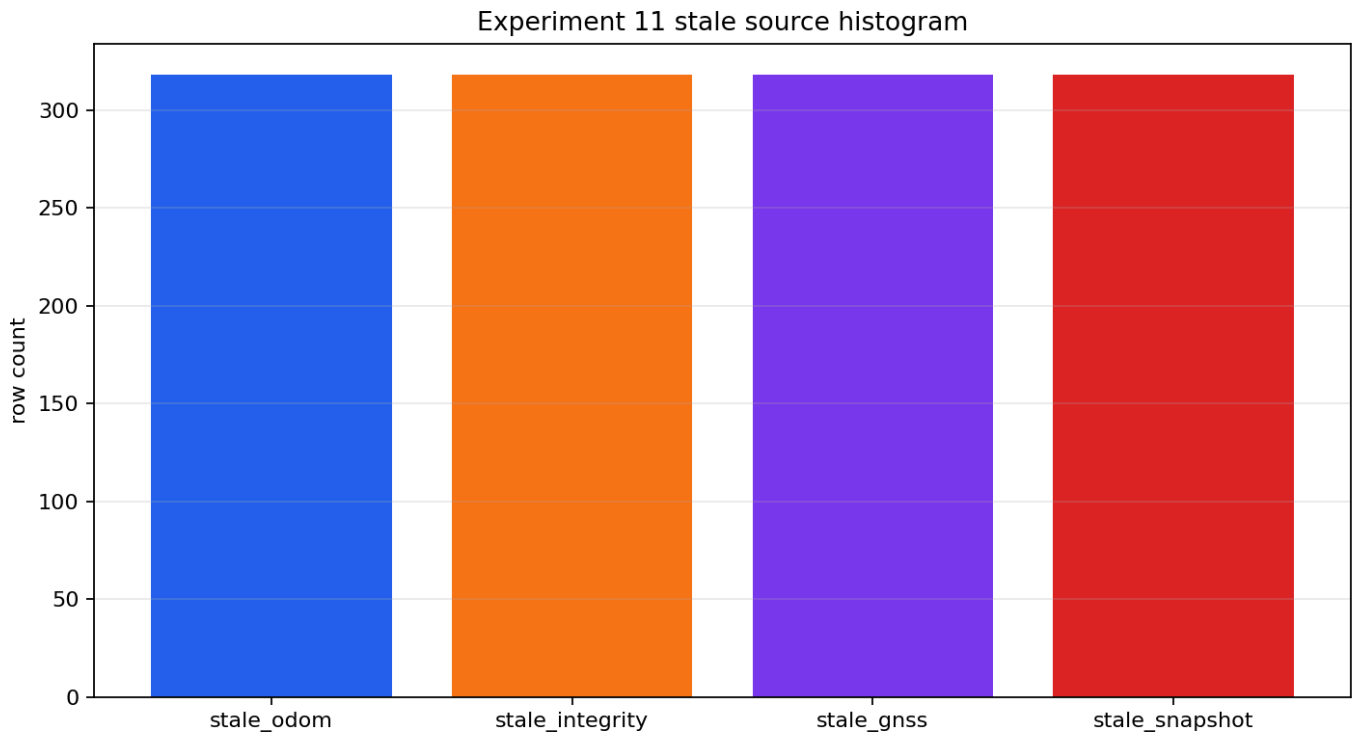
stale variant odomintegrityGNSS epoch snapshot age selected valid/fallback normal rows age 0 selected valid
stale rows source age 2.0 s 0.5 s freshness threshold fallback

Stale reason timeline



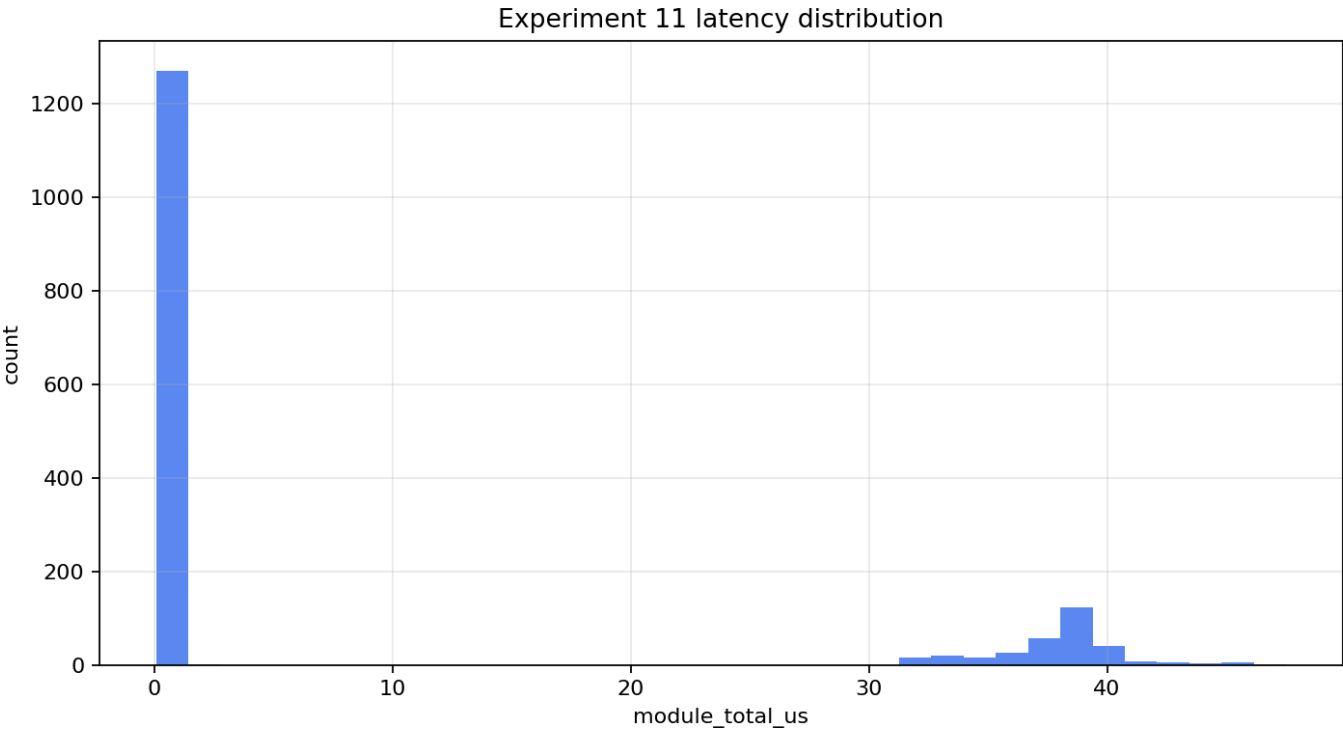
query stale variant fallback reason stale source stale_odomstale_integritystale_gnss_epoch
stale_snapshot reason reason mismatch

Stale source histogram



stale source fallback reason stale reason 318 E11 freshness guard stale source fallback reason

Latency distribution



module_total_us latency E11 normal query fusion advisory stale query freshness guard invalid/fallback p95 39.07 us 10000 us

25.6

Predictor system experiment 11 predictor_stale_snapshot_guard fusion output mode 1590 predictor query stale variant 318 CSV metadata latency profiling E11 PNG IAP log

freshness guard snapshot stale odom stale current integrity stale GNSS epoch stale snapshot normal rows valid FUSION selected result stale rows invalid/fallback finite selected PL fallback reason stale source predictor stale input silent valid snapshot advisory

26. System Predictor query latency stress

system experiment

```
predictor_query_latency_stress
```

```
src/iap/results/predictor_validation/predictor_system_predictor_query_latency_stress_run1
```

26.1

predictor fused_nominal system runtime batch query selected output FUSION /iap/integrity tick batch size 1/10/50/100 query set runtime batch size per-query latency batch total latency

- batch size 1/10/50/100
- batch selected source FUSION selected result valid
- GNSS LiDAR Fusion source validity timeout
- fallback ratio
- per-query latency batch total latency analyzer
- latency NaN/Inf IAP log CSV/JSON/PNG

26.2

run fused_nominal ARAIM preset fusion predictor selected output

experiment	predictor_query_latency_stress

項目	値
araim_experiment	fused_nominal
predictor_output_mode	fusion
start_planner	false
probe_query_set	e12_latency_stress
probe_latency_stress_batch_sizes	1,10,50,100
probe_query_min_period_s	0.0
probe_use_lidar_primitives	true
predictor_debug_max_lidar_primitives	1
validator_required_selected_source	FUSION

実行コマンド

```
RUN_ROOT=/home/dev/ws_iap/src/iap/results/predictor_validation/predictor_system_predictor_query_latency_stress_run1

ros2 launch iap test_predictor.launch.py \
  experiment:=predictor_query_latency_stress \
  start_rviz:=false \
  record_bag:=false \
  predictor_enable_debug_log:=true \
  validator_require_debug_logs:=true \
  run_duration_s:=90 \
  validation_duration_s:=85 \
  predictor_debug_max_lidar_primitives:=1 \
  predictor_export_dir:=$RUN_ROOT/export

python3 src/iap/scripts/dev_predictor/analyze_predictor_system_e12.py \
  --run-dir $RUN_ROOT \
  --fail-on-threshold
```

26.3 実行結果

実行ログ IAP log 確認

Category	項目
export/	test_predictor_query_probe.csvtest_predictor_query_probe_summary.json test_predictor_validation_summary.json
export/	source_selection_debug.csvgnss_epoch_debug.csvgnss_visibility_by_query.csvfallback_reason_by_time.csv
export/	predictor_lidar_debug.csvpredictor_lidar_primitives_debug.csvpredictor_fusion_debug.csv downsampled_map.csv
export/	latency_stress_tick_debug.csvlatency_stress.csvpredictor_e12_analysis_summary.json
profiling/	latency_debug.csv
metadata/	run_manifest.jsonpredictor_launch_config.jsonpredictor_probe_config.json
figures/	E12_latency_vs_query_count.pngE12_p95_p99_latency.pngE12_batch_total_latency.png E12_source_validity_by_batch.png

26.4 結果確認

predictor_e12_analysis_summary.json 確認 passed=truefailures=[]

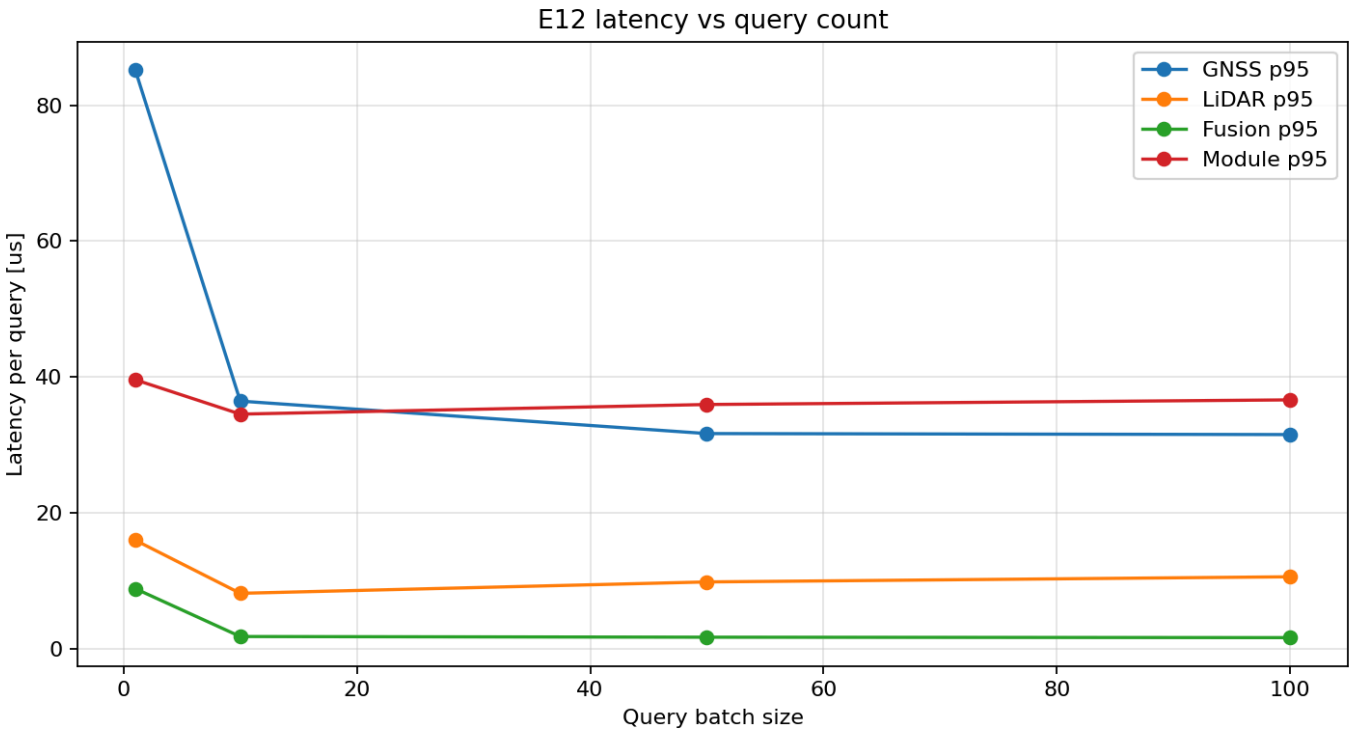
項目	Batch 1	Batch 10	Batch 50	Batch 100	確認
Query count	306	3060	15300	30600	> 0
Selected source FUSION ratio	1.000	1.000	1.000	1.000	> 0.95
Selected valid ratio	1.000	1.000	1.000	1.000	> 0.95
GNSS valid ratio	0.9706	0.9706	0.9706	0.9706	> 0.95
LiDAR valid ratio	1.000	1.000	1.000	1.000	> 0.95
Fused valid ratio	1.000	1.000	1.000	1.000	> 0.95
Fallback ratio	0.000	0.000	0.000	0.000	< 0.05
module_total_us p95	39.59	34.53	35.94	36.62	p99 < 10000 us

	Batch 1	Batch 10	Batch 50	Batch 100	
module_total_us p99	43.53	36.43	37.03	37.89	< 10000 us
batch_total_us p95	289.73	1663.13	8181.04	16296.95	< 200000 us

49266 predictor query 49266 per-query latency rows 1224 batch tick rows batch nonfinite latency rows batch 100 module_total_us p95=36.62 us batch 1 3.0x batch total latency query count 100-query batch p95 16.30 ms

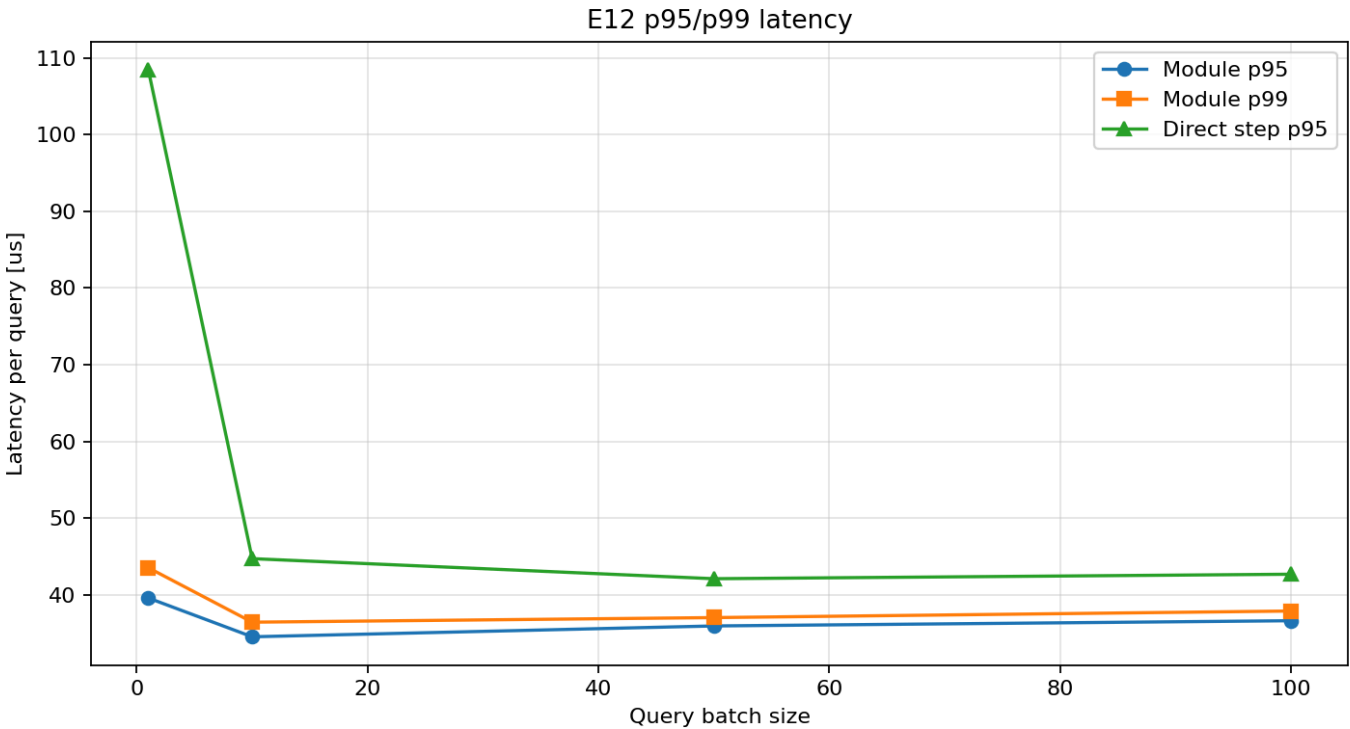
26.5

Latency vs query count



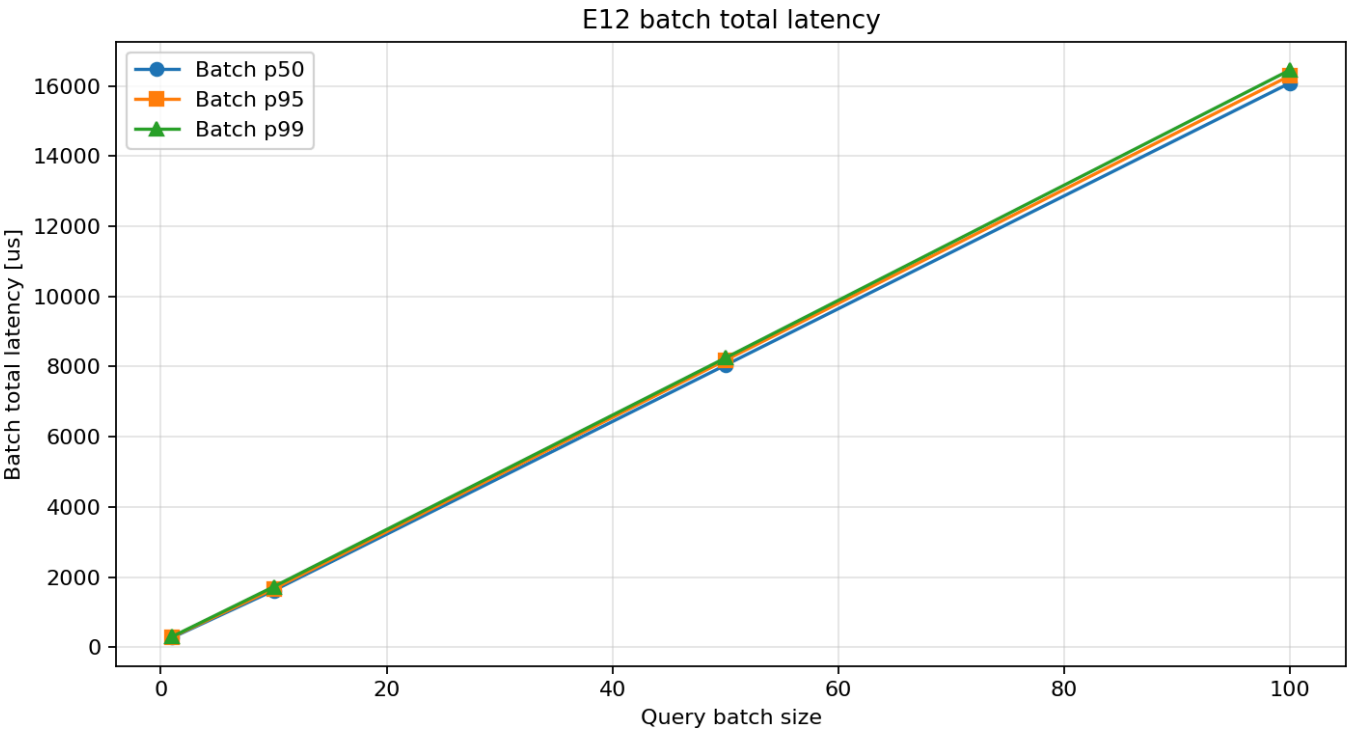
GNSSLiDARFusion module total per-query p95 latency batch size batch size 1 100 module per-query p95 35-40 us batch size

P95/P99 latency



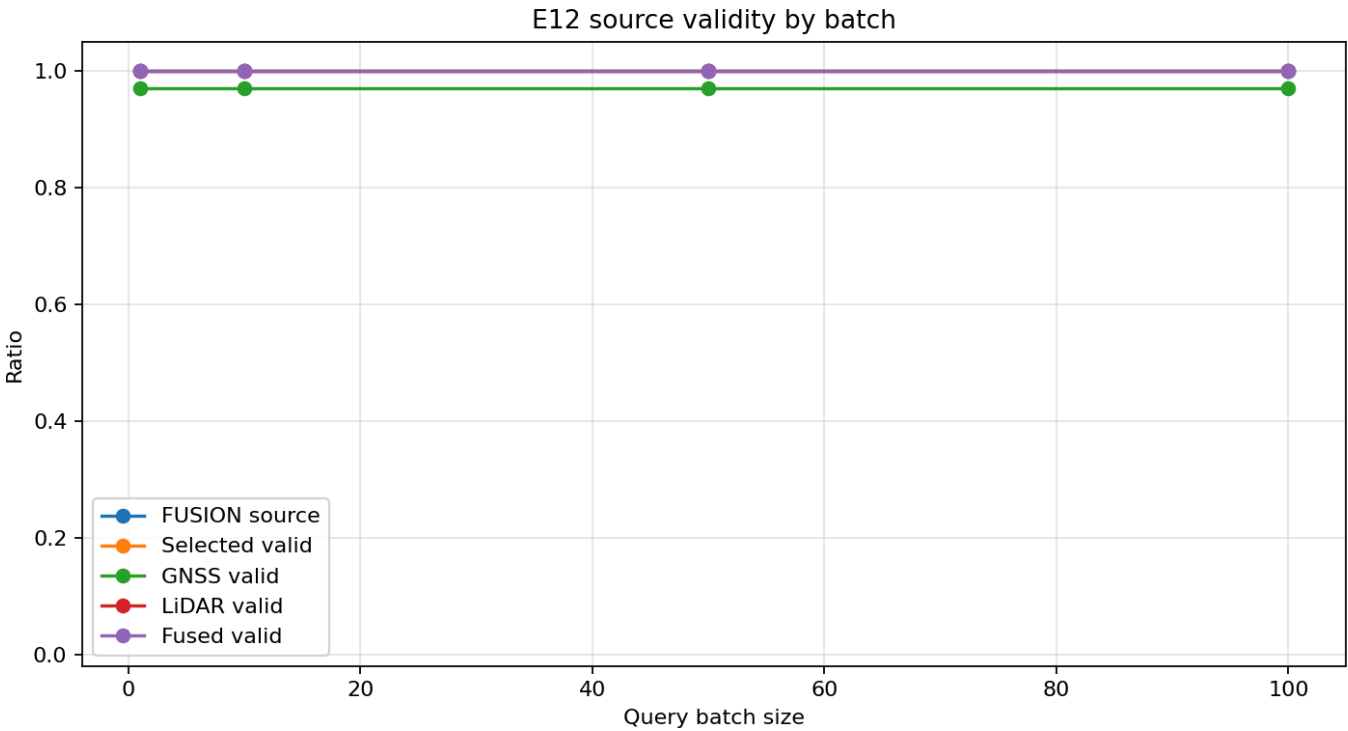
module total p95/p99 latency direct total step p95 batch module p99 44 us 10000 us high-frequency query tail latency

Batch total latency



batch p50/p95/p99 batch total latency query count 100-query batch p95 16296.95 us 200000 us tick 100 advisory query

Source validity by batch



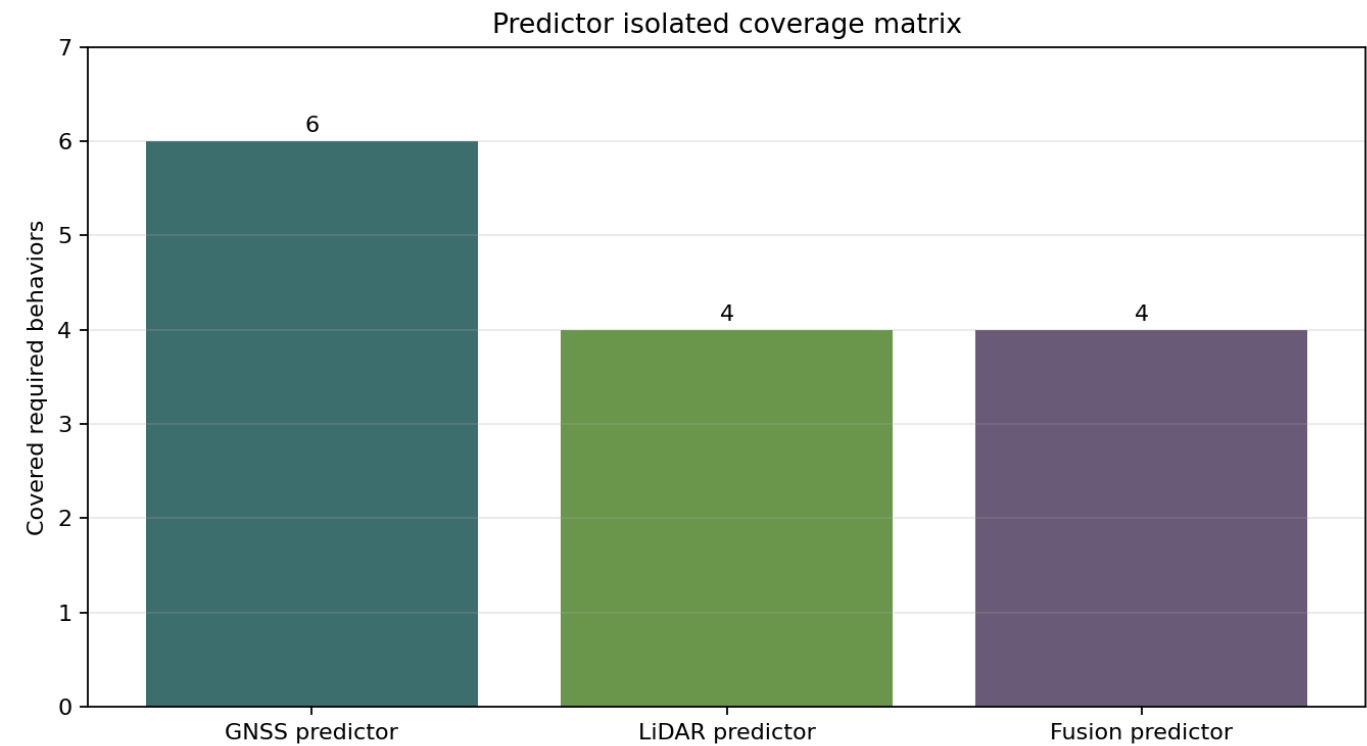
selected source selected valid GNSS valid LiDAR valid fused valid ratio batch selected source FUSION selected/LiDAR/Fused valid ratio 1.0 GNSS valid ratio 0.9706 0.95 stress query source timeout selected output

26.6

Predictor system experiment 12 predictor_query_latency_stress fused_nominal runtime fusion selected output 1/10/50/100 batch sweep 49266 predictor query CSV metadata latency profiling E12 PNG IAP log

batch size selected source fallback batch selected source FUSION ratio selected valid ratio 1.0 fallback ratio 0.0
per-query module tail latency batch 100 module_total_us p99=37.89 us batch total latency query count batch 100
batch_total_us p95=16.30 ms predictor advisory query system runtime batch

27.



```
src/iap/docs/dev_predictor/predictor_isolated_test_coverage_artifacts/
```

colcon_test_isolated_final.log
ctest_isolated_final.log
test_predictor_module_gtest_list.txt
test_lidar_observability_fim_gtest_list.txt
test_predicted_araim_gtest_list.txt
test_predictor_module_gtest_result.json
test_lidar_observability_fim_gtest_result.json
test_predicted_araim_gtest_result.json
predictor_isolated_coverage_matrix.csv
predictor_isolated_coverage_matrix.json
predictor_isolated_test_results_summary.json
predictor_isolated_test_results.png
predictor_isolated_coverage_matrix.png
gnss_open_sky_satellite_geometry.png
gnss_occlusion_visibility_counts.png
lidar_feature_rich_primitives.png
lidar_fallback_diagnostics.png

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lidar_primitive_generation_checks.png
fusion_lambda_addition_schematic.png
gnss_geometry_sweep.csv
gnss_geometry_sweep_pdop_pl.png
gnss_geometry_sweep_skyplots.png
gnss_sigma_sweep.csv
gnss_sigma_sweep_pl.png
gnss_occlusion_pl_degradation.csv
gnss_occlusion_pl_degradation.png
current_advisory_separation.csv
current_vs_advisory_isolated.png
lidar_corridor_degeneracy.csv
lidar_corridor_geometry_topdown.png
lidar_corridor_eigenvalues.png
lidar_corridor_axis_information.png
lidar_primitive_generation_parameters.csv
lidar_primitive_generation_parameter_curves.png
fusion_gate_safety.csv
fusion_gate_selected_vs_gnss.png
fusion_lambda_matrices.json
fusion_lambda_heatmaps.png
fusion_lambda_eigenvalues.png
../../results/predictor_validation/predictor_system_predictor_gnss_open_sky_only_run1/export/test_predictor_query_probe.csv
../../results/predictor_validation/predictor_system_predictor_gnss_open_sky_only_run1/export/test_predictor_query_probe_summary.csv
../../results/predictor_validation/predictor_system_predictor_gnss_open_sky_only_run1/export/predictor_e1_analysis_summary.json
../../results/predictor_validation/predictor_system_predictor_gnss_open_sky_only_run1/export/gnss_epoch_debug.csv
../../results/predictor_validation/predictor_system_predictor_gnss_open_sky_only_run1/export/gnss_visibility_by_query.csv
../../results/predictor_validation/predictor_system_predictor_gnss_open_sky_only_run1/export/source_selection_debug.csv
../../results/predictor_validation/predictor_system_predictor_gnss_open_sky_only_run1/profiling/latency_debug.csv
../../results/predictor_validation/predictor_system_predictor_gnss_open_sky_only_run1/metadata/run_manifest.json
../../results/predictor_validation/predictor_system_predictor_gnss_open_sky_only_run1/figures/E1_selected_pl_timeline.png
../../results/predictor_validation/predictor_system_predictor_gnss_open_sky_only_run1/figures/E1_gnss_geometry_timeline.png
../../results/predictor_validation/predictor_system_predictor_gnss_open_sky_only_run1/figures/E1_current_vs_advisory.png
../../results/predictor_validation/predictor_system_predictor_gnss_open_sky_only_run1/figures/E1_latency_distribution.png
../../results/predictor_validation/predictor_system_predictor_gnss_open_sky_only_run1/figures/E1_query_spatial_map.png
../../results/predictor_validation/predictor_system_predictor_lidar_feature_rich_only_bag/export/test_predictor_query_probe.csv
../../results/predictor_validation/predictor_system_predictor_lidar_feature_rich_only_bag/export/test_predictor_query_probe_summary.csv
../../results/predictor_validation/predictor_system_predictor_lidar_feature_rich_only_bag/export/test_predictor_validation_summary.csv
../../results/predictor_validation/predictor_system_predictor_lidar_feature_rich_only_bag/export/predictor_e2_analysis_summary.csv
../../results/predictor_validation/predictor_system_predictor_lidar_feature_rich_only_bag/export/predictor_lidar_debug.csv
../../results/predictor_validation/predictor_system_predictor_lidar_feature_rich_only_bag/export/predictor_lidar_primitives.csv
../../results/predictor_validation/predictor_system_predictor_lidar_feature_rich_only_bag/export/fallback_reason_by_time.csv
../../results/predictor_validation/predictor_system_predictor_lidar_feature_rich_only_bag/export/downsampled_map.csv
../../results/predictor_validation/predictor_system_predictor_lidar_feature_rich_only_bag/profiling/latency_debug.csv
../../results/predictor_validation/predictor_system_predictor_lidar_feature_rich_only_bag/metadata/run_manifest.json

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../../../../results/predictor_validation/predictor_system_predictor_lidar_feature_rich_only_bag/figures/E2_selected_pl_timeline.png

../../../../results/predictor_validation/predictor_system_predictor_lidar_feature_rich_only_bag/figures/E2_lidar_diagnostics_timeline.png

../../../../results/predictor_validation/predictor_system_predictor_lidar_feature_rich_only_bag/figures/E2_lidar_eigenvalues_timeline.png

../../../../results/predictor_validation/predictor_system_predictor_lidar_feature_rich_only_bag/figures/E2_lidar_primitives_topdown.png

../../../../results/predictor_validation/predictor_system_predictor_lidar_feature_rich_only_bag/figures/E2_lidar_normal_distribution.png

../../../../results/predictor_validation/predictor_system_predictor_fusion_nominal_run1/export/test_predictor_query_probe.csv

../../../../results/predictor_validation/predictor_system_predictor_fusion_nominal_run1/export/test_predictor_query_probe_summary.json

../../../../results/predictor_validation/predictor_system_predictor_fusion_nominal_run1/export/test_predictor_validation_summary.json

../../../../results/predictor_validation/predictor_system_predictor_fusion_nominal_run1/export/predictor_e3_analysis_summary.json

../../../../results/predictor_validation/predictor_system_predictor_fusion_nominal_run1/export/predictor_fusion_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_fusion_nominal_run1/export/predictor_lidar_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_fusion_nominal_run1/export/gnss_visibility_by_query.csv

../../../../results/predictor_validation/predictor_system_predictor_fusion_nominal_run1/export/fallback_reason_by_time.csv

../../../../results/predictor_validation/predictor_system_predictor_fusion_nominal_run1/profiling/latency_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_fusion_nominal_run1/metadata/run_manifest.json

../../../../results/predictor_validation/predictor_system_predictor_fusion_nominal_run1/figures/E3_source_hpl_vpl_timeline.png

../../../../results/predictor_validation/predictor_system_predictor_fusion_nominal_run1/figures/E3_lambda_contribution_timeline.png

../../../../results/predictor_validation/predictor_system_predictor_fusion_nominal_run1/figures/E3_lambda_sum_error_timeline.png

../../../../results/predictor_validation/predictor_system_predictor_fusion_nominal_run1/figures/E3_lambda_condition_timeline.png

../../../../results/predictor_validation/predictor_system_predictor_fusion_nominal_run1/figures/E3_source_selection_histogram.png

../../../../results/predictor_validation/predictor_system_predictor_fusion_nominal_run1/figures/E3_latency_distribution.png

../../../../results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/export/test_predictor_query_probe.csv

../../../../results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/export/test_predictor_query_probe_summary.json

../../../../results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/export/predictor_e4_analysis_summary.json

../../../../results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/export/gnss_visibility_by_query.csv

../../../../results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/export/satellite_used_mask.csv

../../../../results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/export/scenario_region_labels.csv

../../../../results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/export/predictor_lidar_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/export/predictor_lidar_primitives.png

../../../../results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/export/predictor_fusion_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/export/source_selection_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/export/fallback_reason_by_time.csv

../../../../results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/profiling/latency_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/metadata/run_manifest.json

../../../../results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/figures/E4_gnss_degradation_timeline.png

../../../../results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/figures/E4_lidar_support_timeline.png

../../../../results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/figures/E4_gnss_vs_fusion_hpl_vpl.png

../../../../results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/figures/E4_lambda_contribution_timeline.png

../../../../results/predictor_validation/predictor_system_predictor_gnss_degraded_lidar_good_run1/figures/E4_spatial_query_map_colormap.png

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/export/test_predictor_query_probe.csv

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/export/test_predictor_query_probe_summary.json

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/export/test_predictor_validation_summary.json

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/export/predictor_e5_analysis_summary.json

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../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/export/lidar_gate_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/export/gnss_visibility_by_query.csv

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/export/predictor_lidar_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/export/predictor_lidar_primitives_c

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/export/predictor_fusion_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/export/source_selection_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/export/fallback_reason_by_time.csv

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/export/downsampled_map.csv

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/profiling/latency_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/metadata/run_manifest.json

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/metadata/predictor_probe_config.js

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/figures/E5_lidar_degradation_timeli

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/figures/E5_gnss_stability_timeline.

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/figures/E5_fusion_gate_timeline.png

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/figures/E5_selected_vs_gnss_pl.png

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/figures/E5_lidar_primitives_topdown

../../../../results/predictor_validation/predictor_system_predictor_lidar_sparse_gnss_good_run1/figures/E5_lidar_eigenvalues_timeli

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/export/test_predictor_query_pro

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/export/test_predictor_query_pro

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/export/test_predictor_validation

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/export/predictor_e6_analysis_su

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/export/outage_event_times.json

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/export/recovery_metrics.json

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/export/gnss_visibility_by_query

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/export/predictor_lidar_debug.cs

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/export/predictor_lidar_primitiv

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/export/predictor_fusion_debug.c

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/export/source_selection_debug.c

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/export/fallback_reason_by_time.

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/profiling/latency_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/metadata/run_manifest.json

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/metadata/predictor_probe_config

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/figures/E6_outage_window_timeli

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/figures/E6_fallback_reason_time

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/figures/E6_source_selection_tin

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/figures/E6_recovery_latency.png

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/figures/E6_gnss_n_visible_timel

../../../../results/predictor_validation/predictor_system_predictor_gnss_outage_lidar_recovery_run4/figures/E6_selected_pl_timeline

../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/export/test_predictor_query_probe.csv

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../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/export/test_predictor_query_probe_summar
../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/export/predictor_e7_analysis_summary.js
../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/export/invalid_output_audit.json
../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/export/negative_case_rows.csv
../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/export/snapshot_debug.csv
../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/export/current_vs_advisory_debug.csv
../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/export/source_selection_debug.csv
../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/export/fallback_reason_by_time.csv
../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/export/predictor_fusion_debug.csv
../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/export/predictor_lidar_debug.csv
../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/export/gnss_visibility_by_query.csv
../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/profiling/latency_debug.csv
../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/metadata/run_manifest.json
../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/metadata/predictor_probe_config.json
../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/figures/E7_valid_fallback_flags.png
../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/figures/E7_fallback_reason_histogram.pr
../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/figures/E7_selected_pl_timeline.png
../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/figures/E7_current_vs_selected_pl.png
../../../../results/predictor_validation/predictor_system_predictor_no_source_negative_run1/figures/E7_source_selection_timeline.pr
../../../../results/predictor_validation/predictor_system_predictor_current_advisory_separation_run1/export/test_predictor_query_pr
../../../../results/predictor_validation/predictor_system_predictor_current_advisory_separation_run1/export/test_predictor_query_pr
../../../../results/predictor_validation/predictor_system_predictor_current_advisory_separation_run1/export/test_predictor_validati
../../../../results/predictor_validation/predictor_system_predictor_current_advisory_separation_run1/export/predictor_e8_analysis_s
../../../../results/predictor_validation/predictor_system_predictor_current_advisory_separation_run1/export/current_advisory_separa
../../../../results/predictor_validation/predictor_system_predictor_current_advisory_separation_run1/export/current_advisory_variar
../../../../results/predictor_validation/predictor_system_predictor_current_advisory_separation_run1/export/source_selection_debug.
../../../../results/predictor_validation/predictor_system_predictor_current_advisory_separation_run1/export/gnss_visibility_by_quer
../../../../results/predictor_validation/predictor_system_predictor_current_advisory_separation_run1/export/fallback_reason_by_time
../../../../results/predictor_validation/predictor_system_predictor_current_advisory_separation_run1/profiling/latency_debug.csv
../../../../results/predictor_validation/predictor_system_predictor_current_advisory_separation_run1/metadata/run_manifest.json
../../../../results/predictor_validation/predictor_system_predictor_current_advisory_separation_run1/metadata/predictor_probe_conf
../../../../results/predictor_validation/predictor_system_predictor_current_advisory_separation_run1/figures/E8_current_vs_advisory
../../../../results/predictor_validation/predictor_system_predictor_current_advisory_separation_run1/figures/E8_copied_current_flag
../../../../results/predictor_validation/predictor_system_predictor_current_advisory_separation_run1/figures/E8_selected_vs_gnss_ti
../../../../results/predictor_validation/predictor_system_predictor_current_advisory_separation_run1/figures/E8_variant_latency_dis
../../../../results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1/export/gnss_sigma_sweep_system.csv
../../../../results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1/export/predictor_e9_analysis_summar

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[../results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1/scale_1x/export/test_predictor_query_prototype.csv](#)

[../results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1/scale_2x/export/test_predictor_query_prototype.csv](#)

[../results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1/scale_4x/export/test_predictor_query_prototype.csv](#)

[../results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1/scale_1x/export/gnss_visibility_by_scale.csv](#)

[../results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1/scale_2x/export/gnss_visibility_by_scale.csv](#)

[../results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1/scale_4x/export/gnss_visibility_by_scale.csv](#)

[../results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1/scale_1x/profiling/latency_debug.csv](#)

[../results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1/scale_2x/profiling/latency_debug.csv](#)

[../results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1/scale_4x/profiling/latency_debug.csv](#)

[../results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1/scale_1x/metadata/run_manifest.json](#)

[../results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1/scale_2x/metadata/run_manifest.json](#)

[../results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1/scale_4x/metadata/run_manifest.json](#)

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[../results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1/scale_2x/metadata/predictor_probe_config.yaml](#)

[../results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1/scale_4x/metadata/predictor_probe_config.yaml](#)

[../results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1/figures/E9_sigma_vs_pl.png](#)

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[../results/predictor_validation/predictor_system_predictor_gnss_sigma_degradation_run1/figures/E9_latency_by_scale.png](#)

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[../results/predictor_validation/predictor_system_predictor_corridor_lidar_degeneracy_run1/export/test_predictor_query_prototype.csv](#)

[../results/predictor_validation/predictor_system_predictor_corridor_lidar_degeneracy_run1/export/test_predictor_validation_prototype.csv](#)

[../results/predictor_validation/predictor_system_predictor_corridor_lidar_degeneracy_run1/export/predictor_e10_analysis_summary.csv](#)

[../results/predictor_validation/predictor_system_predictor_corridor_lidar_degeneracy_run1/export/lidar_corridor_degeneracy.csv](#)

[../results/predictor_validation/predictor_system_predictor_corridor_lidar_degeneracy_run1/export/predictor_lidar_debug.csv](#)

[../results/predictor_validation/predictor_system_predictor_corridor_lidar_degeneracy_run1/export/predictor_lidar_primitive.csv](#)

[../results/predictor_validation/predictor_system_predictor_corridor_lidar_degeneracy_run1/export/predictor_fusion_debug.csv](#)

[../results/predictor_validation/predictor_system_predictor_corridor_lidar_degeneracy_run1/export/source_selection_debug.csv](#)

[../results/predictor_validation/predictor_system_predictor_corridor_lidar_degeneracy_run1/export/fallback_reason_by_time.csv](#)

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[../results/predictor_validation/predictor_system_predictor_corridor_lidar_degeneracy_run1/profiling/latency_debug.csv](#)

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[../results/predictor_validation/predictor_system_predictor_corridor_lidar_degeneracy_run1/metadata/predictor_probe_config.yaml](#)

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[../results/predictor_validation/predictor_system_predictor_corridor_lidar_degeneracy_run1/figures/E10_lidar_condition_time.png](#)

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[../results/predictor_validation/predictor_system_predictor_corridor_lidar_degeneracy_run1/figures/E10_lidar_gate_timeline.png](#)

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../../../../results/predictor_validation/predictor_system_predictor_corridor_lidar_degeneracy_run1/figures/E10_lidar_eigenvalues.tif

../../../../results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3/export/test_predictor_query_probe.csv

../../../../results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3/export/test_predictor_query_probe_summary.csv

../../../../results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3/export/test_predictor_validation_summary.csv

../../../../results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3/export/predictor_e11_analysis_summary.csv

../../../../results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3/export/stale_snapshot_audit.json

../../../../results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3/export/stale_snapshot_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3/export/predictor_fusion_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3/export/predictor_lidar_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3/export/gnss_visibility_by_query.csv

../../../../results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3/export/source_selection_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3/export/fallback_reason_by_time.csv

../../../../results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3/profiling/latency_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3/metadata/run_manifest.json

../../../../results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3/metadata/predictor_probe_config.json

../../../../results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3/figures/E11_age_vs_validity.png

../../../../results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3/figures/E11_stale_reason_timeline.png

../../../../results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3/figures/E11_stale_source_histogram.png

../../../../results/predictor_validation/predictor_system_predictor_stale_snapshot_guard_run3/figures/E11_latency_distribution.png

../../../../results/predictor_validation/predictor_system_predictor_query_latency_stress_run1/export/test_predictor_query_probe.csv

../../../../results/predictor_validation/predictor_system_predictor_query_latency_stress_run1/export/test_predictor_query_probe_summary.csv

../../../../results/predictor_validation/predictor_system_predictor_query_latency_stress_run1/export/test_predictor_validation_summary.csv

../../../../results/predictor_validation/predictor_system_predictor_query_latency_stress_run1/export/predictor_e12_analysis_summary.csv

../../../../results/predictor_validation/predictor_system_predictor_query_latency_stress_run1/export/latency_stress.csv

../../../../results/predictor_validation/predictor_system_predictor_query_latency_stress_run1/export/latency_stress_tick_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_query_latency_stress_run1/export/source_selection_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_query_latency_stress_run1/export/gnss_visibility_by_query.csv

../../../../results/predictor_validation/predictor_system_predictor_query_latency_stress_run1/export/predictor_lidar_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_query_latency_stress_run1/export/predictor_fusion_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_query_latency_stress_run1/export/fallback_reason_by_time.csv

../../../../results/predictor_validation/predictor_system_predictor_query_latency_stress_run1/profiling/latency_debug.csv

../../../../results/predictor_validation/predictor_system_predictor_query_latency_stress_run1/metadata/run_manifest.json

../../../../results/predictor_validation/predictor_system_predictor_query_latency_stress_run1/metadata/predictor_probe_config.json

../../../../results/predictor_validation/predictor_system_predictor_query_latency_stress_run1/figures/E12_latency_vs_query_count.png

../../../../results/predictor_validation/predictor_system_predictor_query_latency_stress_run1/figures/E12_p95_p99_latency.png

../../../../results/predictor_validation/predictor_system_predictor_query_latency_stress_run1/figures/E12_batch_total_latency.png

../../../../results/predictor_validation/predictor_system_predictor_query_latency_stress_run1/figures/E12_source_validity_by_batch.csv

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Predictor Isolated Test Coverage 3 GTest 39

- GNSS predictor open-sky 8 finite PL 3x3 FIM
- GNSS predictor missing epoch too few satellites excluded satellites map occlusion/skymask
- GNSS geometry sweep sigma sweep geometry degradation measurement noise inflation PDOP PL FIM diagnostics
- GNSS clock Schur complement position-only FIM
- LiDAR predictor feature-rich primitives FIM sparse/degenerate/missing input fallback diagnostics
- LiDAR corridor degeneracy LiDAR fusion gate isolated baseline
- Primitive generation PCA radius support threshold voxel sampling cloud normals
- Fusion predictor $\lambda_{pred} = \lambda_{prior} + \lambda_{gnss} + \lambda_{lidar}$ GNSS-only GNSS+LiDAR invalid prior no-source conservative selected PL query input
- System experiment 1 predictor `predictor_gnss_open_sky_only` launch `/iap/integrity` 2748 GNSS-only query valid fallback ratio 0% IAP log CSV JSON PNG
- System experiment 2 predictor `predictor_lidar_feature_rich_only` launch LiDAR-only advisory query 308 query selected `LIDAR` LiDAR valid ratio 100% median condition 7.72 LiDAR FIM primitive fallback map snapshot PNG
- System experiment 3 predictor `predictor_fusion_nominal` launch Fusion advisory query 292 query selected `FUSION` GNSS/LiDAR/Fused valid ratio 100% $\lambda_{sum_error_p95}=0.0$ Fusion FIM source selection fallback latency PNG
- System experiment 4 predictor `predictor_gnss_degraded_lidar_good` launch planner Fusion advisory query 269 query selected `FUSION` E4 median GNSS HPL/PDOP E1 open-sky baseline LiDAR condition 7.72 $\lambda_{sum_error_p95}=0.0$ GNSS degradation LiDAR support Fusion FIM derived CSV PNG
- System experiment 5 predictor `predictor_lidar_sparse_gnss_good` launch GNSS good LiDAR sparse/degraded Fusion advisory query 529 query selected `FUSION` LiDAR primitive 28 conservative gate violation 0 $\lambda_{sum_error_p95}=0.0$ LiDAR gate GNSS stability Fusion FIM latency PNG
- System experiment 6 predictor `predictor_gnss_outage_lidar_recovery` launch GNSS-required outage 326 query outage window $gnss_valid_ratio=0.0$ selected source NONE fallback reason `gnss:no_gnss_epoch` GNSS selected output valid `FUSION` `recovery_delay_s=0.03` outage/recovery metrics source selection fallback GNSS visibility latency PNG
- System experiment 7 predictor `predictor_no_source_negative` launch no-source negative 530 query selected invalid selected source NONE fallback ratio 100% finite selected PL count 0 copied current flag count 0 invalid output audit negative-case rows snapshot current/advisory copy audit latency PNG
- System experiment 8 predictor `predictor_current_advisory_separation` launch current/advisory separation 1587 analyzer query selected `GNSS` valid current variant 529 selected HPL/VPL GNSS HPL/VPL artificial current copied count 0 current/advisory separation CSV variant summary latency PNG
- System experiment 9 predictor `predictor_gnss_sigma_degradation` launch GNSS sigma sweep 1x/2x/4x run 530 query selected source GNSS fallback ratio 0% $effective_sigma_mean$ GNSS/selected HPL/VPL scale GNSS λ_{trace} scale sweep summary CSV per-scale logs metadata latency PNG
- System experiment 10 predictor `predictor_corridor_lidar_degeneracy` launch corridor LiDAR degeneracy 529 query median along information 1.0857 cross information 1148.4353 median $weak_axis_ratio=0.000945$ median $degeneracy_score=1057.7632$ `lidar_allowed_for_fusion=false` ratio 100% corridor degeneracy CSV LiDAR/Fusion debug map snapshot latency PNG
- System experiment 11 predictor `predictor_stale_snapshot_guard` launch stale snapshot guard 1590 query `normal/stale_odom/stale_integrity/stale_gnss/stale_snapshot` variant normal rows valid `FUSION` stale rows invalid fallback finite stale selected PL count 0 stale reason 318 reason mismatch stale audit stale debug source selection latency PNG
- System experiment 12 predictor `predictor_query_latency_stress` launch high-frequency query latency stress 1/10/50/100 batch sweep 49266 query batch selected source FUSION ratio selected valid ratio 100% fallback ratio 0% batch 100 `module_total_us` p99=37.89 us `batch_total_us` p95=16.30 ms latency stress CSV batch tick debug source/FIM debug metadata PNG

Predictor isolated advisory query system experiments isolated baseline